Data Management

GIT: A version control system

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The importance of version control

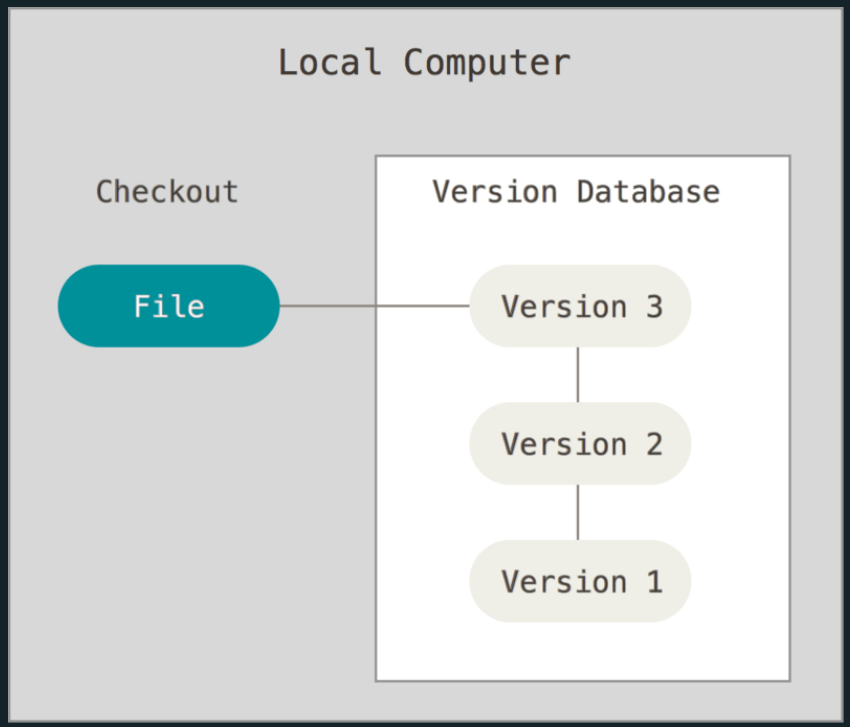
What is version control?

Version control is a way to keep track of changes to code, text, and documents. And data and outputs.

- It gives you an organized revision history
- It lets you experiment *without fear*
- It lets you go back and forth between many different versions of the same file, and see a list of the differences
- It makes (the technical aspects of) collaboration a breeze
- It lets you and your collaborators work on different versions and then merge them

From local to distributed version control system

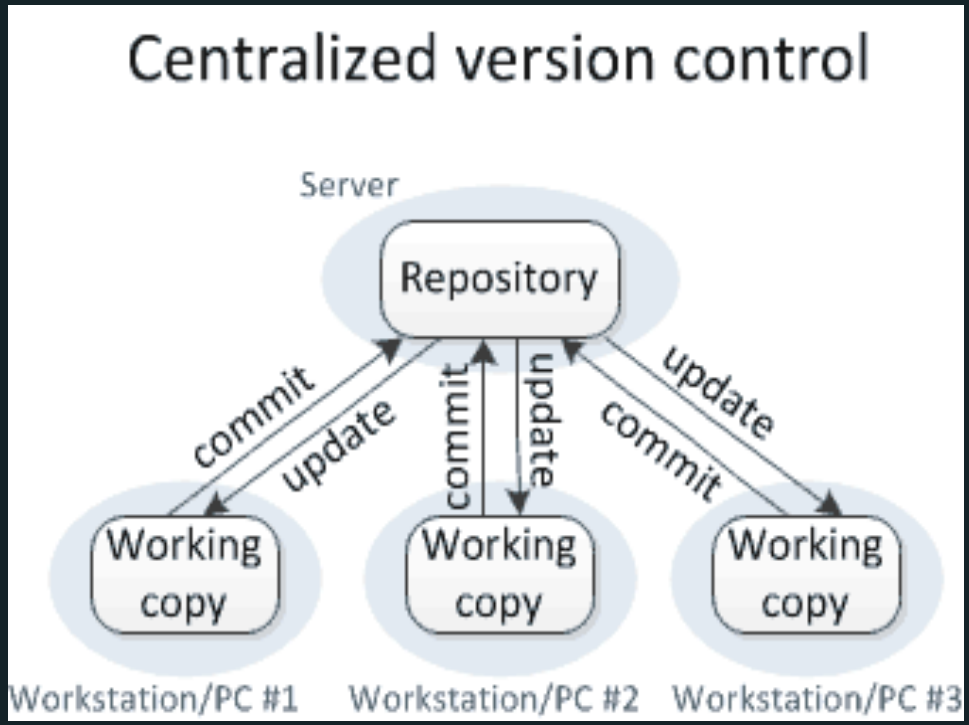
- **Local**: everything is on your computer



- No collaboration
- Not possible to retrieve files if the local machine crashes

From local to distributed version control system

- **Centralized**:
  - all files on 1 server
  - many collaborators checkout files

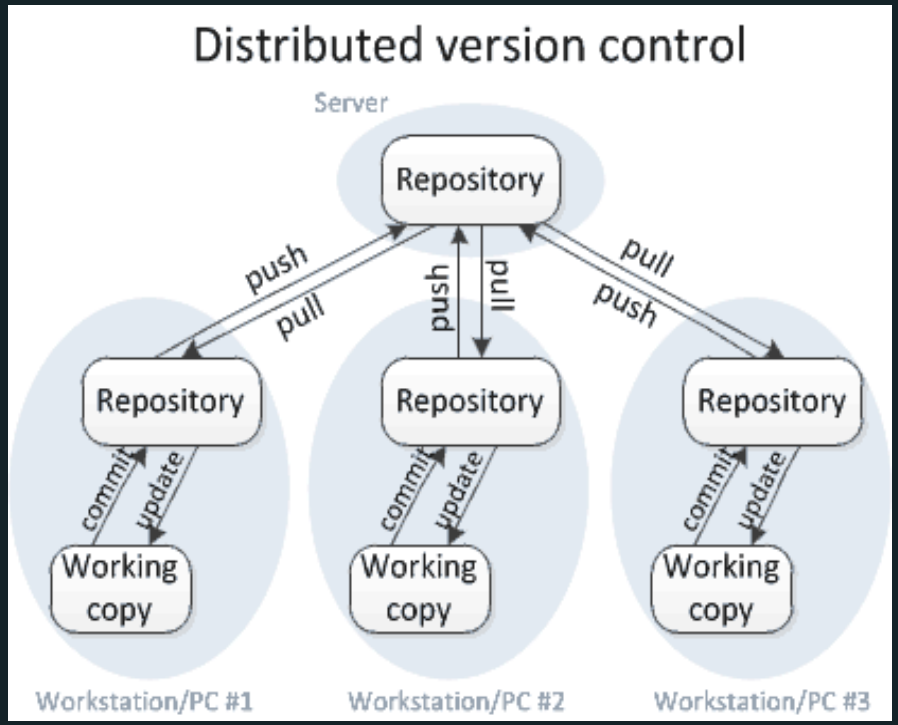


\+ Collaboration

– Not possible to retrieve files if the central server crashes

From local to distributed version control system

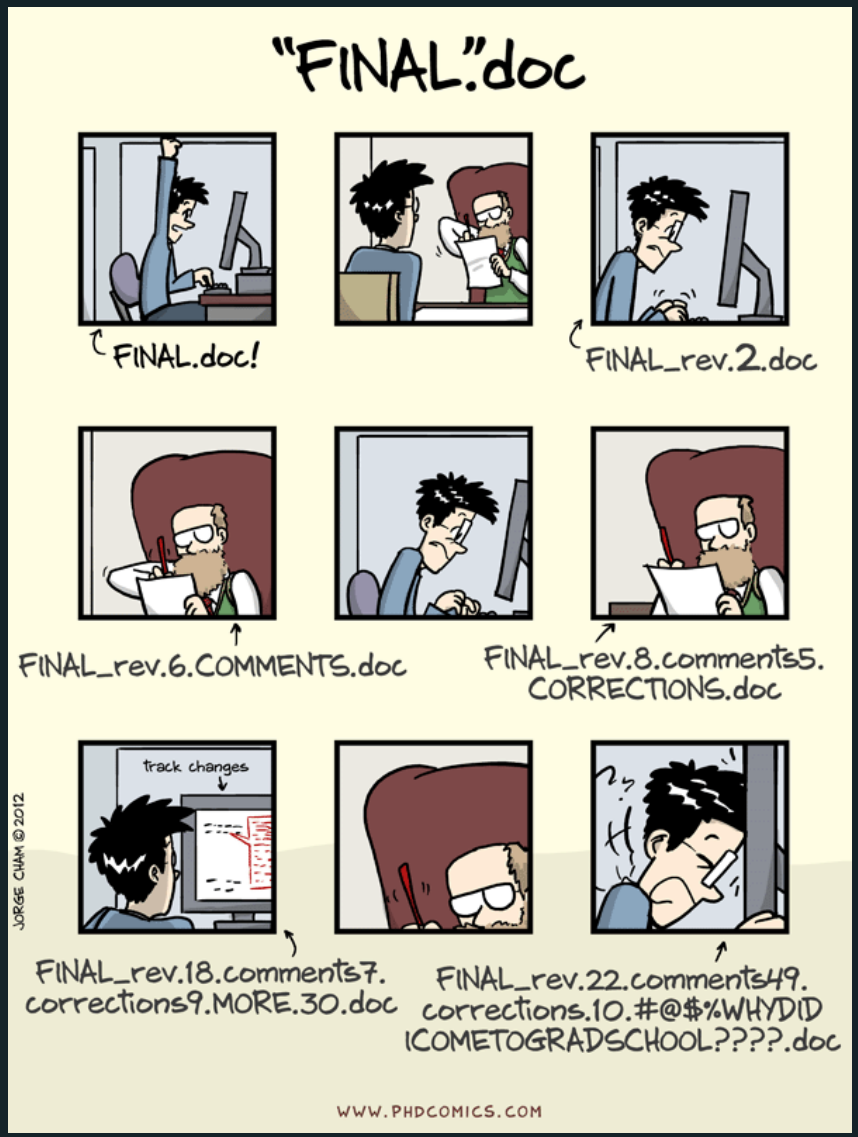
- **Distributed**:
  - one or more servers
  - many collaborators



+ Collaboration

+ Each user has their own repository and a working copy

Why bother?



Also git vs. Dropbox from a researcher's perspective

[CCL] Version control system

Enables **coordinatation** → no code change is lost or accidentally overwriten.

Provides an organized **sharing** platform → *open source* & documentation

⇒ key tool from our **project management** perspective

⇒ widely used in a companies / not enough in research:

- Software development
- Scientific researcher
- Anything involving coding (even latex)

Git(Hub)

This is Git

Source: https://xkcd.com/1597/

Git(Hub): a solution

- Git:
  - Git is a **distributed version control system**. *(Wait, what?)*
  - *Okay, try this:* Imagine if Dropbox and the "Track changes" feature in MS Word had a baby. Git would be that baby.
  - most popular *open source* version control system out there.
- GitHub
  - GitHub = **online hosting platform** that provides an array of services built on top of the Git system.
    - (Similar platforms include Bitbucket and GitLab.)

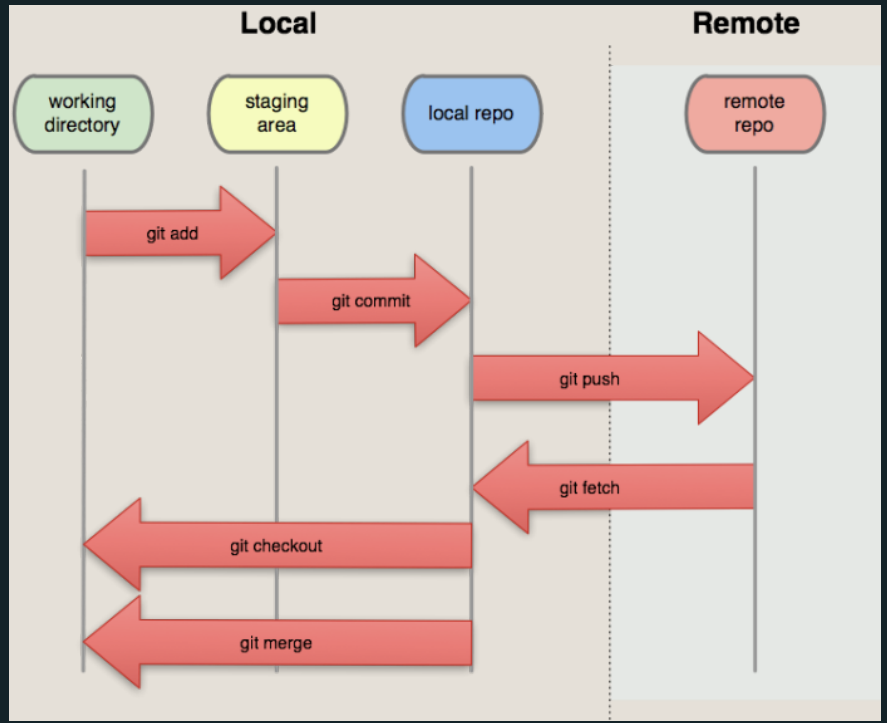
Git vs. Github

- It's important to realize that Git and GitHub are distinct things.
- We don't *need* GitHub to use Git... But it will make our lives so much easier.

→ There is a learning curve, but I promise you it's worth it.

Git model

1. You do work in your **working directory**
2. Then you add it to your **staging area**
3. Once you've staged all you changes for one discrete task, **commit** a snapshot of the staging area
4. If you have a remote repository, **push your commit**

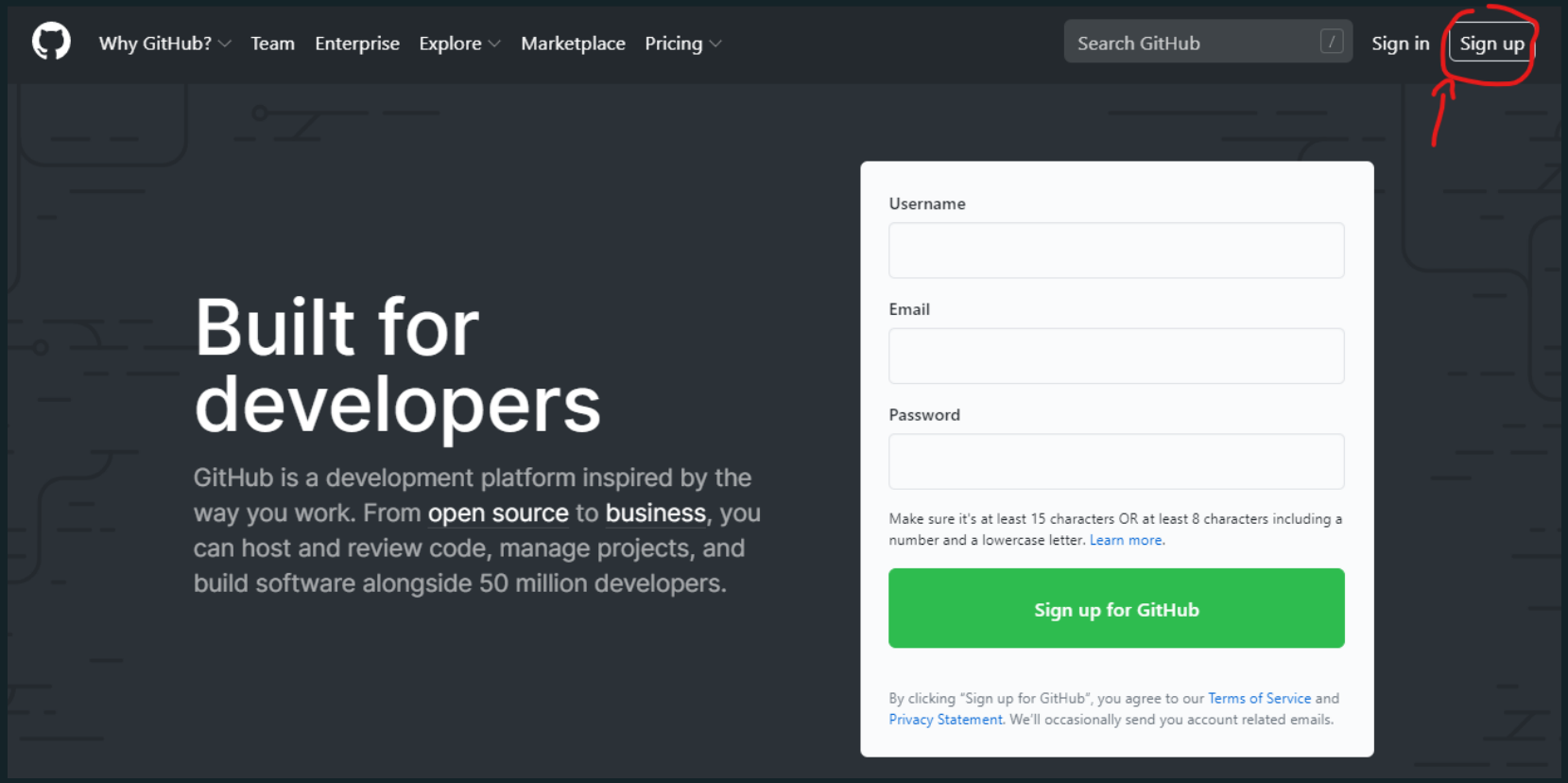


Getting started on a project

Where we create our first repository!

[Task 1] Setup GitHub account

- Navigate to GitHub's homepage + "Sign Up"
  - Go through the account setting steps ("Verify your email address"...)



Navigate to GitHub's homepage. Navigate to "Sign Up" in the top right hand side of the page.

[Task 2] Getting started with Git(Hub)

1. Install Git (Linux, Mac, Windows) if not already installed
2. Git comes with a command line interface (powerful!).
3. You might want to add a *graphical interface* to make things easier:
   - GitHub desktop
   - You can link it with your GitHub account

[Task 3] Your first (local) repository

Let's look at an example using GitHub desktop

1. Open GitHub Desktop and select File/New repository
2. Choose the name and the local directory to use
3. Start working in the directory, i.e.
    - Create some .txt file with some text
    - Commit it
    - Make a modification, and commit again: look at the changes!

[Hint] What actually is the Git repository?

- The Git local repository is associated with a particular directory
- Open the directory in your Git interface to see your options
- Git stores all its workings in that directory in a hidden subfolder called “.git”

3 special options:

- **README.md**: description of the directory
- **.gitignore**: what should be ignored by the tracking systel
- **licence** → open source?

[Hint] What should I include?

1. At a minimum:
   - Code (.do, .py, .R, .m, .jl, and so on)
   - Text files (.txt)
   - LATEX documents (.tex)
2. I also recommend:
   - Raw .csv datasets, if small (<10 MB)
3. These are binary files, so you can't see differences between versions. I recommend including them anyway.
   - PDF files
   - Word, Excel, PowerPoint files
4. Some people also include all datasets.
   - Note that GitHub doesn't allow files larger than 100 MB, or projects with total size larger than 1 GB.

For datasets, look into Git Large File Storage.

[Hint] What should I exclude?

In order to avoid driving your collaborators crazy, you must tell Git to ignore the junk files using a file called .gitignore. It looks like this:

- Junk created by LaTeX: *.synctex.gz, *.out *.log
- Junk created by Python: *.pyc

Best practice: use .gitignore to explicitly exclude everything that you don't want to include, and commit .gitignore like any other regular file.

GitHub maintains a list of standard .gitignore files for many common languages.

Backbone of git: Commits & branches

Where we commit ourselves (locally)!

Commits: saving a snapshot

"One discrete task" = a collection of changes, across multiple files (or not), that does *one thing*.

Examples:

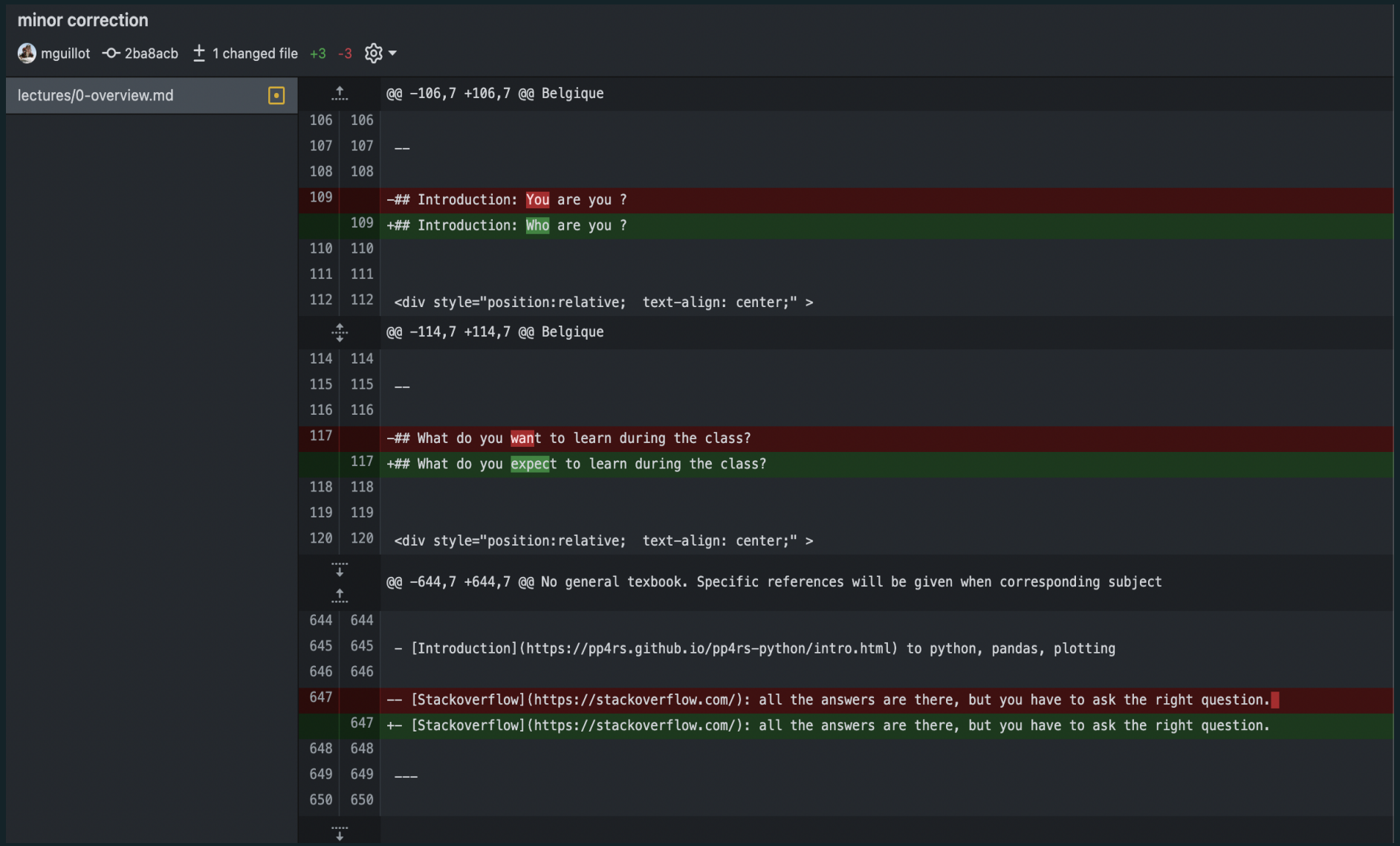
- Change the formatting of a variable from string to numeric, and treat it properly across multiple scripts
- Change your regression specification in code, in the output, and in your paper and supporting documentation
- Add a new function

Before you commit

- Your code should run properly → run tests
- No compilation erros (in Latex for example)
- Output should be consistent inside the commit (including comments)

But it's better to have *frequent commits* (that might have small mistakes) than to have *giant, infrequent* commits.

Viewing changes when committing



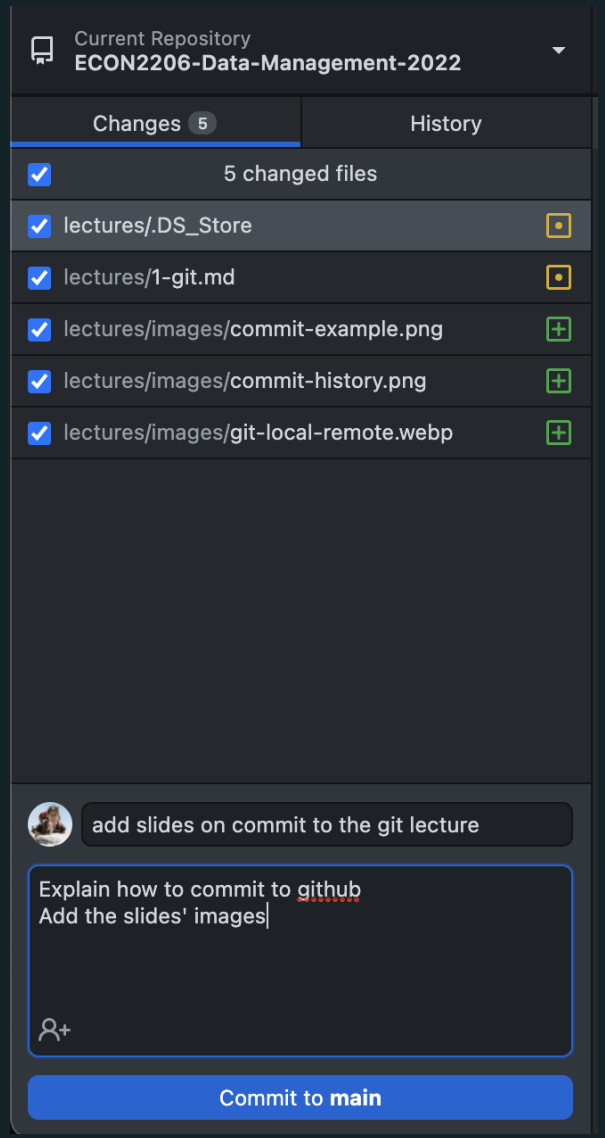
Commit message

Examples:

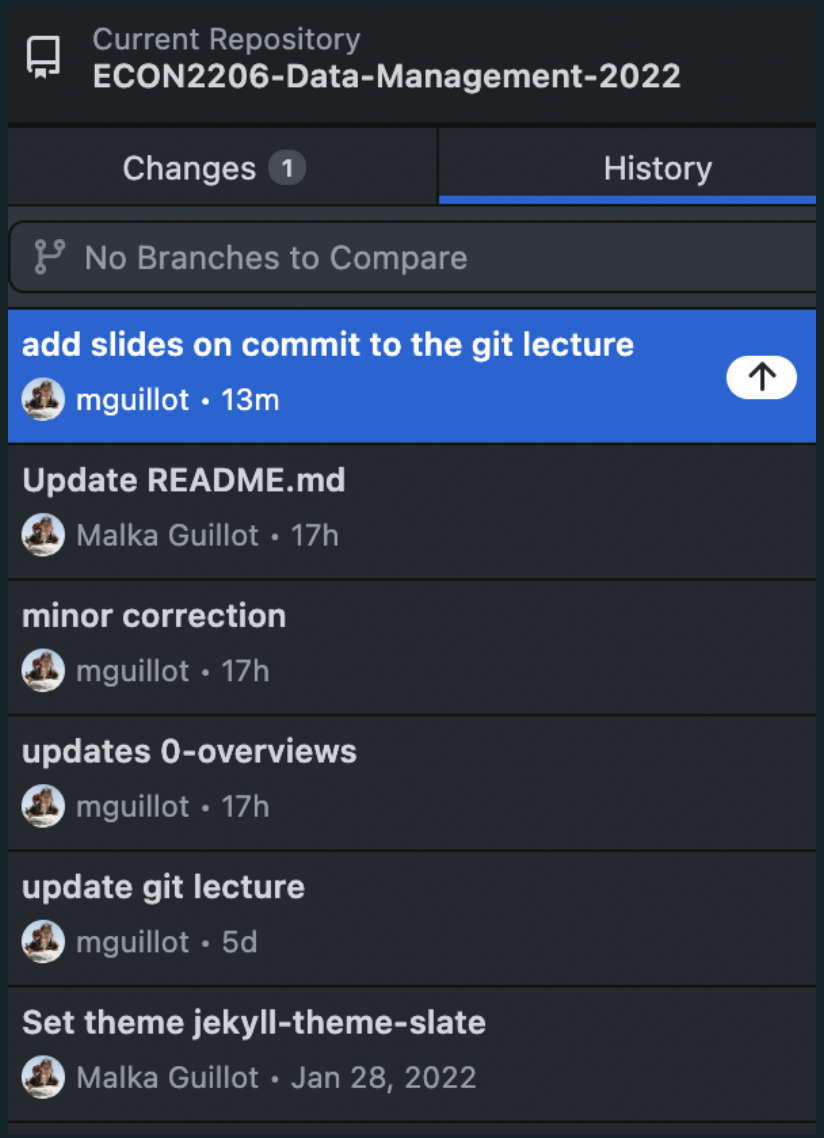
- “Change the formatting of start date variable from string to date format”
- “Add year dummies to regression specification”

→ The more detail, the more your future self will thank you.

Commit message: example

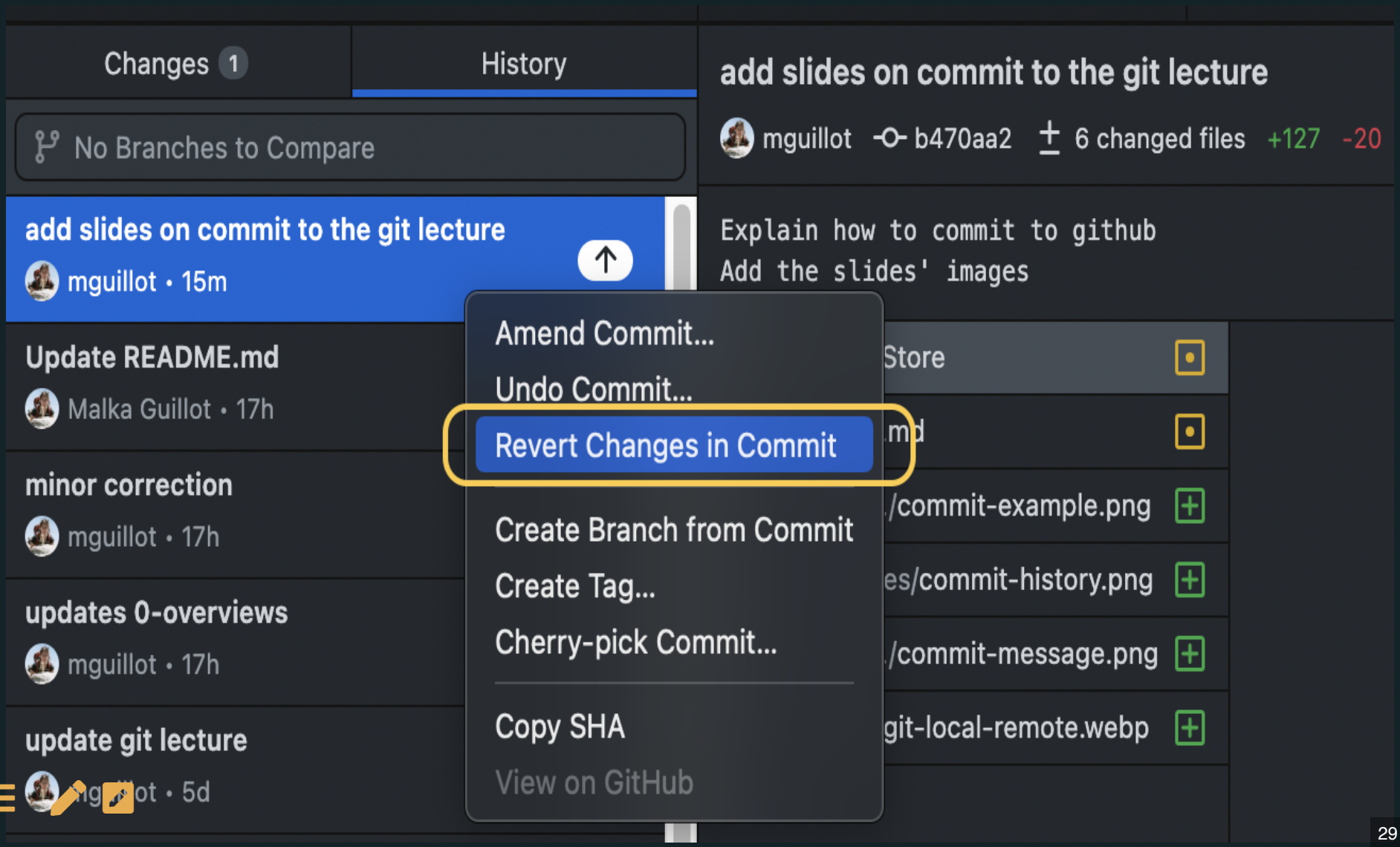


Viewing commit history



When things go wrong: go back in time

What happens when a commit was a mistake? **Revert it**, to make a new commit that undoes it.



This can happen!

Source: https://xkcd.com/1296/

Branches: trying things out

Branches are the most powerful part of Git

- By default, all the work you do goes into the “master” branch
- Want to experiment? Start a new branch
  - You can switch between branches, and make commits to either branch
- If your experiment works out, commit and merge back into the master branch
  - If there are conflicts between the commits you’ve made on the two branches, Git will ask you to resolve them
  - This is easiest with a graphical interface like GitKraken
  - Only works with binary files
  - If your experiment doesn’t work out, delete the new branch painlessly

Keeping it local vs. using a remote repository

Git doesn't require a remote repository. You can run it 100% on your computer, with no connection to an outside server.

- Useful if you have restrictions on your code (e.g. confidential health data)
- A remote repository helps
  - keep things backed up seamlessly,
  - collaborate with others
- You can push all your branches to the remote repository, or only some of them
- Big companies often have an internal git server

Collaborating

Where we open ourselves to others and go remote!

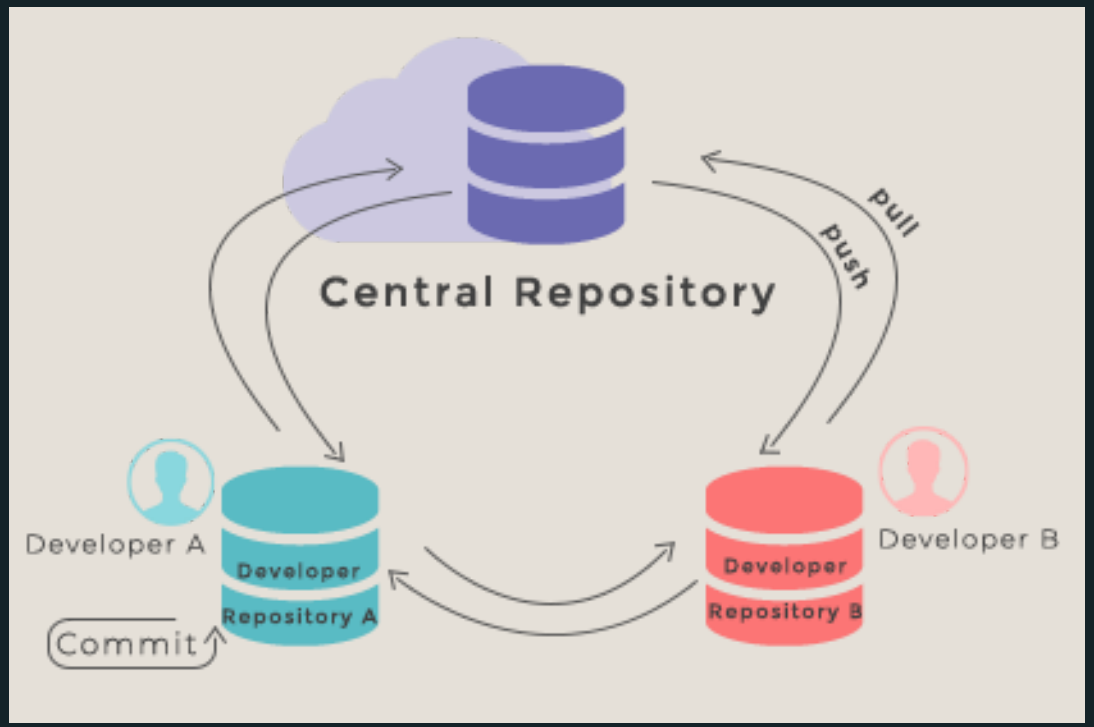
Interacting with the remote directory

The remote repository is on a server, and holds a record of your commits and branches

You push to the remote repository to save all your commits

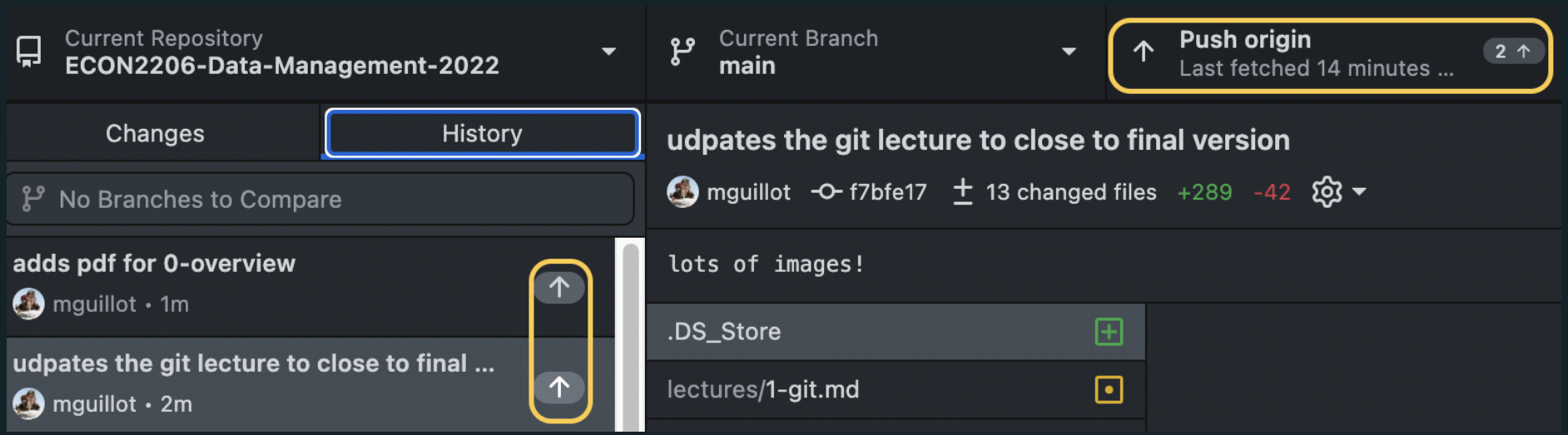
- You pull from the remote repository to load all new commits
- Always commit before pushing or pulling
- If what you’re doing is an experiment, make a new branch to avoid any trouble for your coauthor
- If there are conflicts between your commits and your colleagues’s commits, Git will ask you to resolve them

Basic workflow: push - pull



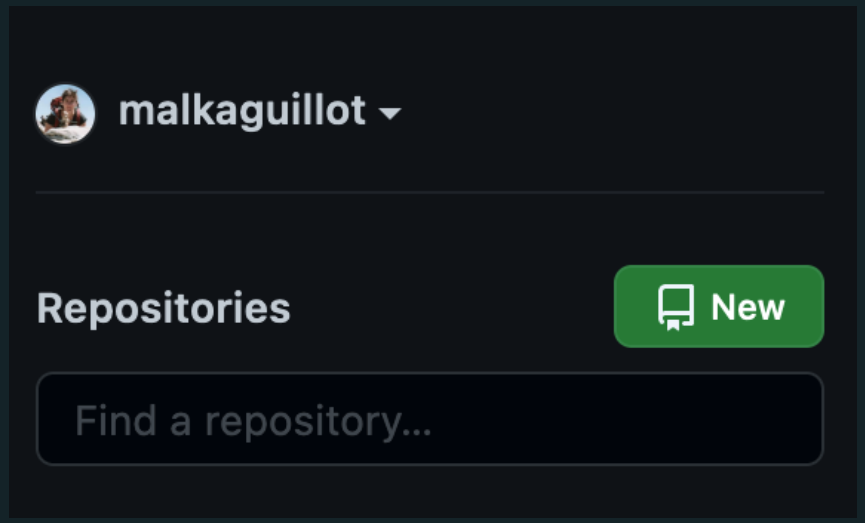
This is what happens between your computer (local) and your repository (remote).

Pushing to the remote repository (GitHub Desktop)



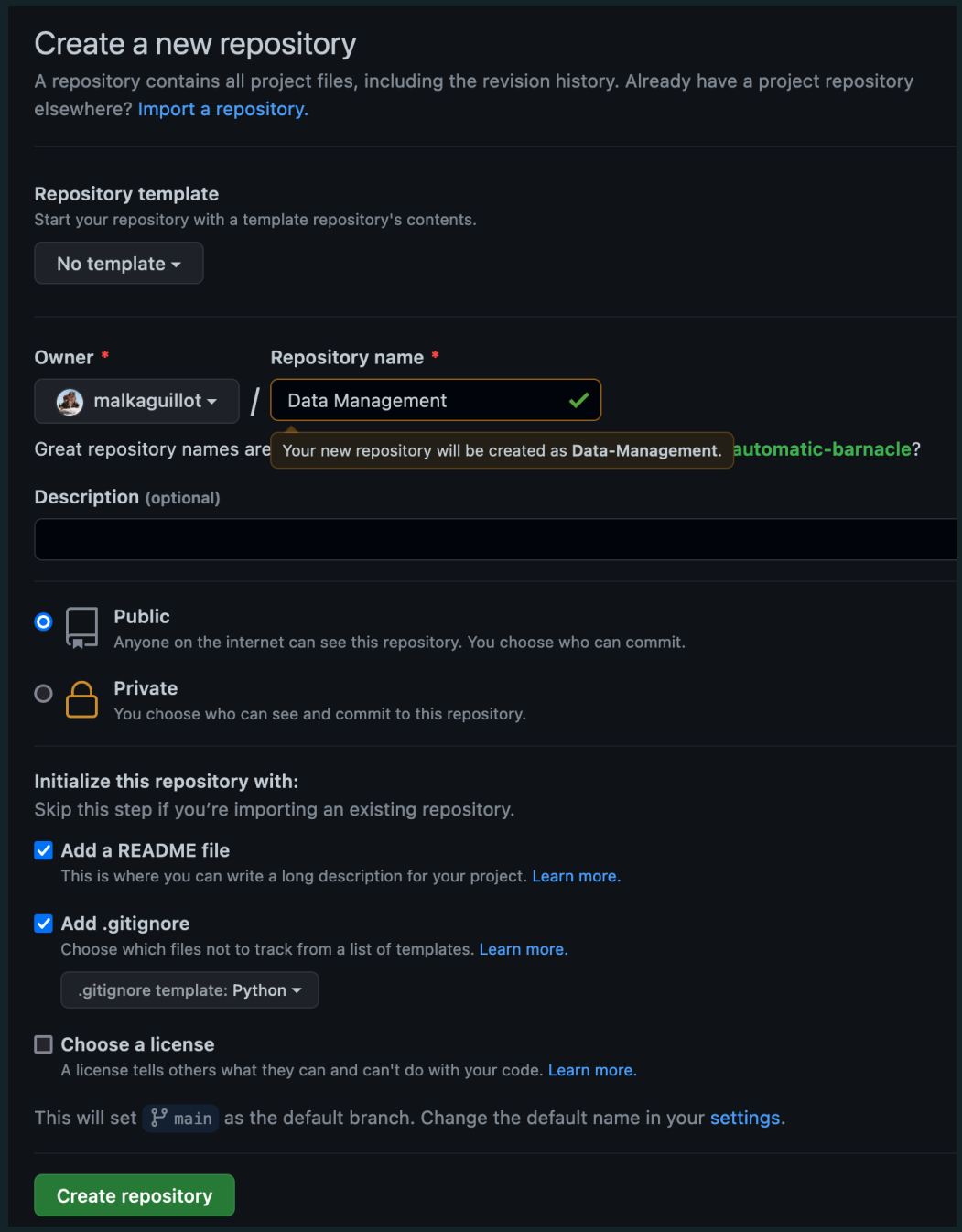
Sending my commits to the internet!

Create a remote repository



- Make sure you click the box to initialize it with a README
- gitignore → python template
- licence

Create a remote repository

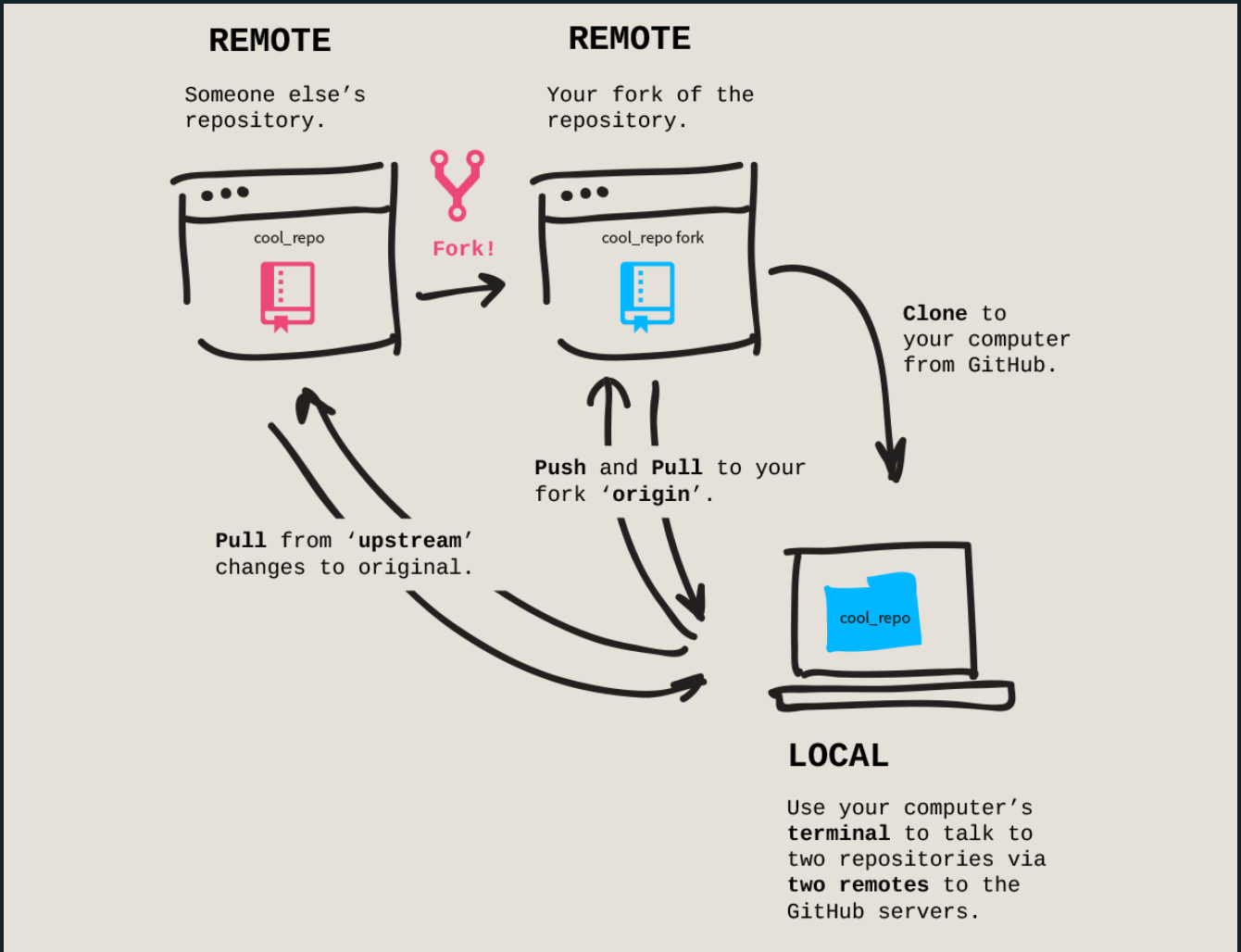


The README.md

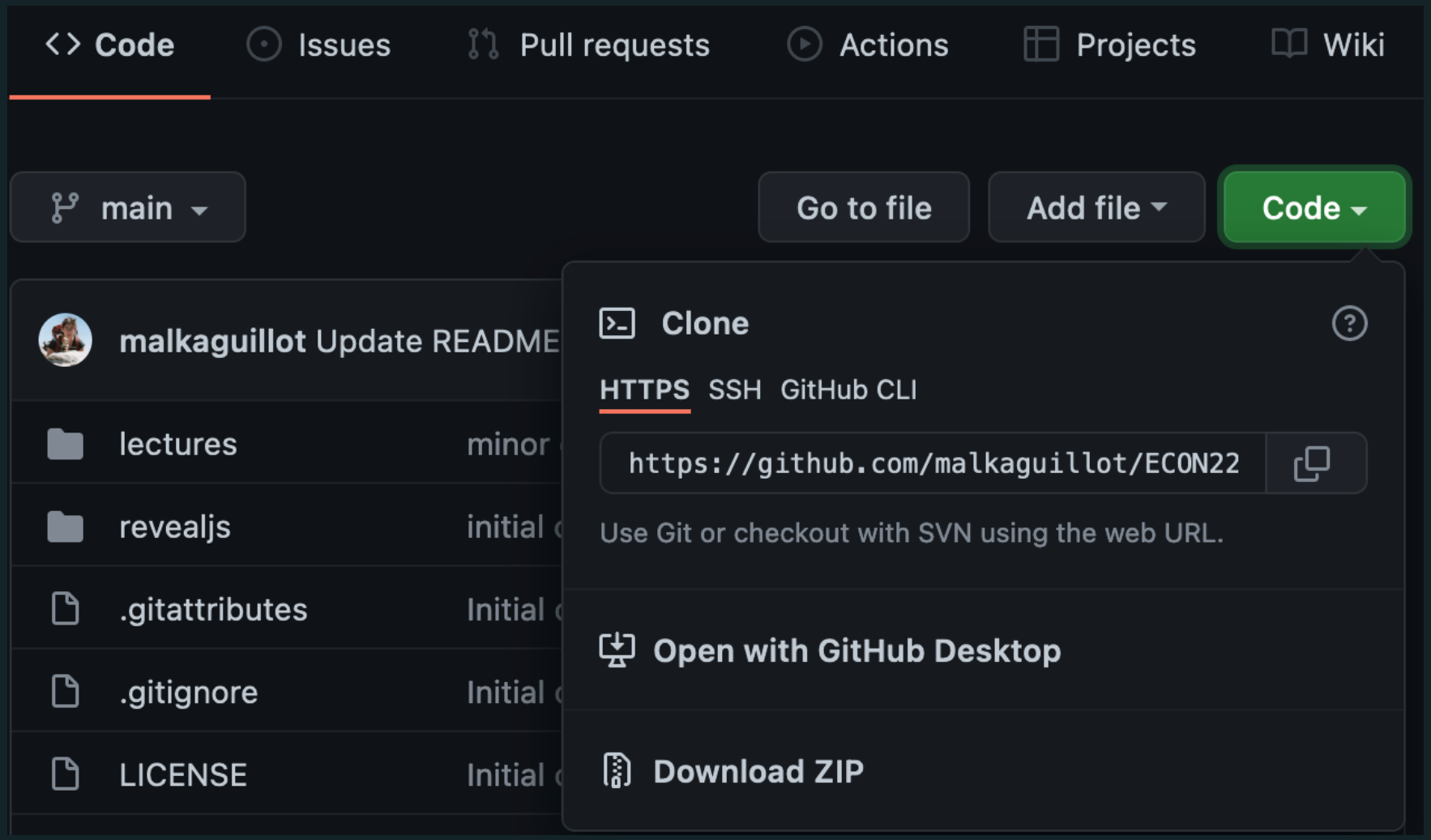
- Very important file!
  - Objective: communicate important information about your project
- A markdown file
- Markdown?= lightweight markup language
  - The guide
  - The syntax

Not only useful for README: for eg., these slides are written in markdown!

Basic work: clone or fork ?



Cloning a repo

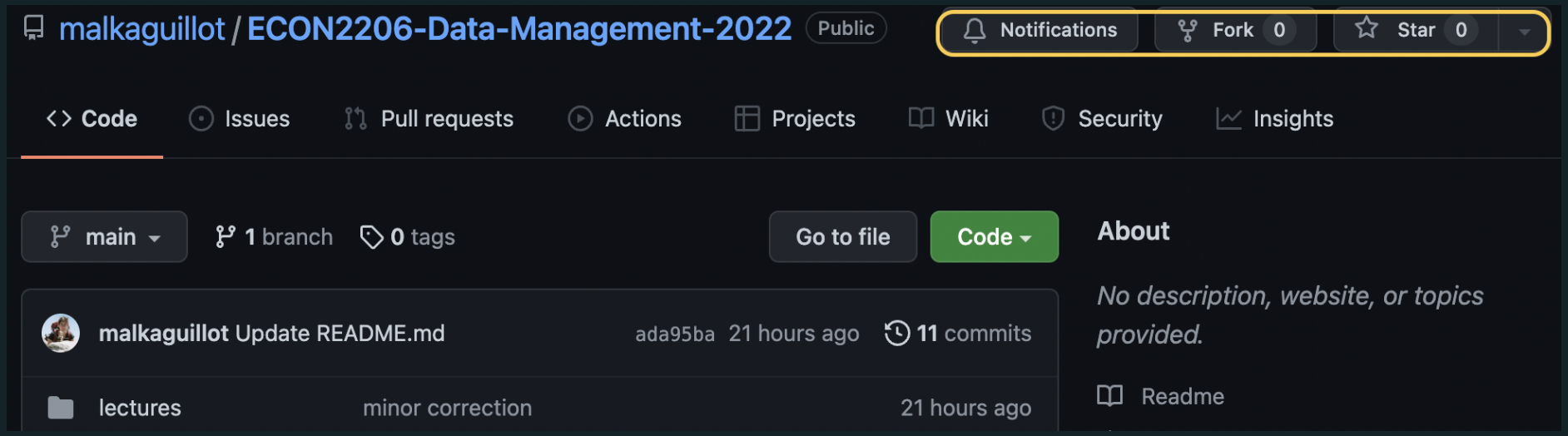


Git Challenge 1

- Create an example repository on your GitHub account (including a readme).
- git clone this repository to your computer. Go to this directory.
- Create three files named file1.txt, file2.txt, and file3.txt in your local repository.
- Stage, commit, and push file1.txt to your remote repository. Refresh the URL on your GitHub page. Do you see your commit?
- Stage, commit, and push file2.txt and file3.txt to your remote repository as a single commit.

Navigating GitHub

Example: our course repository



- **Notification**: Notify you when there are changes or conversations in the repo.
- **Star**: Add this repo to a list of repos that appear in your feed. Think of this as "favoriting" a repo.
- **Fork**: Make a copy of this repository in your own account. → Useful if you are not directly involved with a project but want to build on top of someone else's code.

Git challenge 2 (using GitHub desktop):

- Fork the course repository
- Change the **upstream repository**
  - *In repository settings*: change the "Primary remote repository" to my repo HTTPS addres
- Create a folder **sandbox**: this is were you are going to work!
- Open the .gitignore (you can create it still)
- add on a new line: **sandbox/*** : this will ignore the content of the sandbox when working with the remote => no conflict !
- create a toy file in the sandbox
- In the meantime, I make a commit
- Then can you fetch my commit?

Epilogue

Want more of this?

Let's learn one day how to use the command line interface!

How to interact with the materials?

- Set up GitHub
- Fork the class repository (-> your remote repository)
- Clone your repository on your computer (-> your local repository)
- Add an upstream origin (mine)
- Work in the *sandbox* folder
  - this way, you can fetch my updates

References

- Extensive git manual: https://happygitwithr.com/
- git - the simple guide
- github cheatsheet https://education.github.com/git-cheat-sheet-education.pdf
- interactive tutorial https://gitimmersion.com/index.html
- interactive tutorial on git branching https://learngitbranching.js.org/?locale=fr_FR
- In case it goes wrong: http://ohshitgit.com/

For next week:

- Get confortable with using **Git(Hub)**
  - practice with the challenges
  - go over references
  - work on the interactive tutorials
- **Python** installation
  - Install Anaconda, try out to run python in a Jupyter notebook and spyder
  - See installation guide link
  - Wait for next week's introduction by Michel!
    - Basics of python's syntax: Learn Python
      - less Classes and Objects + Modules and Packages.

Data Management

Introduction

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HEC Liège | ECON2306

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    - Data lifecycle
    - Goals of data analysis
2. The tools
    - python
    - git
3. Organisation & logistics

Introduction: Who are we?

Teaching assistant	Lecturer
Michel Coppee	Malka Guillot
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Bât. N1 Economie (bureau 33a)
rue Louvrex 14
4000 Liège
Belgique

Who am I?

PhD in economics from the Paris School of Economics

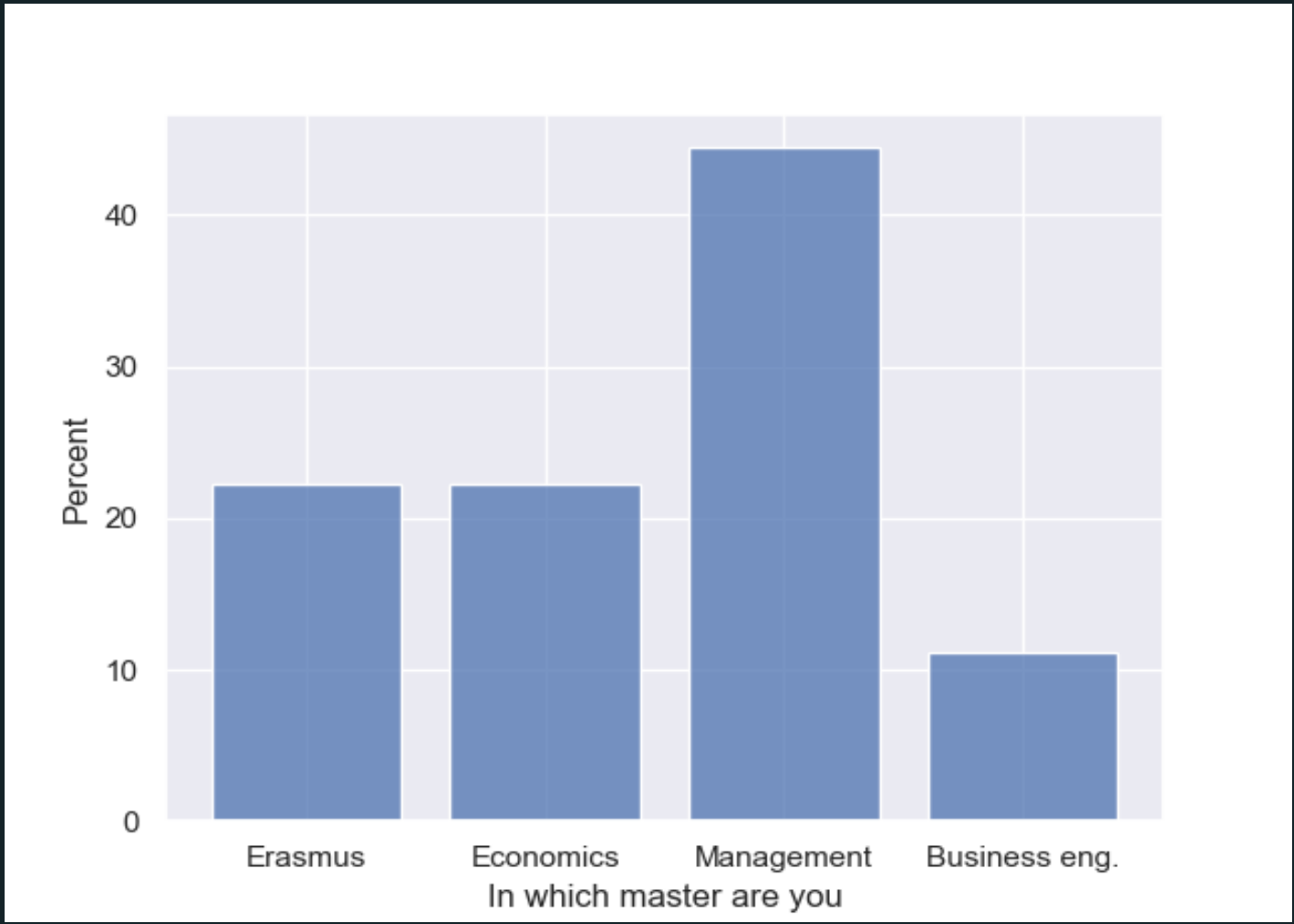
Postdoc at ETH

Assistant professor in applied micro economics at HEC Liège

Interested in **public economics** questions: inequality and taxation

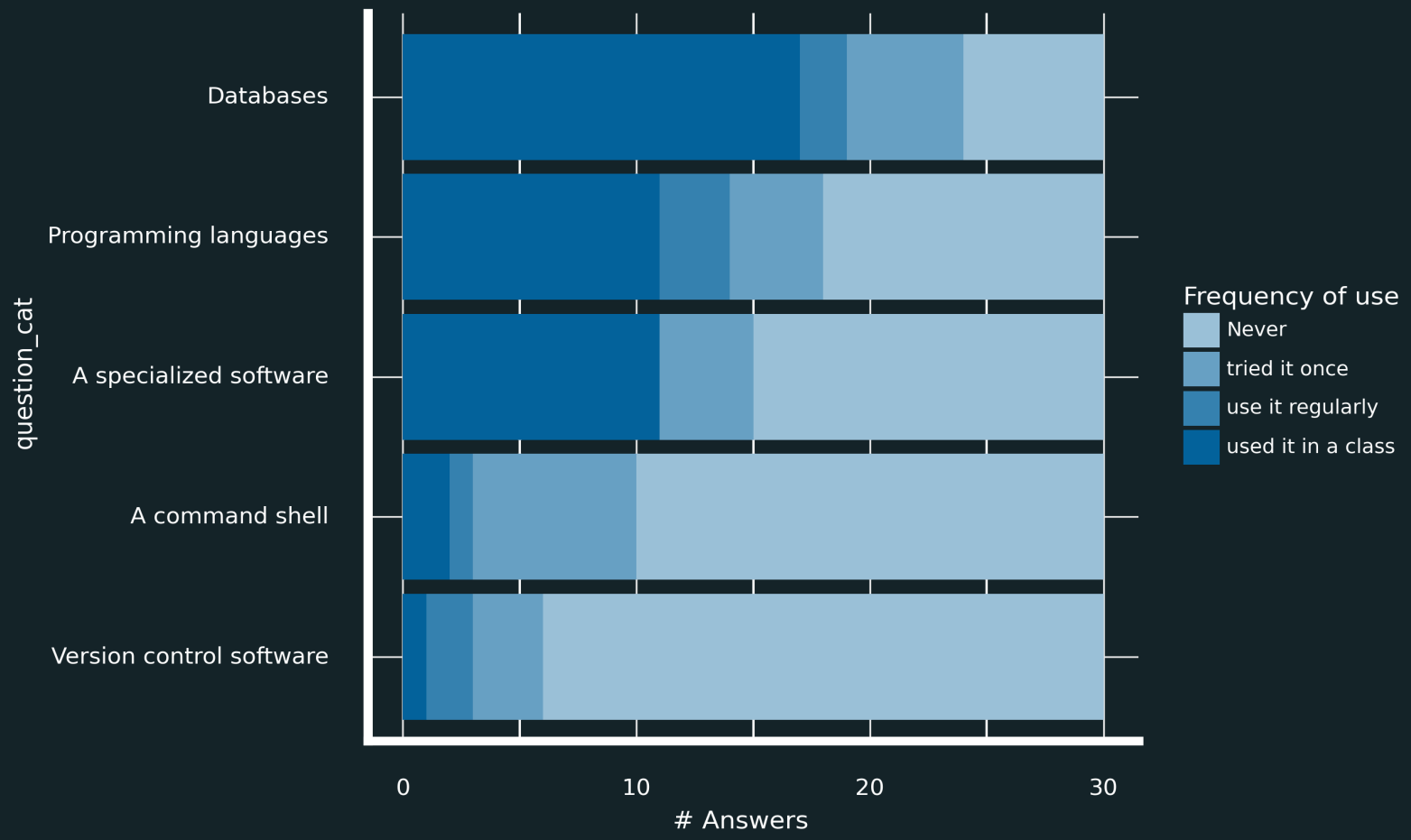
Using the standard econometric toolbox + natural language processing + machine learning

Introduction: Who are you ?



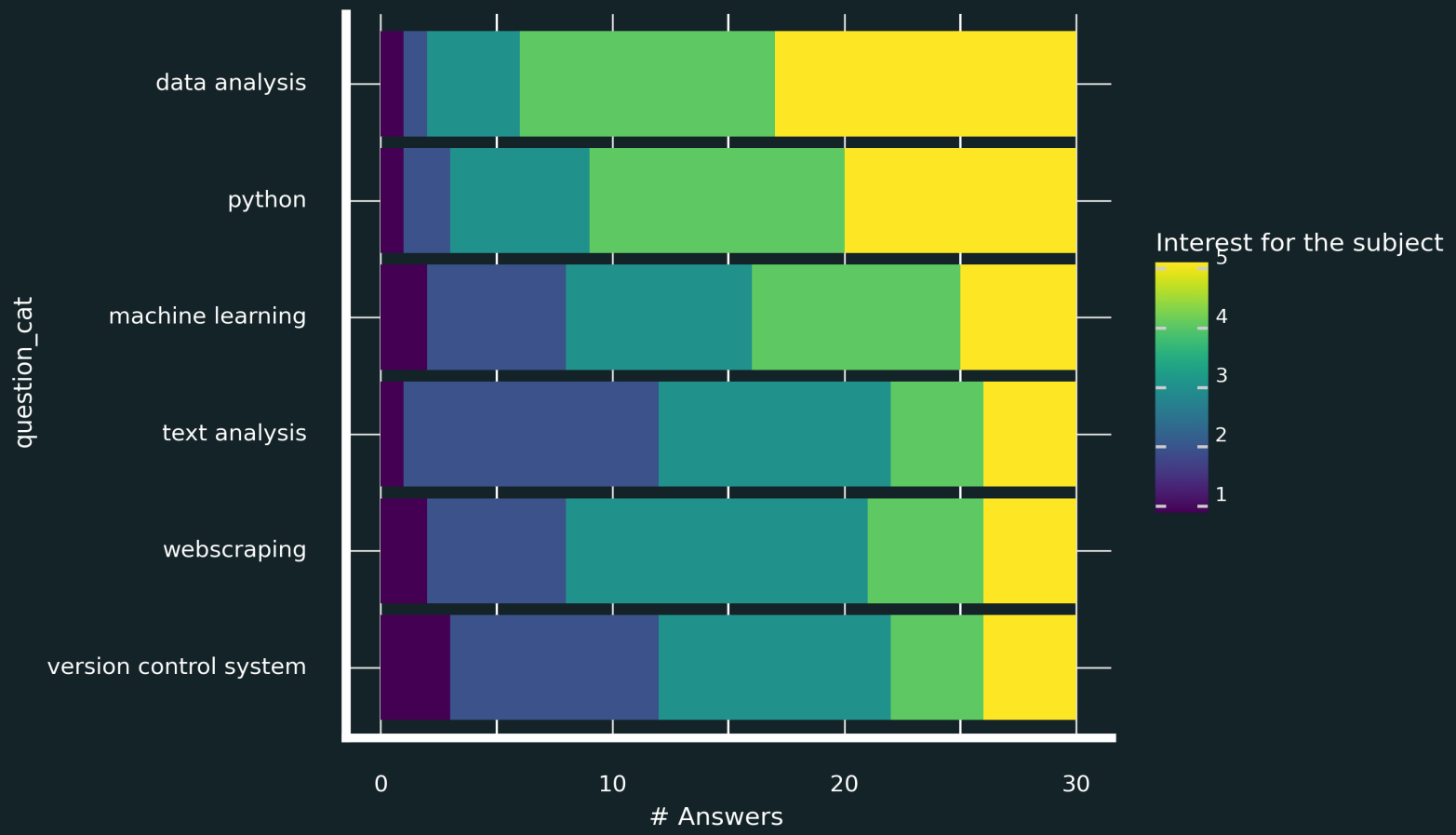
Source: pre-class survey

Your programming experience



Source: pre-class survey

Which part of the class interest you most?



Source: pre-class survey

Why have you chosen this class?

Source: pre-class survey

Course objectives

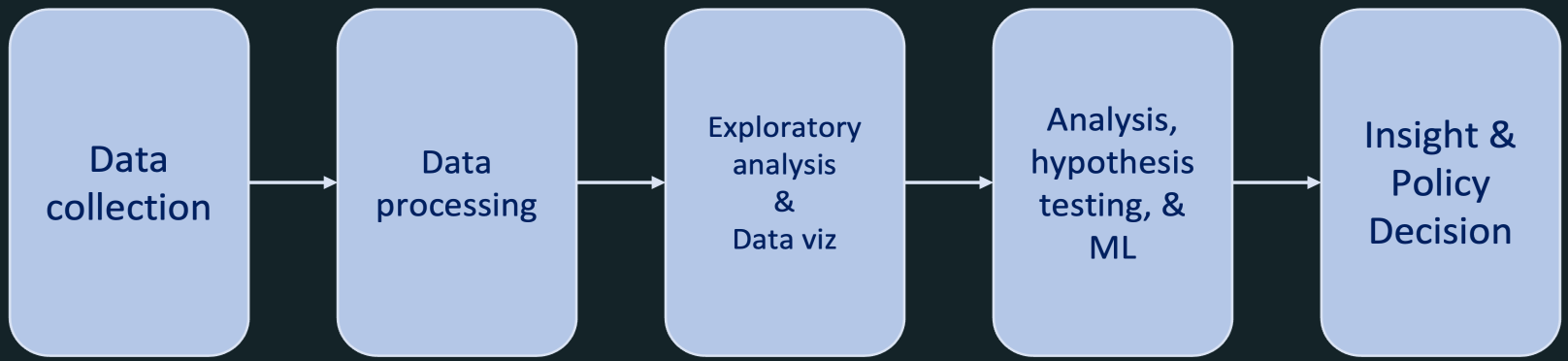
From acquisition of data to data analysis

The class focuses concepts & skills related to the management of data, that are central for the **exploitation** of data.

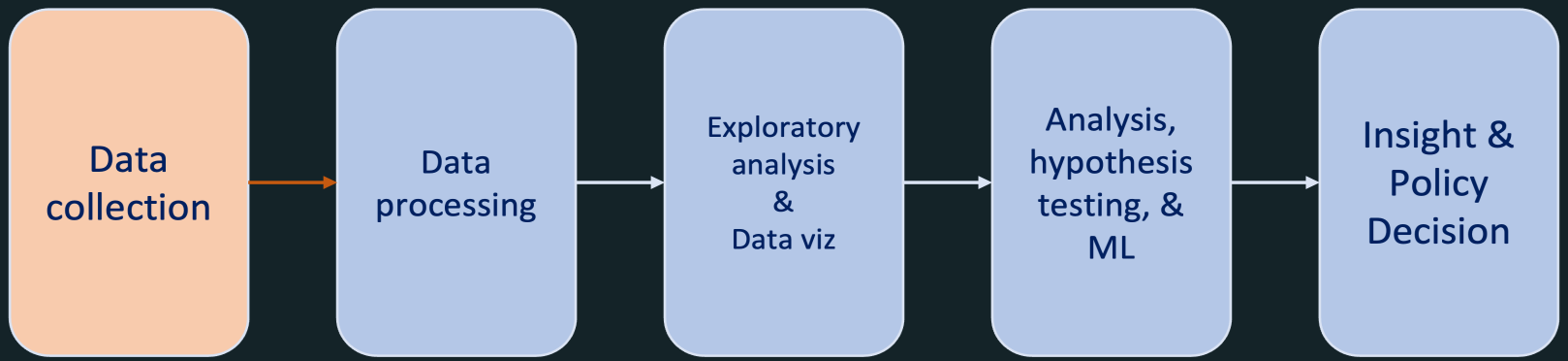
Goals:

- Equip you with the standard datascience toolkit.
- Put it to work on a real-world project.

Data lifecycle



Data lifecycle



Looking for data?

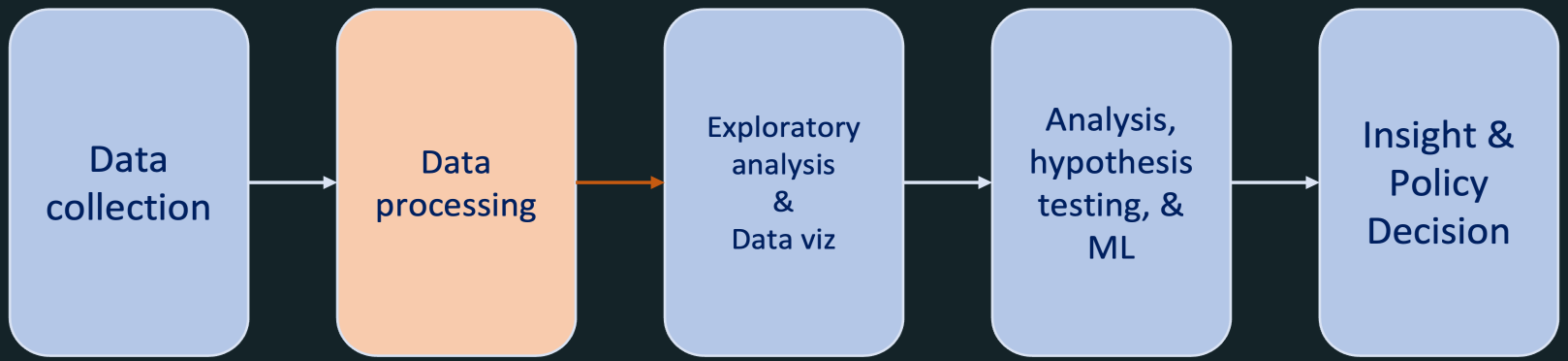
- Search engines:
  - google datasetsearch
  - https://fr.statista.com/
- Institutional reposities
  - Open data Liège
  - The official portal for European data
  - OECD data
  - Statbel and Open data Belgium
  - Open data France (also here)
- List of resources:
  - Common-built archive
  - Personal lists here, here or here
  - Registry of Open Data on AWS

You do not find what you are looking for?

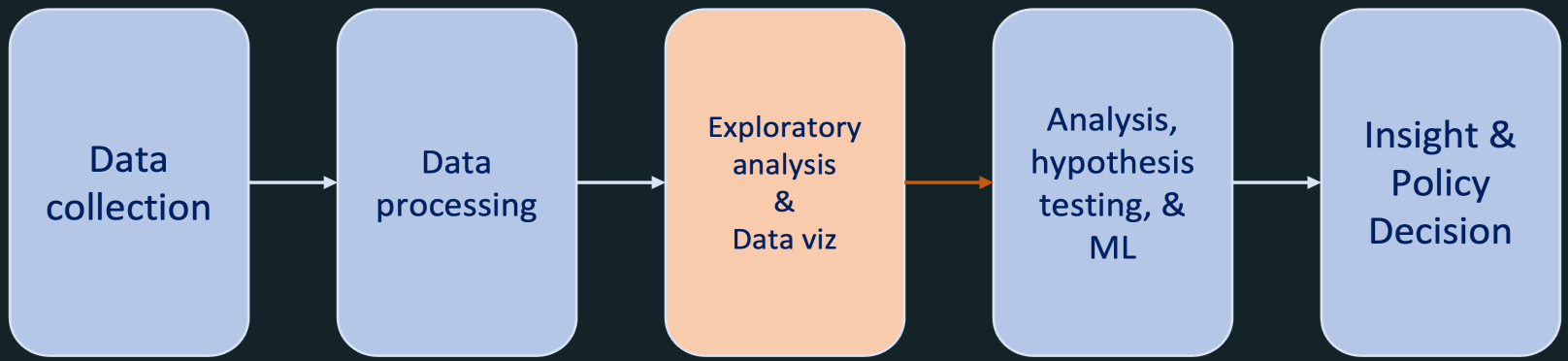
Some solutions:

- Data collection
  - webscraping
  - asking for the data
  - administrative sources
- Implement a survey

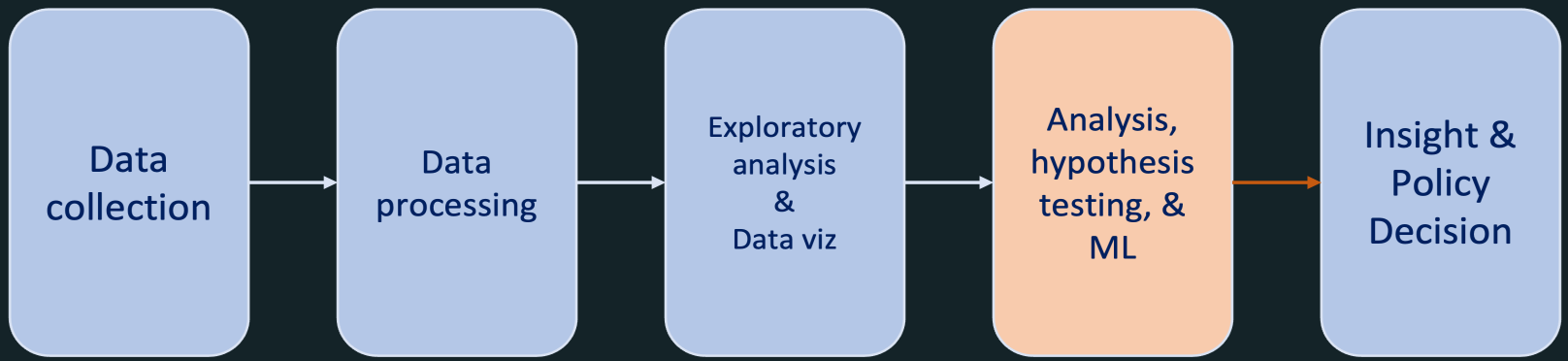
Data lifecycle



Data lifecycle



Data lifecycle



Analysing data

When we analyze data, what is our ultimate goal?

- Description: Simply characterize observed patterns in the data.
- Causation: Learn about causal relationships: If we change X, how will Y change?
- Prediction: Be able to guess the value of one variable from other information.

Analysing data

Examples of a (research) question in each category:

- Description: Is wealth inequality increasing faster in the U.S. than in Europe?
- Causation: Does Medicaid coverage reduce the risk of bankruptcy?
- Prediction: Can nighttime satellite imagery be used as a real-time indicator of GDP?

Discerning which type of goal you have is critical for:

- Choosing methods:
  - Distinct approaches are required to achieve different goals.
- Interpreting results:
  - Mistaking one goal for another can lead your audience to make very bad decisions.

The data generating process

What's shared across all goals of data analysis.

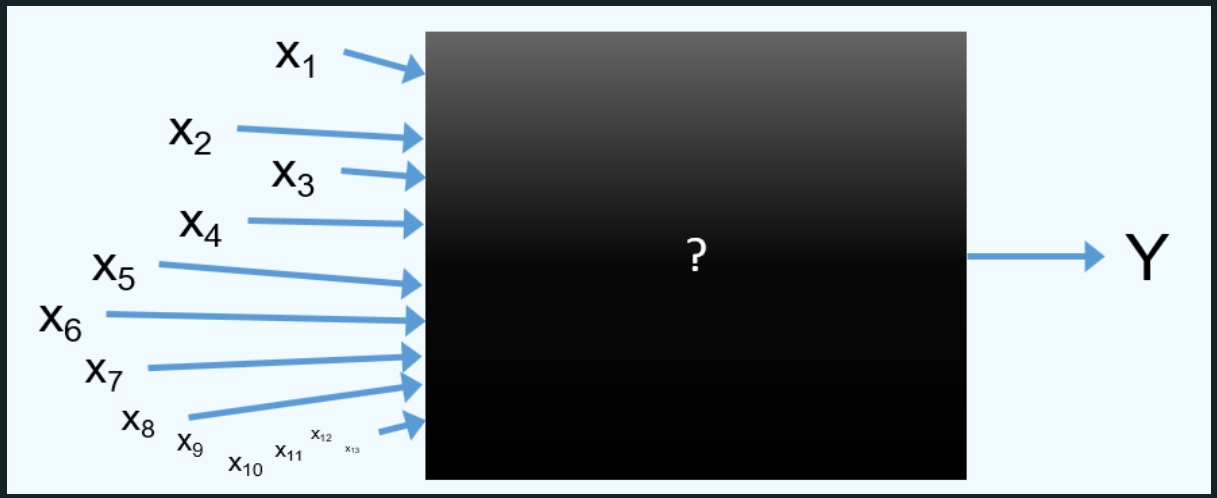
We start with an **outcome variable** called Y. We want to know:

- Where do the values of Y come from?
- How does Y relate to other variables?

The data generating process (DGP) in the real world determines the values of Y

The data generating process

The true DGP is unknowable, a black box:



We interpret the picture with a function:

$$Y = f(x_1, x_2, \cdots, x_N)$$

Questions:

- What are all these inputs $x_1, x_2, \cdots$?
- What is the functional form? (How does the box work?)

Types of goals:

1. Descriptive: Document observed relationships between X and Y.
   - Does not not necessarily tell us about the true DGP.
   - Helps us understand facts about the world.
2. Causal: Try to understand one piece of how the box works (the true DGP)
   - When you change one factor, how does it change the result?
   - Helps us make decisions about what to do
     - (in policy, business, personal life).
3. Predictive: Create your own box to try to match the output.
   - Doesn't matter if it works the same, or if you have the correct inputs.
   - Only matters how closely your box produces the same result.
   - Helps us know what's likely to happen in a new situation.

What methods should we use for each goal?

1. Descriptive: Exploratory analysis and regression
    - Covering soon!
2. Causal: Econometrics.
    - See your other classes.
3. Predictive: Statistical learning / machine learning.
    - Introduction in a few weeks!

Example

How did the under-developed and agricultural Switzerland from the 1850s become such an economically successful country?

A research question in history... where data are central

- Timing of the development of certain economic sectors?
- Geographical distribution of the firms?
- Interaction with public policies ?

Example: REFLEX project

- *Approach*: The Swiss Commercial Registry (1883-)

№ 1 — II. Theil 1883 IIe Partie — № 1

Schweizerisches Handelsamtsblatt

Feuille officielle suisse du commerce — Foglio ufficiale svizzero di commercio

Bern, 4. Januar — Berne, le 4 Janvier — Berna, li 4 Gennaio

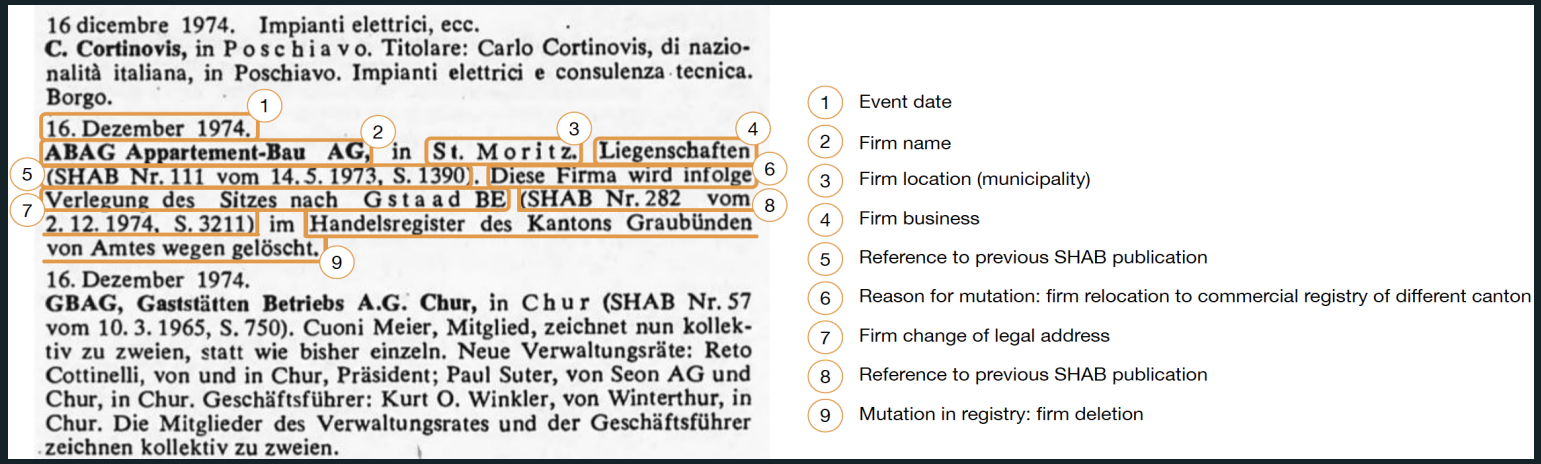
Publikationsorgan der eidgenössischen Departemente für Finanzen, Zoll und Handel
Organe de publicité des Départements fédéraux des Finances, des Péages et du Commerce
Organo di Pubblicità dei Dipartimenti federali per le Finanze, i Dazi ed il Commercio

Jährlicher Abonnementspreis Fr. 5. — Abonnemente nehmen alle Postämter sowie die Expedition des *Schweiz. Handelsamtsblattes* in Bern entgegen.
Abonnement annuel Fr. 5. — On s'abonne auprès des bureaux de poste et à l'expédition de la *Feuille officielle suisse du commerce* à Berne.
Prezzo delle associazioni Fr. 5. — Associazioni presso gli uffizi postali ed alla spedizione del *Foglio ufficiale svizzero di commercio* a Berna.

- *Methods*:
  - Transforming the pdf into structured data using:
    - google vision
    - Open AI
    - deep learning

Example: REFLEX project

- *Objectives*:



- *Challenges*:
  - Correcting for OCR mistakes
  - Building a firm identifyer
  - 3 languages
  - ...

Backbone of the class

1. The **skills**:
    - Data collection
    - Data manipulation:
        - Cleaning, Pipelines, data structure
    - Data vizualisation
    - Data modelling
2. The **tools**:
    - python
    - git
3. The **concepts**:
    - *Project management*: documenting, sharing & managing code
    - *Reproducibility*

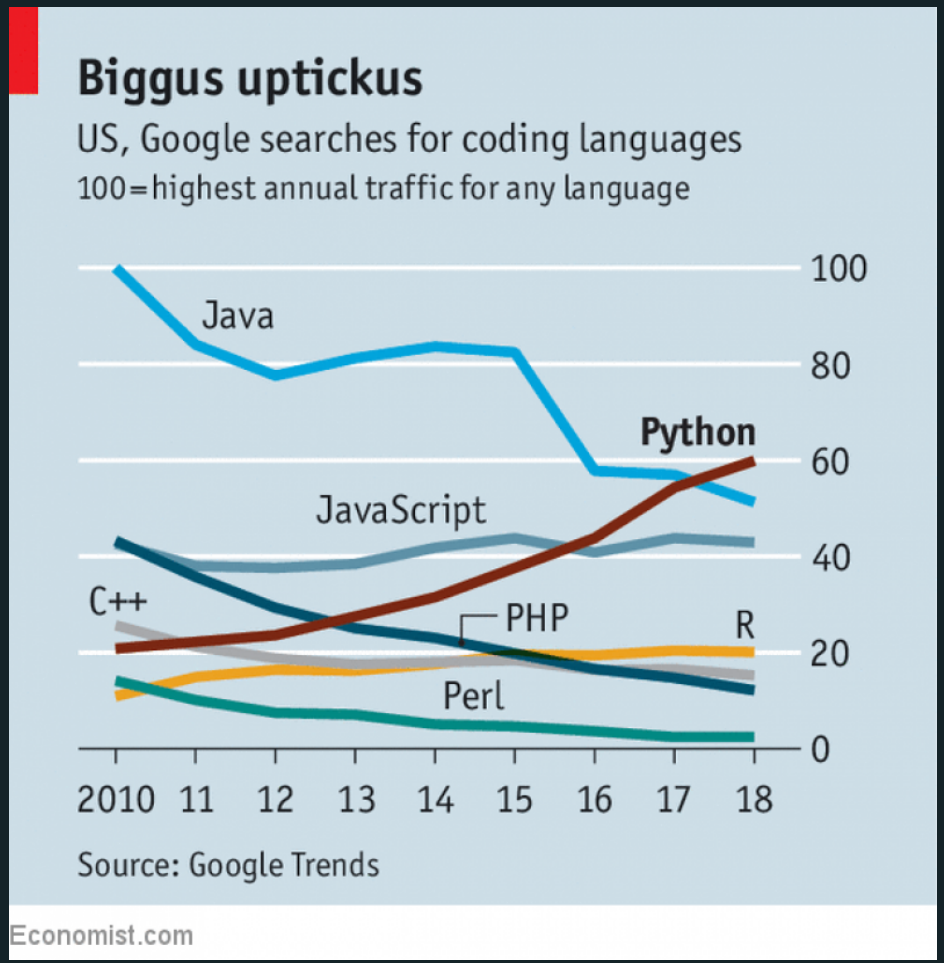
Public targeted: **anyone using data for projects**.

What this course **is**, and *is not*

- **It is**:
  - **Applied** and oriented towards practice;
  - **General** overview of different techniques - what they are and how to use them.
  - **Data analysis** in general, not restricted to a research or a field (economics, political science).
  - In python.
- *It is not*:
  - **Computer science**. We're not coding up models from scratch.
  - **Mathematical statistics**. We're not deriving the functions by hand.

The tools: python & git

Why Python?


Biggus uptickus
US, Google searches for coding languages
100=highest annual traffic for any language
Java
Python
JavaScript
C++
PHP
R
Perl
100
80
60
40
20
0
2010
11
12
13
14
15
16
17
18
Source: Google Trends
Economist.com

Why choose Python for this class?

- Easy-to-use, wide-spread and popular programming language
- created by Guido Van Rossum (first released it in 1991 as Python 0.9.0.)
- General-purpose language
  - One of the core languages of scientific computing
- Elegant syntax

Python is a *high-level general-purpose* programming language that emphasizes *readabilty* and *extensibility*.

A high-level programming language

	High-level programming language	Low-level programming language
1.	It is programmer friendly language.	It is a machine friendly language.
2.	High level language is less memory efficient.	Low level language is high memory efficient.
3.	It is easy to understand.	It is tough to understand.
4.	Debugging is easy.	Debugging is complex comparatively.
5.	It is simple to maintain.	It is complex to maintain comparatively.
6.	It is portable.	It is non-portable.
7.	It can run on any platform.	It is machine-dependent.
8.	It needs compiler or interpreter for translation.	It needs assembler for translation.
9.	It is used widely for programming.	It is not commonly used now-a-days in programming.

A general-purpose programming language

- software development (alternative to Java, C, Fortran)
- data analysis
- good for webistes with data applications

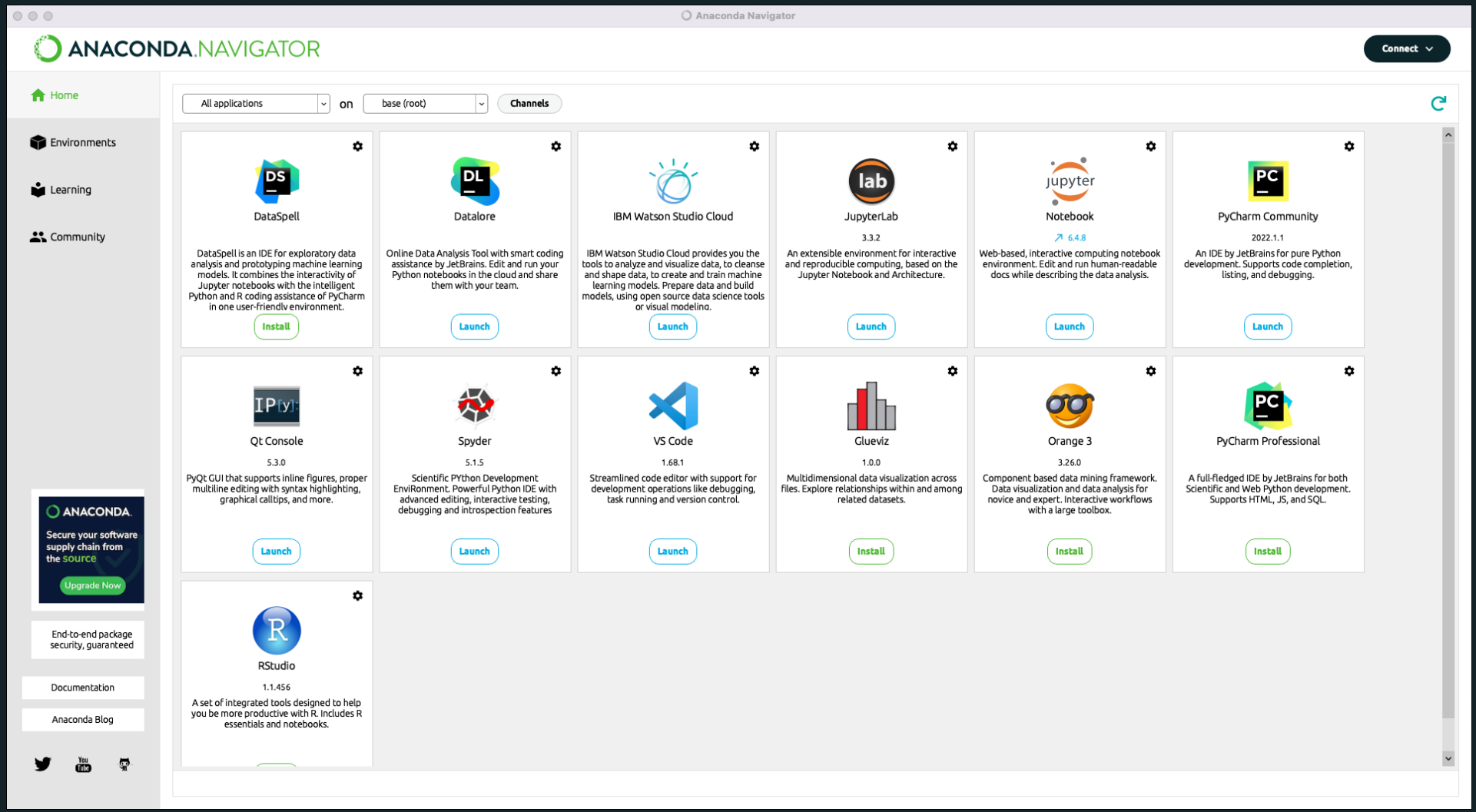
... That consequently sports a huge library

- Web interaction
  - HTTP requests (requests)
  - Scraping web pages (Beautifulsoup4)
  - Performing Browser automations (Selenium)
- Data manipulation: Pandas
- Data visualisation:
  - Generic visualization matplotlib
  - Statistical data visualization seaborn
- Machine learning: scikit-learn
- Statistics: statsmodels
- Natural Language Procession nltk, spacy

Using Python

Anaconda	Jupyter notebook	Visual studio code
a convenient all-in-one install	for homerwork	for longer code

→ Anaconda



Visual studio code & Jupyter notebook are two development environments from the Anaconda set up.

Main python packages

Task	Package
Webscraping	beautiful soup
Data management	
Visualisation	
Web application	
Machine Learning	
Natural language processing	NLTK &

Why git?

- we need a way to make version control and collaboration easier
  - *Version control*: managing changes to code, and easily going back and forth between older and new versions of a code base
- tool used in research & industry

Course organisation & logistics

How does the class work? Spirit

Sessions are designed to be interactive

- mix of live *coding* & *exercises*
- we want to get you comfortable usings your computing environment to solve problems
  - bring your laptop!
  - we expect you have completed the installation guide and have all software installed.
  - ask questions!

How does the class work? Details

- **Mondays**: 1 hour / week
  - 1 hour practice, often an open house session around the class project
- **Wednesdays**: 3 hours / week
  - 2 hours of lecture + practice
    - sometimes the frontier between theory and practice will be fuzzy.
  - 1 hour of "individual work"
    - volontarly
    - you can work at your pace on the course content and directly ask questions to the professor

Online Course Materials

- Website:
  - with the course architecture and content
  - https://malkaguillot.github.io/ECON2206-Data-Management/
- Github repository :
  - https://github.com/malkaguillot/ECON2206-Data-Management-2023
  - includes slides and tutorials
  - Can be tested under a temporarary computing environment:

- lola:
  - For homeworks
  - Course announcement and forum

[Evaluation Policy]

- **4 problem sets**:
  - should be given back as jupyter notebooks in PDF format on lola.
  - **=20%**
  - The problem sets are simple exercises designed to help students to "get their hands in the data & code".
- **Participation in class = 5%**:
- **Course project = 75%**

cf. details

[Course project] Objectives

- The **basics**:
  - End-to-end data project using Python
    - From collection to modelling through visualisation
    - Should answer an open-question with data
  - Group project (2 people; 3 of odd no. of students)
- Use what you learn in this course to **solve a non-trivial real-world question/problem** using a data analysis
- **Deepen one aspect** of the course:
  - Three dimensions must be present in the project:
    - data retrieval, visualisation or modelling;
  - To a greater or lesser extent depending on the project.
  - One dimension must go further than the others

[Course project] Example of web application

→ Some examples in various sector:

- Finance:
  - The Yield Curve
- Health
  - Opioid epidemic in the US
- Transportation:
  - Uber rides
- Energy consumption
- Research project

→ Be creative, have fun!

[Course project] Other example

- Modelling hotel prices ;
- Mapping of cycling practices using Villo!'s data
- Mapping social and economic inequality in Wallonie
- What is the parliament talking about?

What about you?

1 minute to think about a potential field of application.

- Present yourself
- Specify 1 or 2 domain of interest with possible data analysis
  - Can be academic: green finance, social inequality
  - or not: sport, important topic

[Course project] Requirements

1. **Data retrieval**
   - Standard: Get some data from the web, data cleaning
   - Advanced: Data collection with webscraping
2. **Data visualisation**
   - Standard: Propose at least 4 exhibits (graphs or tables)
   - Advanced: More elaborated figures, such as maps, or unsupervised learning representations
3. **Data modelling**
   - Standard:
     - Supervised ML approach: compare at least 2 models or NLP
   - Advanced:
     - Both a ML model and some NLP (for example)

[Course project] Requirements

- Submission format:
  - Invite @malkaguillot and @MichelCop to collaborate on your GitHub repository by the due date.
- Abide by **good coding practices**:
  - code must be in split into meaningful sub-files:
  - code must be documented;
- Requirement for **reproducibility**:
  - Code should replicate
  - Project must be available on GitHub

[Course project] Evaluation: 75% =

Indicative weights

- Project management = 5% (G)
  - reproducibility, github, readme
- Project relevance = 10% (G)
  - Does the project respond to an interesting/important question?
- The 3 dimensions of the class = 30% (G)
  - Data retrieval
  - Data visualisation
  - Data modelling
- One dimension that goes further = 10% (G)
- Oral presentation = 15% (I)
- Group feedback = 5% (I)

Course Communication

- Us → you
  - Course communication will be done through lola's forum
- You → us
  - We will be available
    - During the breaks, after the class.
    - Michel Copée can answer questions about lectures, notebooks, assignments, and projects
  - **Personal question**:
    - face-to-face interaction > email
  - **General interest question**:
    - forum > email

References?

No general texbook. Specific references will be given when corresponding subjects are tackled.

- List on the course website
- Online books:
  - Coding for Economists
  - Python for Economics and Business Research introduction to python, pandas, plotting
- Stackoverflow: all the answers are there, but you have to ask the right question.

Troubleshooting

- Use the course forum to share & find answers
- Let's try to make this a **fun collaborative experience** for everyone

Data Management

Working with data

Malka Guillot

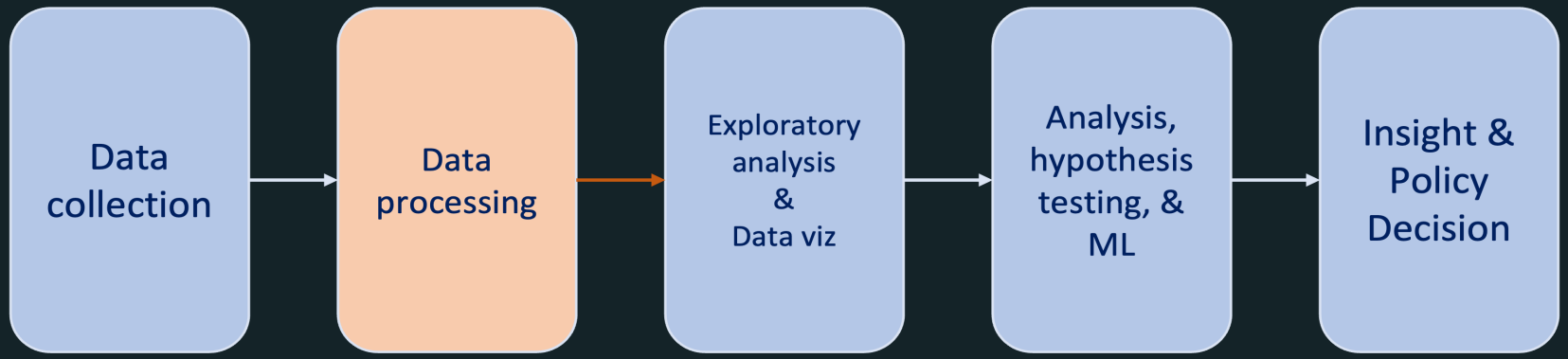
HEC Liège | ECON2306

Table of contents

1. What is data?
2. Data operations
3. Exploratory data analysis

What is data?

Data lifecycle



Objective: get structured & tidy data

Data wrangling (data munging)

= The process of transforming raw data to a set of data tables that can be used for a variety of downstream purposes such as analytics

Tidy data

Principles

1. Each observation forms a row
2. Each variable forms a column
3. Each type of observational unit forms a table.
4. Each observation has a unique identifier (ID)

Advantages:

- **easier** to work with.
- finding errors and issues with data are usually easier with tidy data tables
- more transparent → helps other users to understand
- easy to extend:
  - new observations added as new rows;
  - new variables as new columns.

Structured data

Example of messy data

	Treatment A	Treatment B
John Smith	-	2
Jane Doe	16	11
Mary Johnson	3	1

Example of tidy data

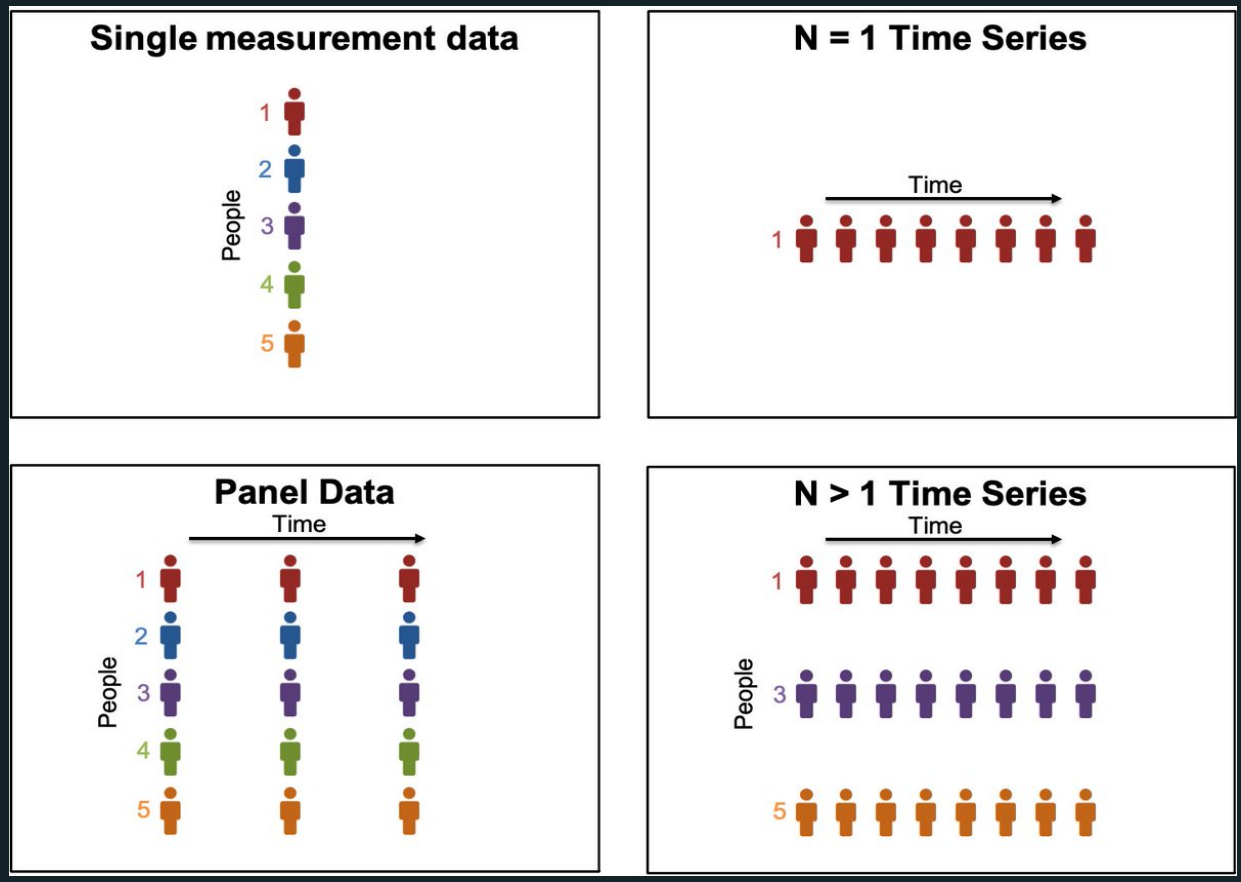
Name	Treatment	Result
John Smith	a	-
Jane Doe	a	16
Mary Johnson	a	3
John Smith	b	2
Jane Doe	b	11
Mary Johnson	b	1

Unstructured data?

⇒ we will be mainly working with structured data + learning how to go from unstructured to structured

Data structures

- cross-sectional data
- time series data
- panel or longitudinal data
  - balanced or not



Variables types

What type of variable do you know?

Variables

- Quantitative vs. qualitative variables
- Stock vs. flow variables

Data cleaning

Filter out duplicates

- duplicates: some observations appearing more than once in the data.
- May be the result of human error or the features of data source

Missing values

- Need to be identified & should be counted
- Potential selection bias: is data missing at random?
- Solutions:
  - Restrict the analysis to observations with non-missing values for **all** variables
  - Imputation : Fill in some value for the missing values, such as the mean or median value.

Extreme values

Substantially larger or smaller values for one or a handful of observations.

- Need conscious decision.
- Is this an error? (drop or replace)
- Is this not an error but not part of what we want to talk about? (drop)
- Is this an integral feature of the data? (keep)

Data wrangling: common steps

1. Write a code - it can be repeated and improved later
2. Understand the structure of the dataset:
    - create data tables, recognize links. Draw a schema.
3. Start by looking into the data table(s) to spot issues
4. Store data in tidy data tables.
5. Get each variable in an appropriate format
6. Have a description of variables
7. Make sure values are in meaningful ranges; correct non-admissible values or set them as missing
8. Identify missing values and store them in an appropriate format. Make edits if needed.
9. Document every step of data cleaning

Data operations

Tidying messy datasets

- **Objective** :
  - Prepare data in a standardized way prior to the analysis.
- **Tool** :
  - pandas package of handling data
  - os package of path

```
import pandas as pd
```

0. The Pandas DataFrame

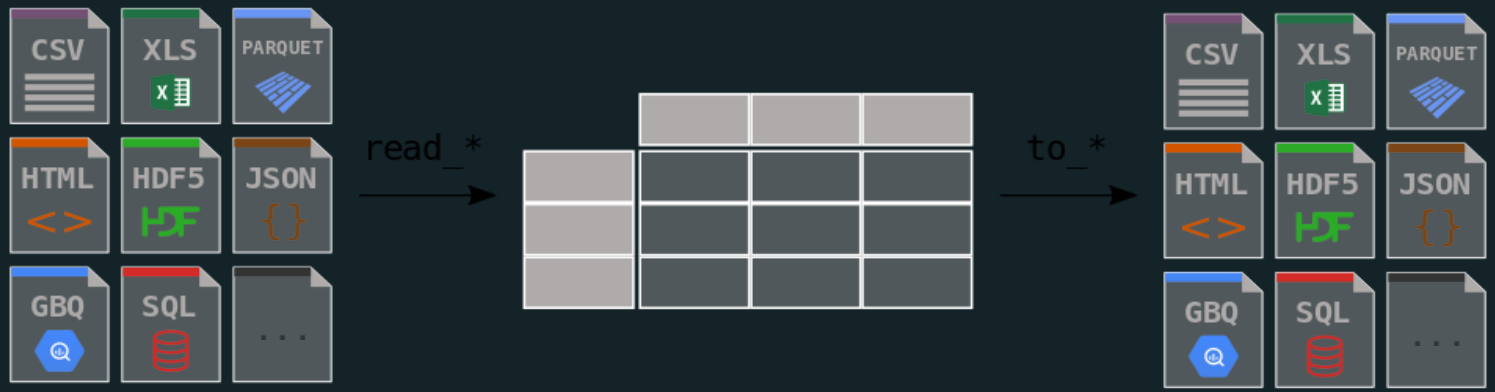
Let's create a table by hand

```
df = pd.DataFrame(
    {
        "Name": [
            "Braund, Mr. Owen Harris",
            "Allen, Mr. William Henry",
            "Bonnell, Miss. Elizabeth",
        ],
        "Age": [22, 35, 58],
        "Sex": ["male", "male", "female"],
    }
)
```

	Name	Age	Sex
0	Braund, Mr. Owen Harris	22	male
1	Allen, Mr. William Henry	35	male
2	Bonnell, Miss. Elizabeth	58	female

Each column in a DataFrame is a Series: cf. exercise

0. The Pandas DataFrame



Importing data

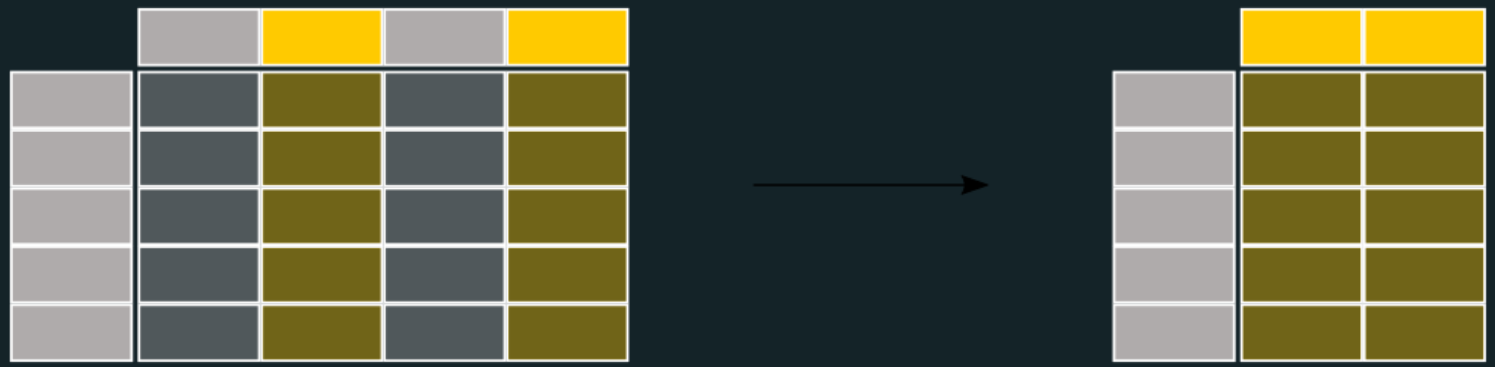
```
titanic = pd.read_csv("data/titanic.csv")
```

Exporting data

```
titanic.to_excel("titanic.xlsx", sheet_name="passengers", index=False)
```

1. Select / slicing

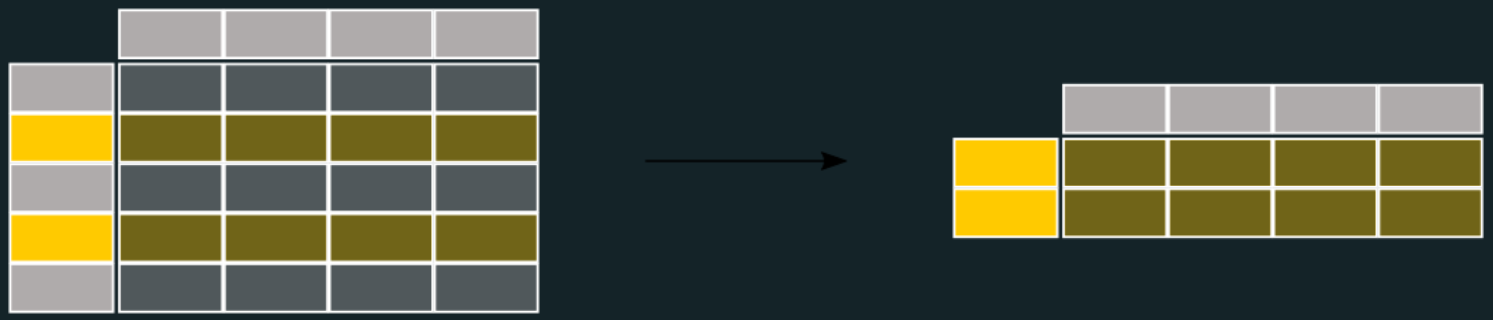
Select only some of the columns:



```
df["Name"]
df[["Name", "Age"]]
```

1. Select / slicing

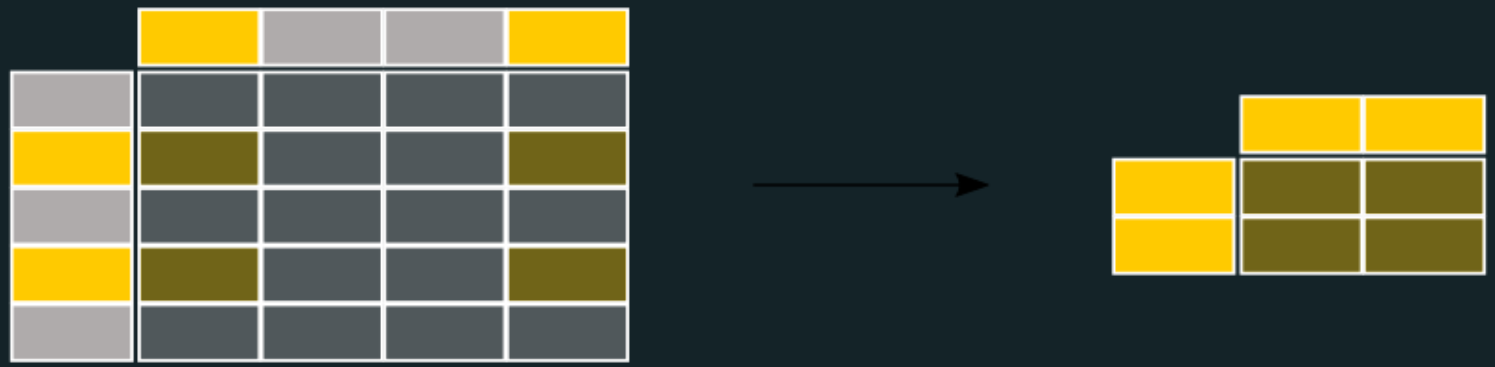
Select only some of the rows:



```
above_35 = titanic[titanic["Age"] > 35]
above_35.head()
```

1. Select / slicing

Select a combinations of rows and columns:



```
adult_names = titanic.loc[titanic["Age"] > 35, "Name"]
```

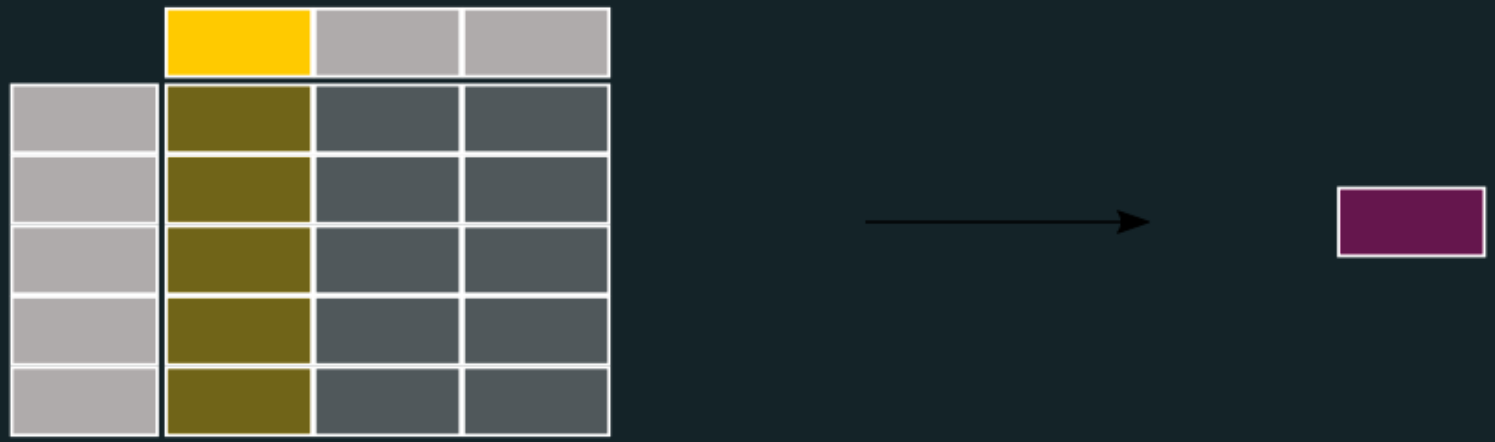
Challenge

Take a few minutes to :

1. Download this dataset and save it into a folder
2. Open a jupyter notebook in the same folder
3. Import the dataset in your notebook
4. Visualise the data (you can use df.head())
5. Create a new dataframe with only the country and points columns
6. Select all observations that are from Cyprus

2. Aggregate / reduce

Combine values across a column into a single value



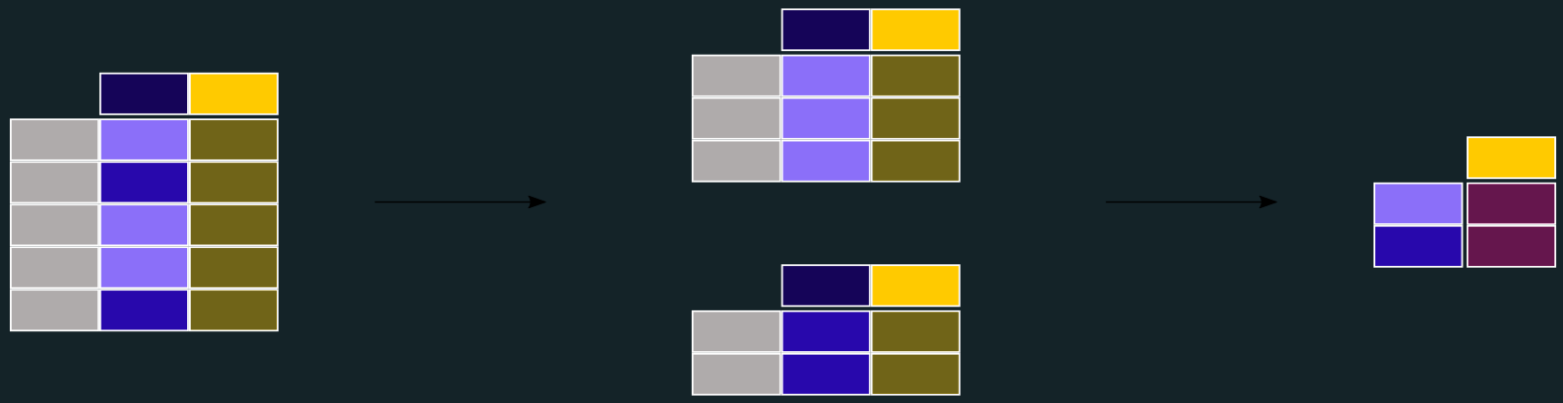
```
titanic["Age"].mean()
titanic[["Age", "Fare"]].median()
```

3. Map

Apply a function to every row, possibly creating more or fewer columns

4. Group by

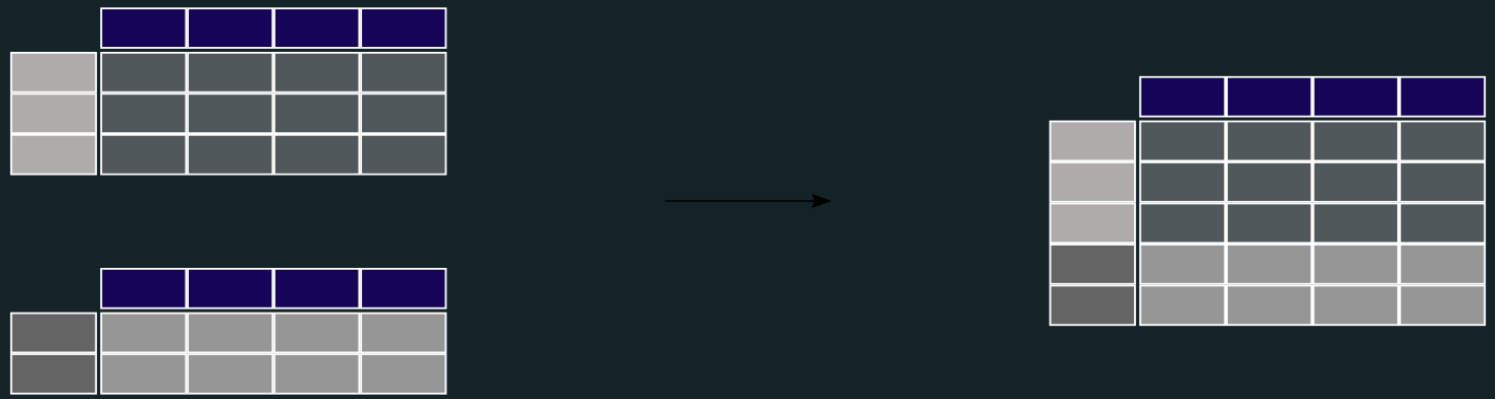
Apply a function to every row, possibly creating more or fewer columns



```
titanic[["Sex", "Age"]].groupby("Sex").mean()
```

5. Combine data from multiple tables

Concatenation



Merge

Exploratory data analysis

Preliminaries: cleaning data

1. Look at the data
2. Transform the variables into a known type
3. The type matters for what we do with them
    - For **aggregation**
        - flows ⇒ summmed
        - stocks ⇒ averaged
    - For **plotting the distribution**:
        - Qualitative ⇒ bar chart with frequencies
        - Quantitative ⇒ histogram

5 reasons to do EDA!

1. To check data cleaning (part of iterative process)
2. To guide subsequent analysis (for further analysis)
3. To give context of the results of subsequent analysis (for interpretation)
4. To ask additional questions (for specifying the (research) question)
5. Offer simple, but possibly important answers to questions.

Summary statistics + graphics

Summary statistics

- For any given variable, a *statistic* is a meaningful number that we can compute from a dataset.
- Basic *summary* statistics describe the most important features of distributions of variables.
- Example?

Distribution of a variable

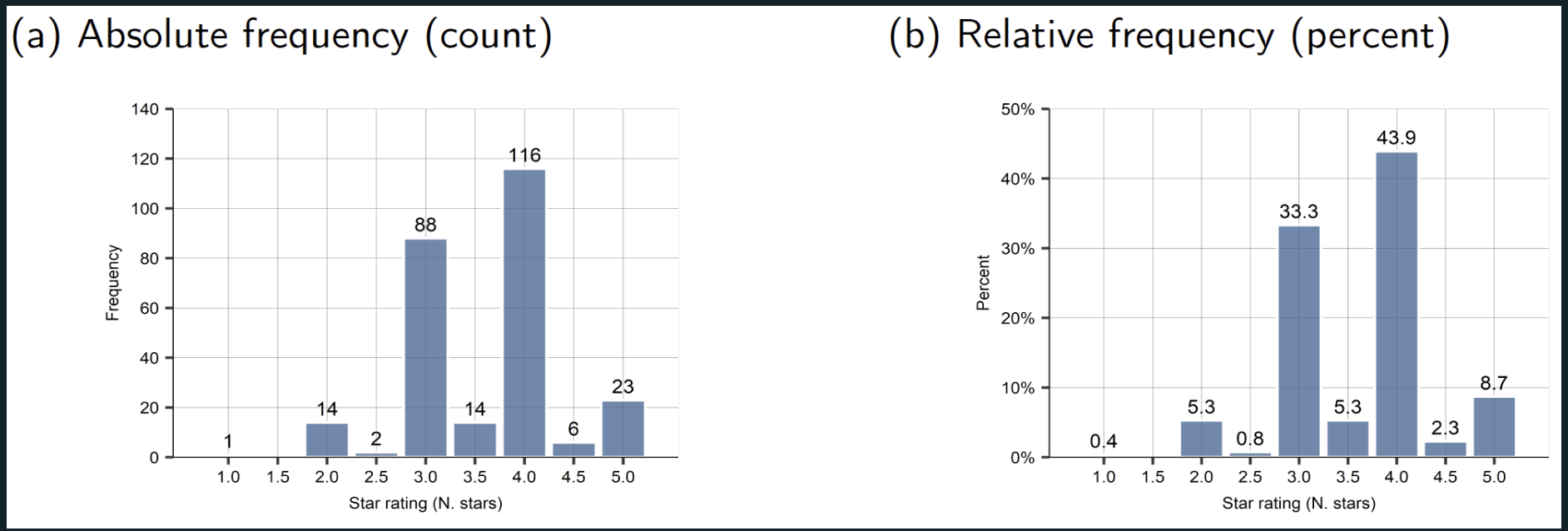
- All variables have a **distribution**
- The distribution of a variable tells the **frequency of each value** of the variable in the data
  - Absolute frequencies (number of observations)
  - Relative frequencies (percent of observations)
- Beware of **missing values**: proportion can be relative to all observations OR only observations with non-missing values (usual choice).

Histograms

Histogram reveals important properties of a distribution.

- Number and location of **modes**:
  - peaks in the distribution that stand out from their immediate neighborhood.
- Approximate regions for center and **tails**
- Symmetric or not
  - Asymmetric distributions have a long (left or right) tail
- **Extreme values**: values that are very different from the rest.

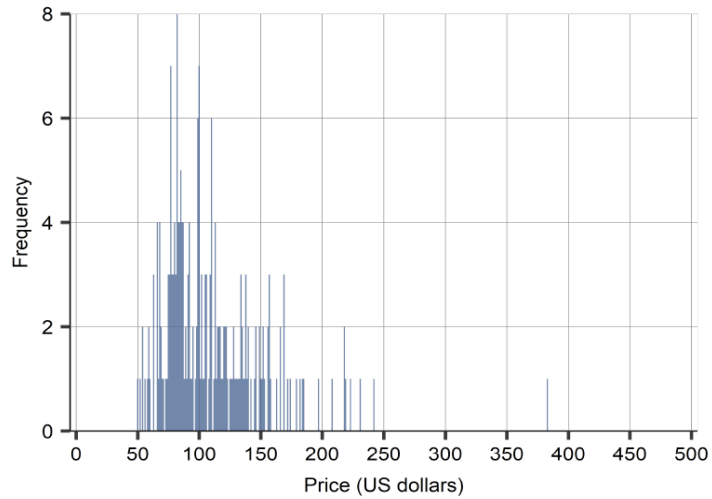
Hotel rating histograms



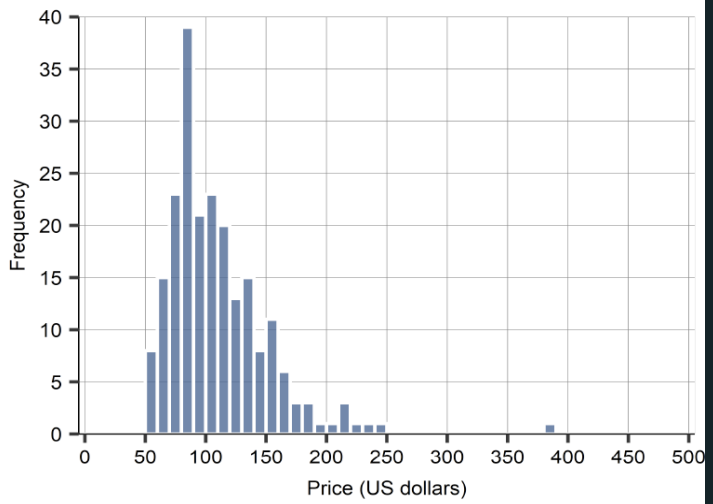
Source: hotels-vienna dataset. Vienna, Hotels only, for a 2017 November weekday

Hotel price histograms

(a) Histogram: individual values



(b) Histogram: 20$ bins



Source: hotels-vienna dataset. Vienna, Hotels only, for a 2017 November weekday

Theoretical distributions

Theoretical distributions can be helpful

- Have well-known properties!
- If variable in our data well approximated by a theoretical distribution –> attribute properties to the variable
- Real life, many variables surprisingly close to theoretical distributions.
- Will be useful when generalizing from data

Normal distribution

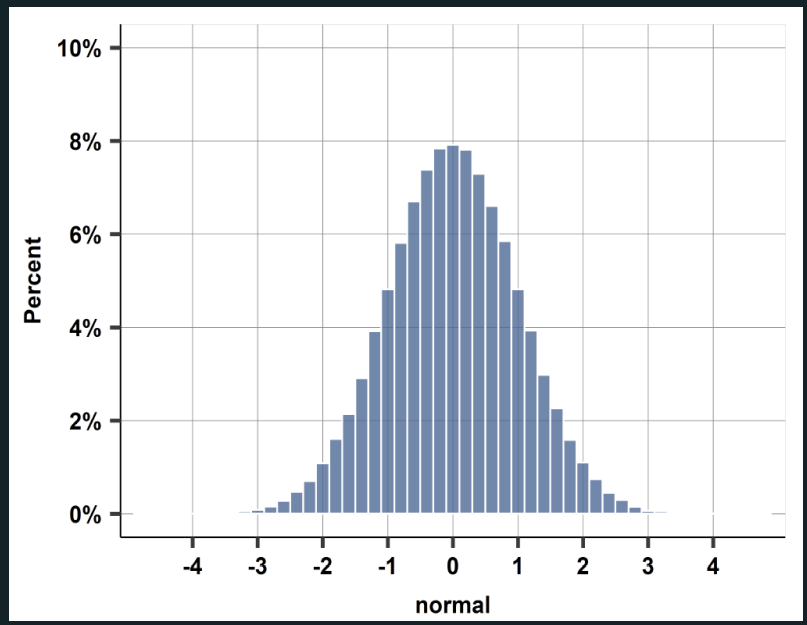
Histogram is bell-shaped

Distribution is captured by two parameters:

- μ is the mean
- σ is the variance

Symmetric = median, mean (and mode) are the same.

Example: height of people, IQs, ect.



The log-normal distribution

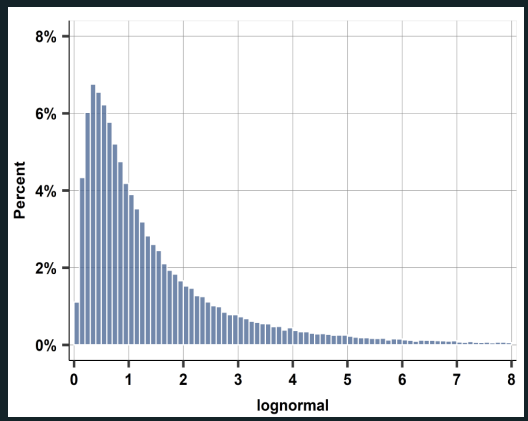
Asymmetrically distributed with long right tails.

Steps:

- start from a normally distributed r.v. (x),
- transform it: (e^x) and
- the resulting variable is distributed log-normal.

Always non-negative

Example distributions of income, or firm size.



Income and log-income

Figure: income

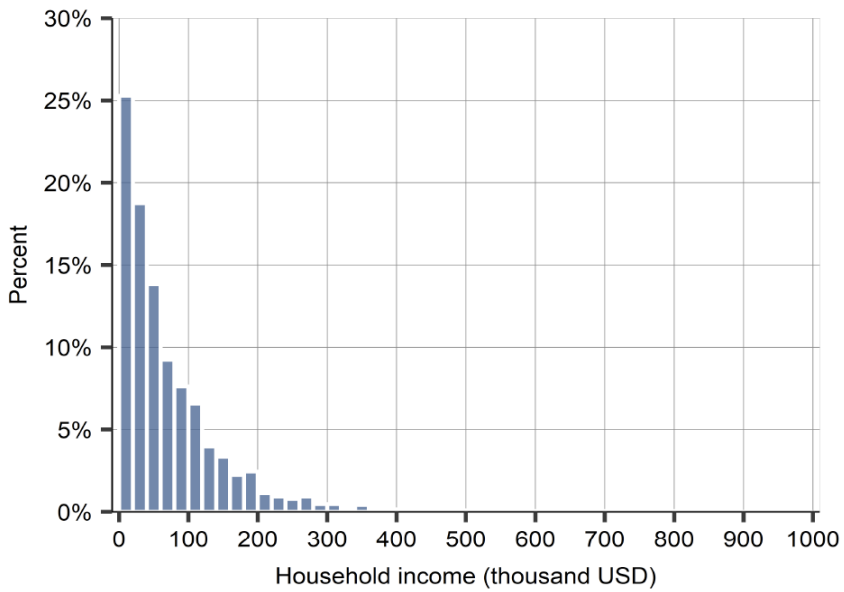
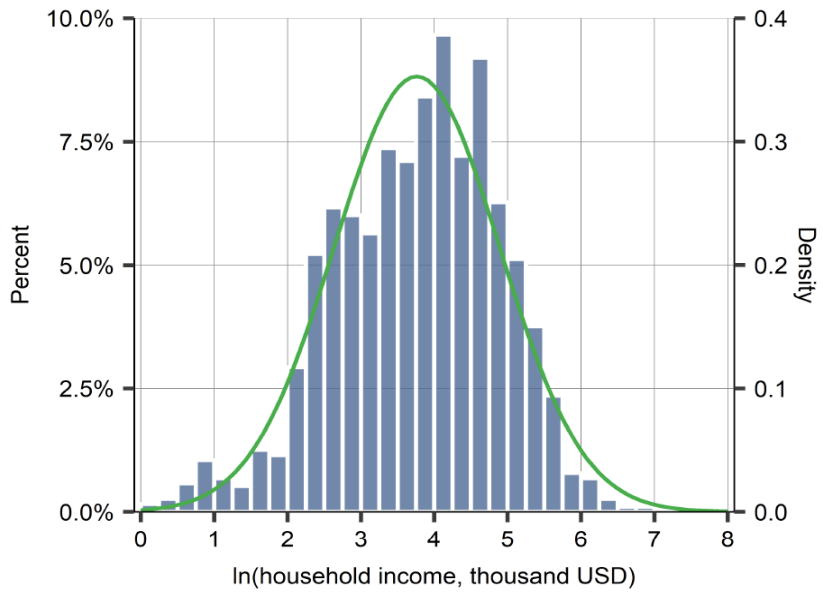


Figure: log income



The power law distribution

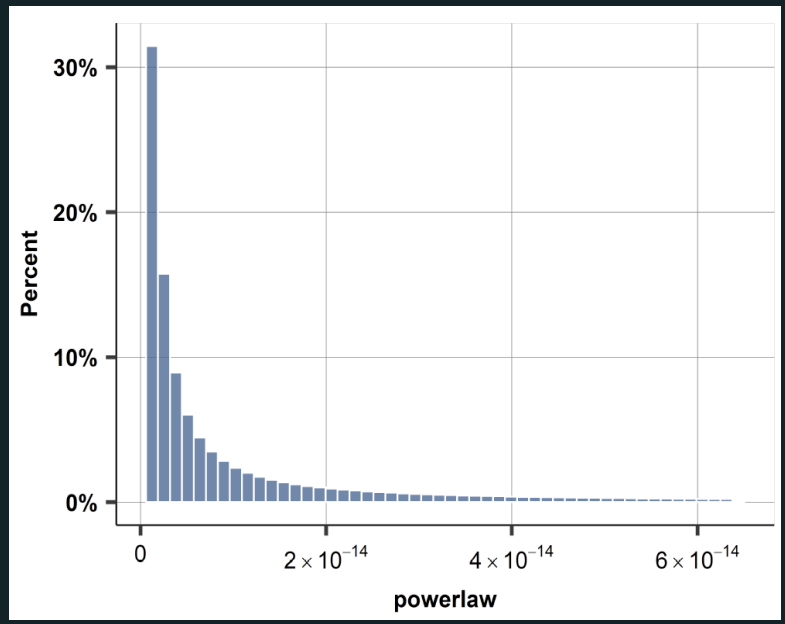
Also called as Pareto distribution

Very large extreme values - well approximated

Relative frequency of close-by values are the same along large and small values

Real world: many examples, but often not the whole distribution

Example: frequency of words, city population, wealth



Next

- Implementation using the introduction to pandas notebook
- More on summary statistics in the comparison & correlation lecture slides
- More on data visualisation principles and practice later.

Data Management

Comparison and correlation

Malka Guillot

HEC Liège | ECON2306

Motivation

- **Are larger companies better managed?**
- To answer such question, we need:
  - data (cf. **previously**)
  - **statistics**
    - summary measures? Interpretations?

In short: comparison & conditioning

- 2 variables:
  - x and y
- Objective:
  - Uncover the patterns of association between x and y
- We compare y, by x values
  - ie. we **condition** y on x (or y given x)
  - y= outcome variable

Table of contents

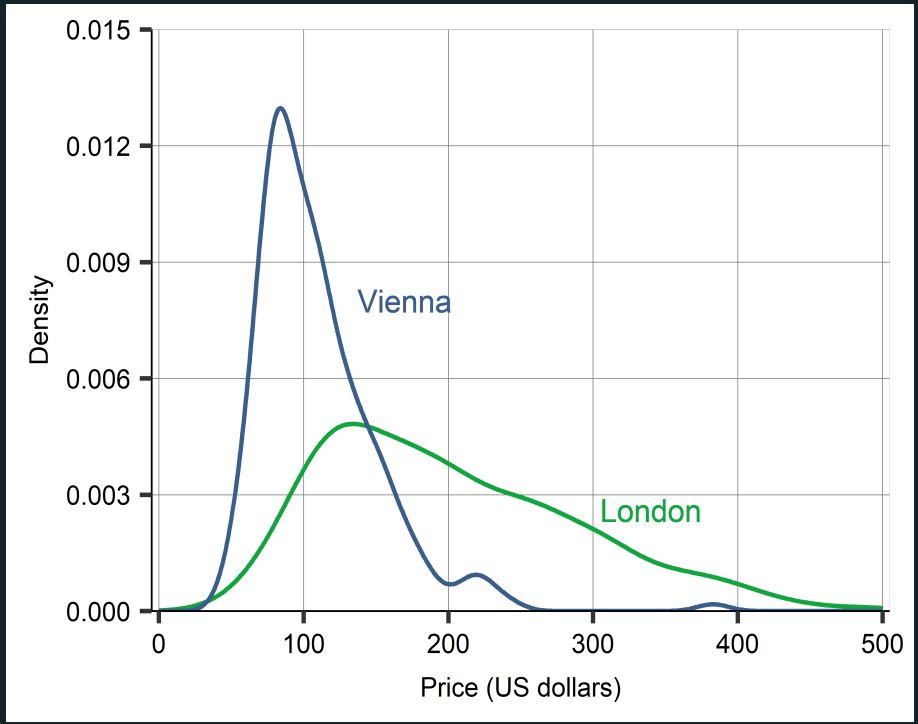
1. Conditional distributions
2. Exemple: Management quality and firm size
3. Correlation & Covariance

Comparisons and conditional distributions

- The conditional distribution of a variable is the distribution of the outcome variable given the conditioning variable.

If the conditioning variable is qualitative (or binary)

  - Comparing histograms



Conditional statistic

- Conditional mean= mean of a variable for each value of the conditioning variable
- The conditional expectation of a variable y given x is:

$$E[y|x]$$

- This is a function
- In the case x is categorical:
  - for a value of x, the cond. exp. gives the expeted value (mean, average) of a y for observations that have that value of x

Conditional and joint distributions of 2 quantitative variables

- 2 variables $\Rightarrow$ many values
- The joint distribution of 2 variables shows the probabilities (frequencies) of each value combination of the 2 variables.

$\Rightarrow$ Scatter plot

(binned) Scatter plot

- a 2D graph with the values of each of the 2 variables measured on its 2 axes
  - Scatter: Each dot correspond to 1 observation
  - Binscatter: averages of y by bins of x (based on quantiles)

Scatter	Binscatter
When dataset is *small*	For larger samples: we bin values

Management quality and firm size

- Management quality and firm size:
  - describing patterns of association
  - **Whether, and to what extent, larger firms are better managed?**
- Answering this question can help understand why some firms are better managed than others.
- Data from the World Management Survey

Measuring management quality

- Interviews by CEO/senior managers, based on that a score is given
- Each score is an assessment of management practices in a particular domain:
  - tracking and reviewing performance or
  - time horizon and breadth of targets, etc
- Measured on a scale of 1 (worst practice) to 5 (best practice).
- Management quality is = average of 18 scores.

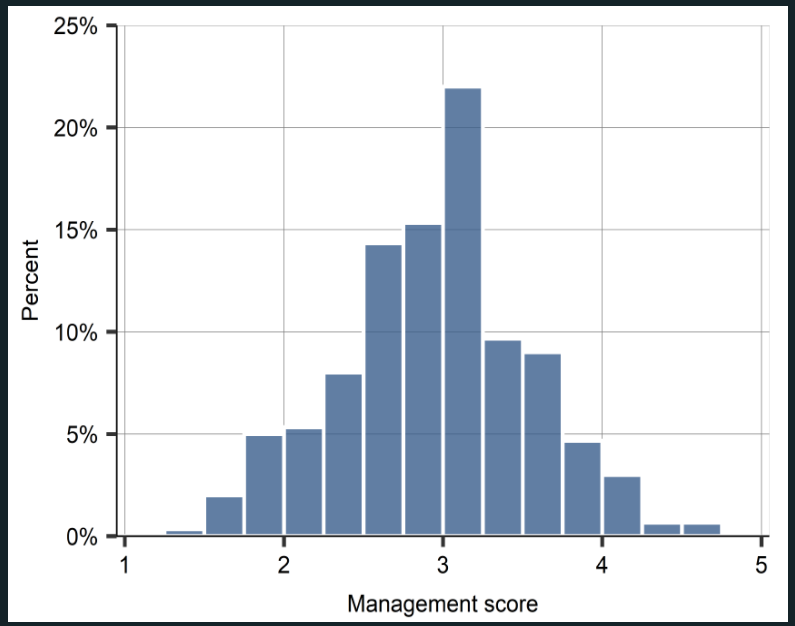
Management quality and firm size

Data

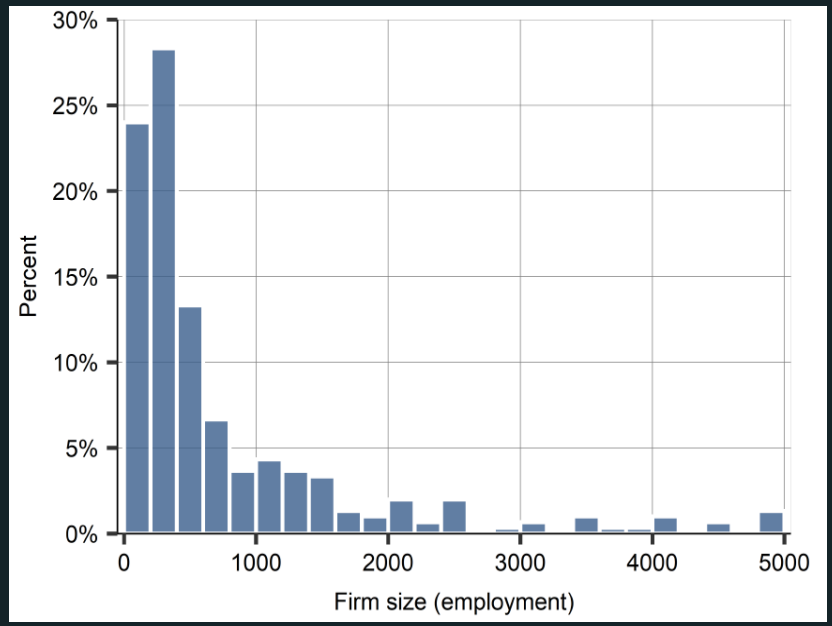
- Cross-sectionnal data of Mexican firms from the 2013 wage survey
- Sample: Only firms with 100-5000 employees ($N = 300$)
- $y =$ quality of management
- $x =$ firm size (number of employees)

Management quality and firm size

Histograms



(a) Management score



(b) Firm size (number of employees

Management quality and firm size

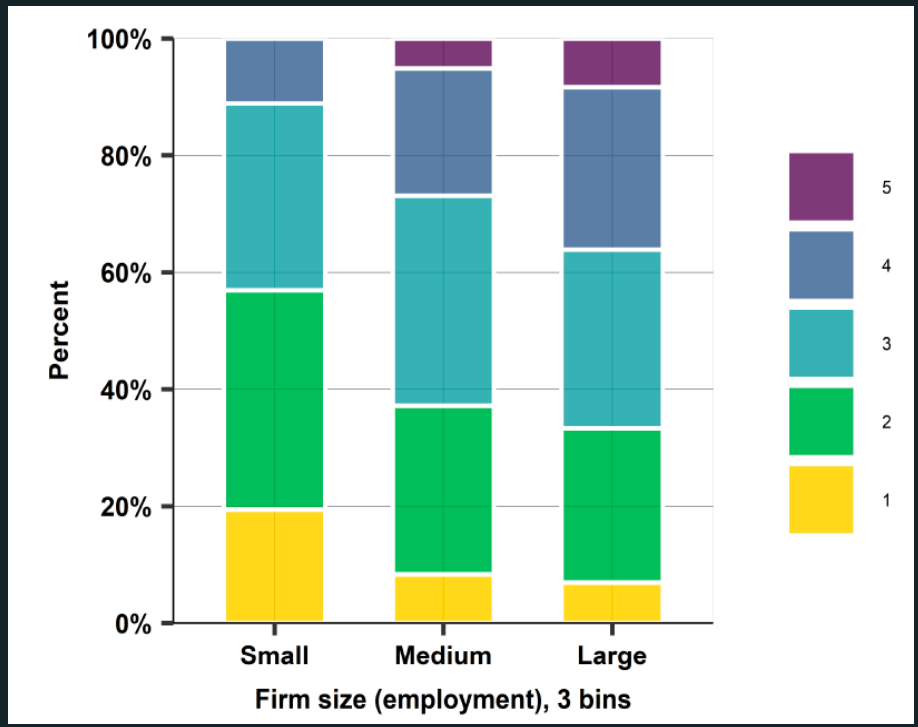
Conditional probabilities in data

- 3 bins of firm size
  - small: 100–199, N=72
  - medium: 200–999, N=156
  - large: 1000, N=72
- For each score variable we have 15 conditional probabilities
  - the probability of each of the 5 values of y by each of the three values of x
  - e.g. $P(y = 1|x = small)$

Management quality and firm size

Conditional probabilities

- Lean management score 1–5
- Firm size: small, medium, large
- Conditional probability:
  - $p(y = 1|x = small) = 20\%$.
  - $p(y = 5|x = large) = 10\%$.
- Shows a pattern of association



Management quality and firm size

Conditional statistic: conditional mean.

Mean given firm size:

- Mean management score is
  - For small firms: 2.68
  - For medium firms: 2.94
  - For large firms: 3.18
- First simple evidence:
  - **Larger firms have better management**

Management quality and firm size

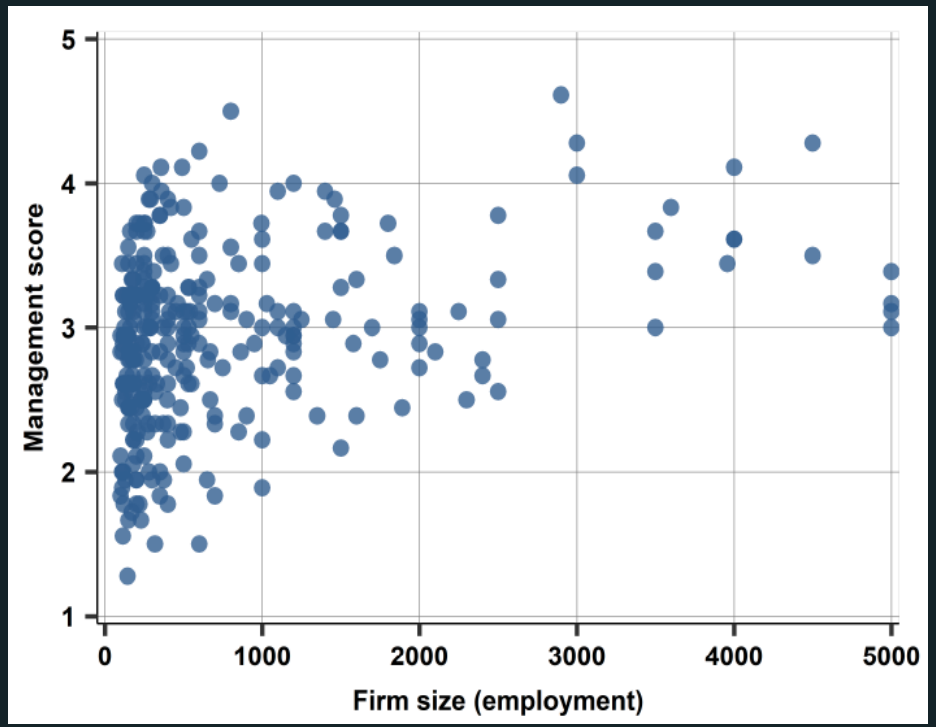
Joint distribution

- How is management quality related to the firm size?
  - $y =$ management score
  - $x =$ employment
- Graphical analysis:
  1. scatterplot
  2. bin scatter

Management quality and firm size

Scatterplot

- Both x- and y- axis quantitative
- Firm size: small, medium, large
- Each dot is an observation:
- Full information on association



Management quality and firm size

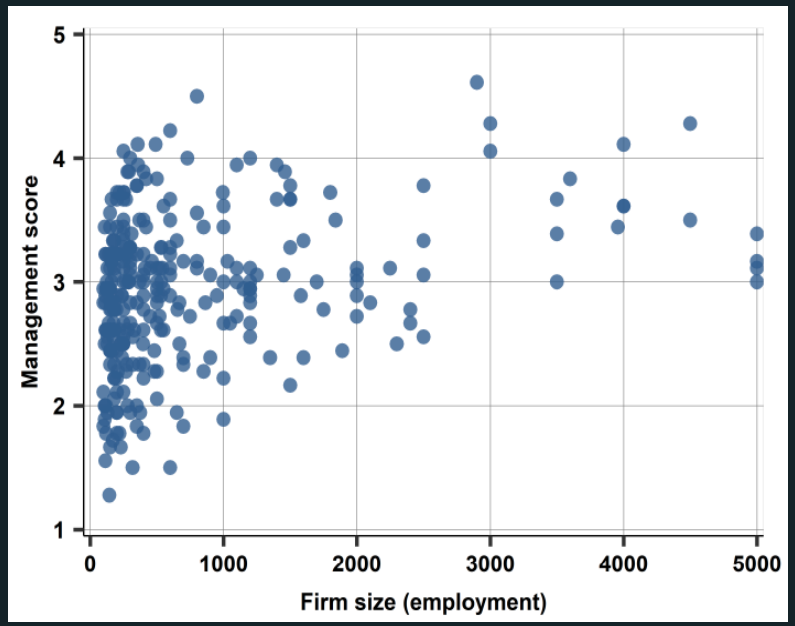
Bin scatter plot

1. Divide x into 10 bins with similar nb. of observations (deciles)
2. Calculate the mean of y conditional on the 10 bins of x.
3. Plot the previous average on the y-axis with bin values on the x-axis
    - i.e. Average management score as a point corresponding to the mean in the employment bin (e.g., 110 for the 100–120 bin).

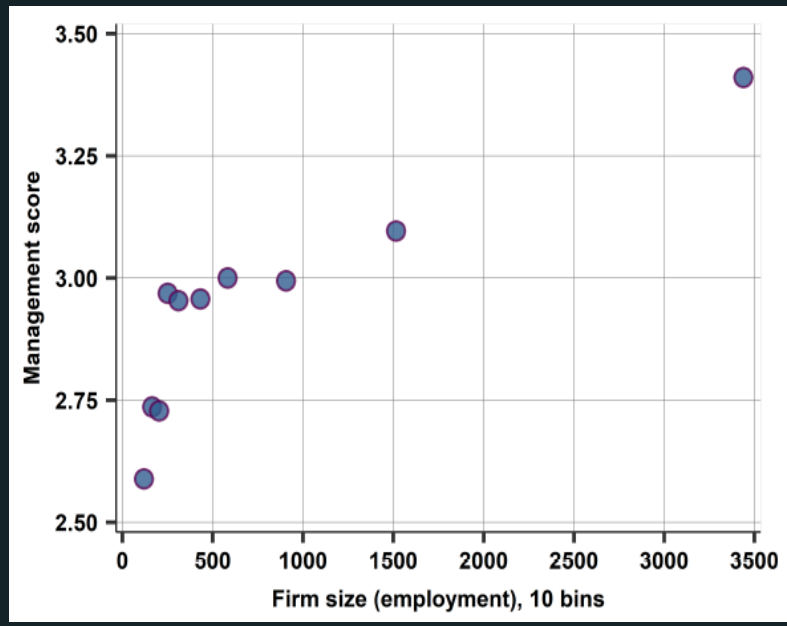
Dots NOT equally spread out - more frequent where more observations!

Management quality and firm size

Joint distributions



(a) Scatterplot



(b) 10 bin-scatter

Dependence and independence

- y is independent of x when the distribution of y does not depend on the conditionning on x
- y is dependent of x when the distribution of y depends on the conditionning on x
  - may take many forms

Mean Dependence

- mean-dependence:
  - conditional expectation $E[y|x]$ varies with the value of x.
  - the extent to which conditional expectations (means) differ.
- measured by covariance and correlation coefficient

Covariance

$$Cov(x, y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{n}$$

Correlation coefficient

$$Corr(x, y) = \frac{Cov(x, y)}{Std(x)Std(y)}$$

- The correlation coefficient is the standardized version of the Covariance
- sum over the observations: $i = 1, \cdots, n$

Management quality and firm size

Correlation of management quality and firm size by industry

Industry	Correlation	# Observations
Auto	0.50	26
Chemicals	0.05	69
Electronics	0.33	24
Food, drinks, tobacco	0.05	34
Materials, metals	0.32	50
Textile, apparel	0.29	43
Wood, furniture, paper	0.28	29
Other	0.44	25
All	**0.30**	**300**

Summary: correlation?

- The correlation coefficient captures a simple measure of mean dependence.
- Qualitative variables:
  - Summarize conditional probabilities (frequencies).
- Quantitative variables:
  - Scatterplots offer a visual insight to the pattern of the relationship.

Data Management

Supervised Machine Learning: Regressions

Malka Guillot

HEC Liège | ECON2306

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2. Overview of ML process
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4. Pre-processing
5. Regression algorithms

References

- Békés & Kézdi (2022) 13 & 14
- Introduction to Statistical Learning chap 1. & 2 & 5.1
- Kleinberg, Ludwig, Mullainathan, and Obermeyer (2015), "Prediction Policy Problems." American Economic Review, 105 (5), pp. 491-95.
- Mullainathan and Spiess (2017), "Machine Learning: An Applied Econometric Approach", Journal of Economic Perspectives, 31 (2), pp. 87-106,

Introduction

[Motivation] Setting

- **Output variable** (or dependent variable) y
  - Dependent variable, target variable, outcomes, labels
- Set of **input variables** $X = (x_1, x_2, \cdots, x_n)$
  - Explanatory variables, independent variables, predictors
- Data generating process

$$f : \mathcal{X} \to \mathcal{Y}$$

$$\mathcal{X} \in \mathbb{R}^{n \times p}, \mathcal{Y} \in \mathbb{R}^{p}$$

Statistical model to approach the DGP

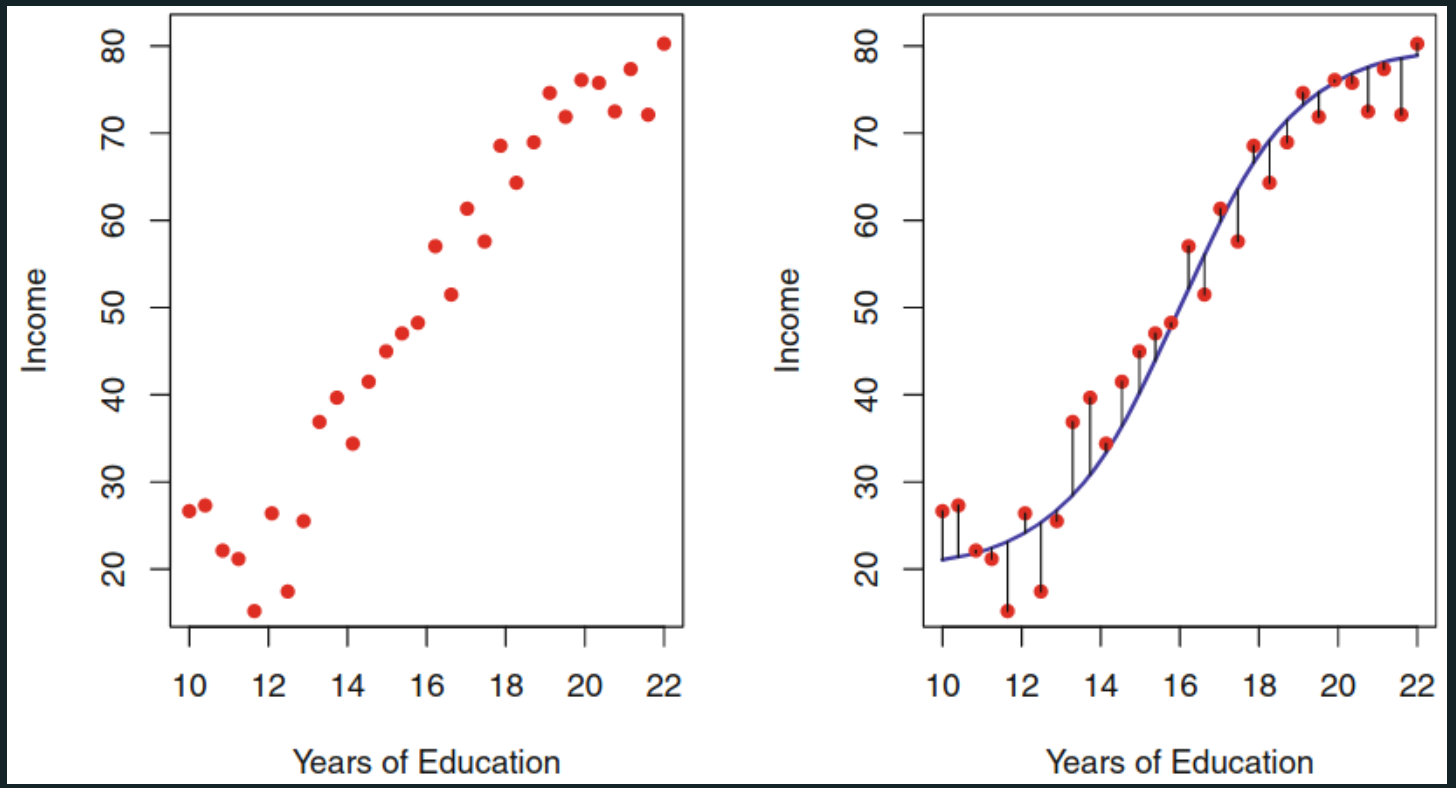
Concretely, finding $f(.)$ s.t.

$$Y = f(X) + \epsilon$$

- $f(X)$ is an unknown function of a matrix of predictors $X = (x_1, \cdots, x_p)$,
- y: a scalar outcome variable
- an error term ϵ with mean zero.
- While X and y are known, $f(\cdot)$ is unknown.

Goal of statistical learning: to utilize a set of approaches to estimate the "best" $f(\cdot)$ for the problem at hand.

Example: income as a function of education



[Motivation] Goals of data analysis

- Description: Document observed relationships between X and y
- Causation: Learn about causal relationships: If we change X, how will y change?
- Prediction: Be able to guess the value of y from the X variables.

$\Rightarrow$ Uncovering the data generating process

[Description] Unsupervised learning

Estimating functions without the aid of outcome variable y.

- We only observe X and want to learn something about its structure
- **Clustering**: Partition data into homogeneous groups based on X
- **Dimensionality reduction** (e.g. PCA)

[Description] Modelling the mean-dependence

- Going further than the correlation analysis
  - Cf. example on management quality and firm size
- Regression analysis uncovers mean-dependence between two variables, controlling for other varialbes:
  - comparing average values of a **dependent variable** (y)
  - for observations that are different in the set of **explanatory variables** (X).
  - Doing so uncovers the pattern of association between y and X.

$\Rightarrow$ Econometrics

[Prediction] Guess the values of y from the X variables

- Estimating functions with **known observations** and **outcome** data.
  - We observe data on y and X
    - $y =$ labels
    - $X =$ predictors
  - We want to learn the mapping $\hat{y} = \hat{f}(X)$
- We predict y by $\hat{y} = \hat{f}(X)$
  - Useful for observations for which y is unknown and X is known

$\Rightarrow$ Supervised machine learning

Supervised Machine learning and/or Econometrics

- **Common objective**: to build a predictive model, for a variable of interest, using explanatory variables (or features)
- **Different cultures**:
  - *E*: probabilistic models designed to describe economic phenomena
  - *ML*: algorithms search through a set of possible prediction models that capture the best relationship

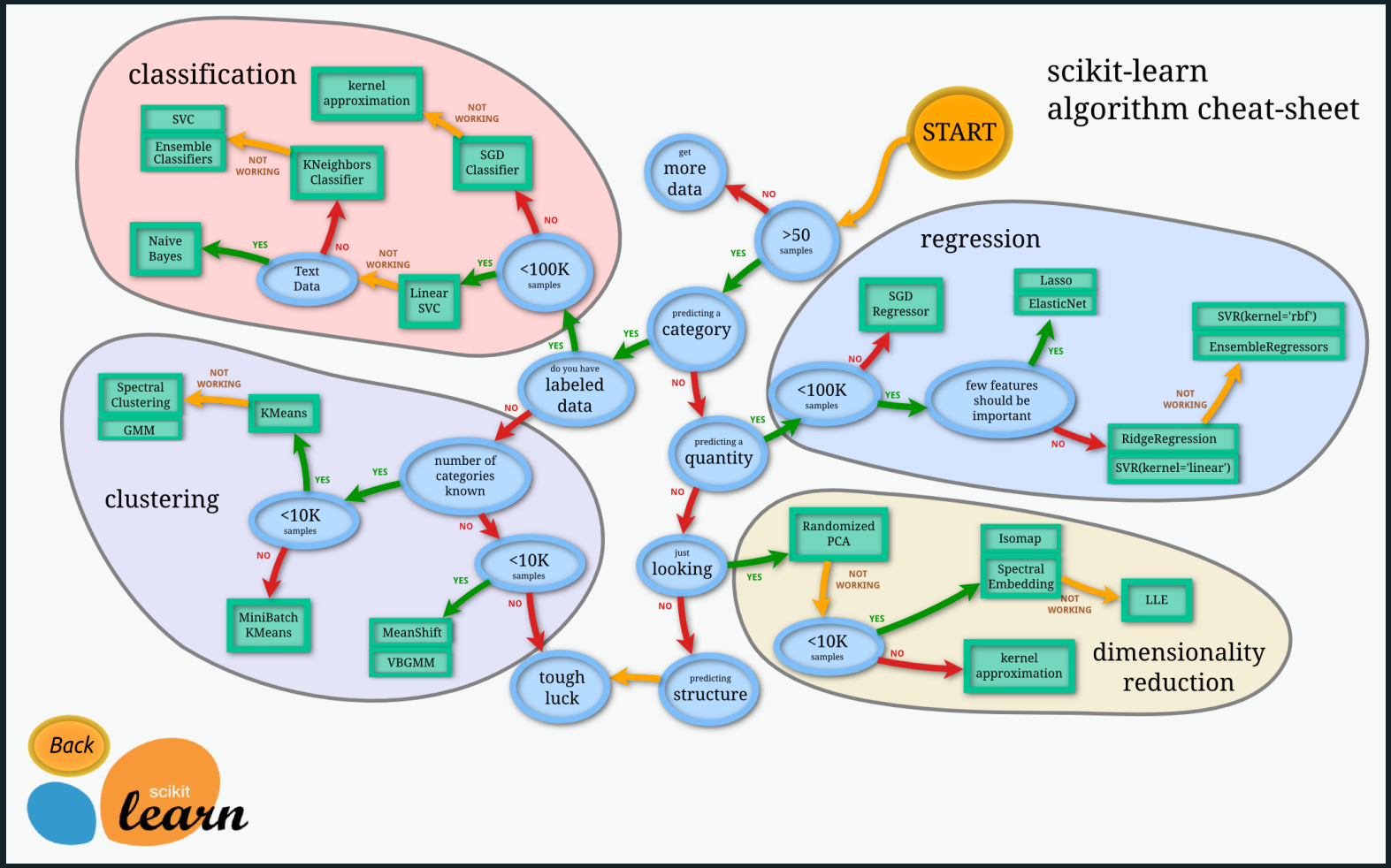
Charpentier A., Flachaire, E. & Ly, A. (2018). Econometrics and Machine Learning. *Economics and Statistics*, 505-506, 147–169.

Prediction setup

- y continuous/quantitative (eg. price)
  - **Regression** problem
- y discrete/binary (eg. default loan, banckrupcy)
  - Probability prediction
  - **Classification** problem
- Time series prediction

Key idea: systematically combine estimation and model selection

Panorama of the models



Source: scikit-learn

Panorama of the model eco-system

- Econometrics: statsmodels
- Machine Learning: scikit-learn
- Deep Learning: keras, pytorch, TensorFlow...

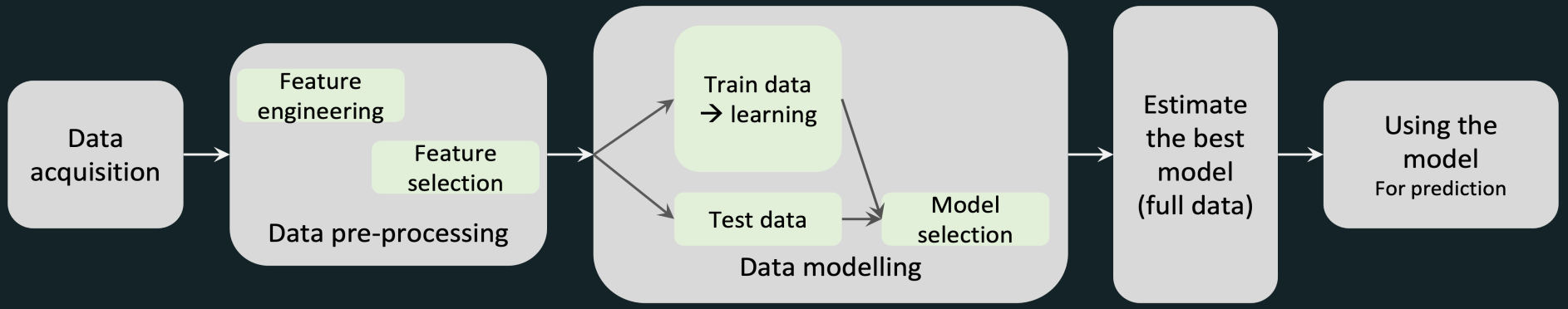
Overview of ML process

How does ML work?

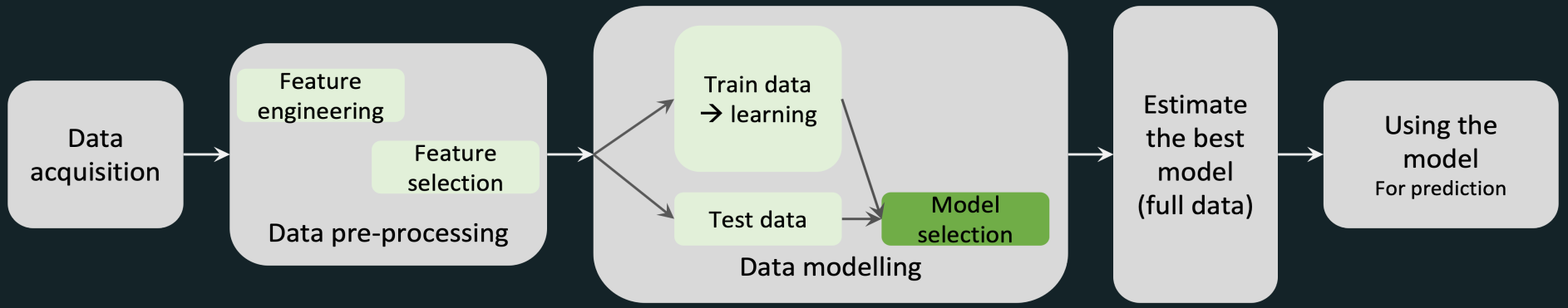
- Econometrics:
  - estimation procedure gives a single set of estimated coefficients
- Machine learning is different
  - Search through a set of possible prediction models
    - Which one does capture best the relationship?
- Estimation of the model on a **training set**
- Test for the goodness of fit on the **test set** (and avoid overfitting)

Typical supervised ML procedure

ML procedure



How to choose between models?



Reducible and irreducible error

$\hat{f}(X) = \hat{y}$ estimated function

$f(X) + \epsilon = \hat{y}$ true function

- **Reducible error**: $\hat{f}$ is used to estimate f, but not perfect
  - Accuracy can be improved by adding more features
- **Irreducible error**: ϵ = all other features that can be used to predict f
  - Unobserved $\rightarrow$ irreducible

Reducible and irreducible error

$$
\begin{aligned}
E(y - \hat{y})^2 &= E[f(X) + \epsilon - \hat{f}(X)]^2 \\
&= \underbrace{[f(X) - \hat{f}(X)]^2}_{Reducible} + \overbrace{Var(\epsilon)}^{Irreducible}
\end{aligned}
$$

⇒ **Objective**: estimating f with the aim of minimizing the reducible error

The Prediction error

- Predicted value $\hat{y}_j$ for target observation j
- Actual value y_j for target observation j
- Prediction error

$$e_j = \hat{y}_j - y_j$$

- Error = predicted alue - actual value

Mean Squared Error (MSE)

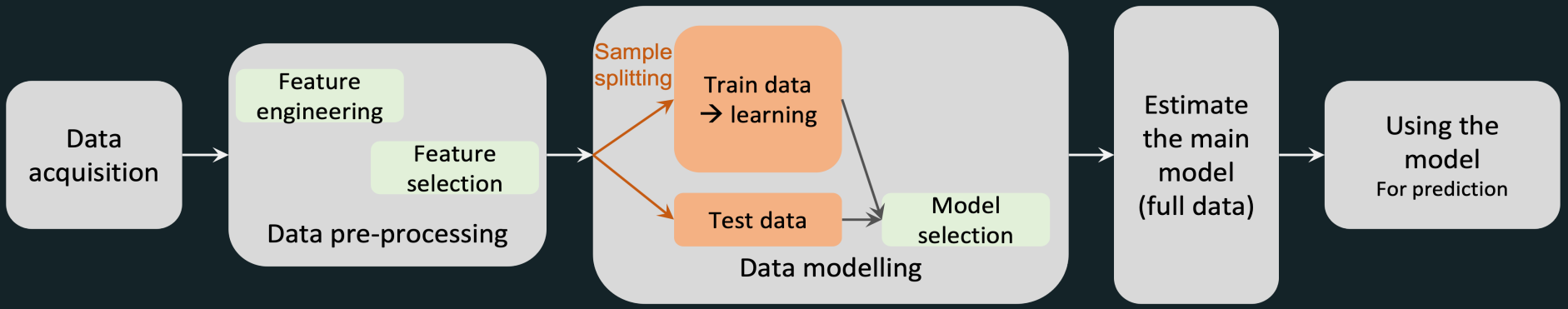
$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{f}(x_i))^2$$

- **Regression setting**: the **mean squared error** is a metric of how well a model fits the data.
- In sample: computed on the training set

$\Rightarrow$ **Training MSE**

Sample splitting approach to measuring model fit

Sample splitting



Resampling methods

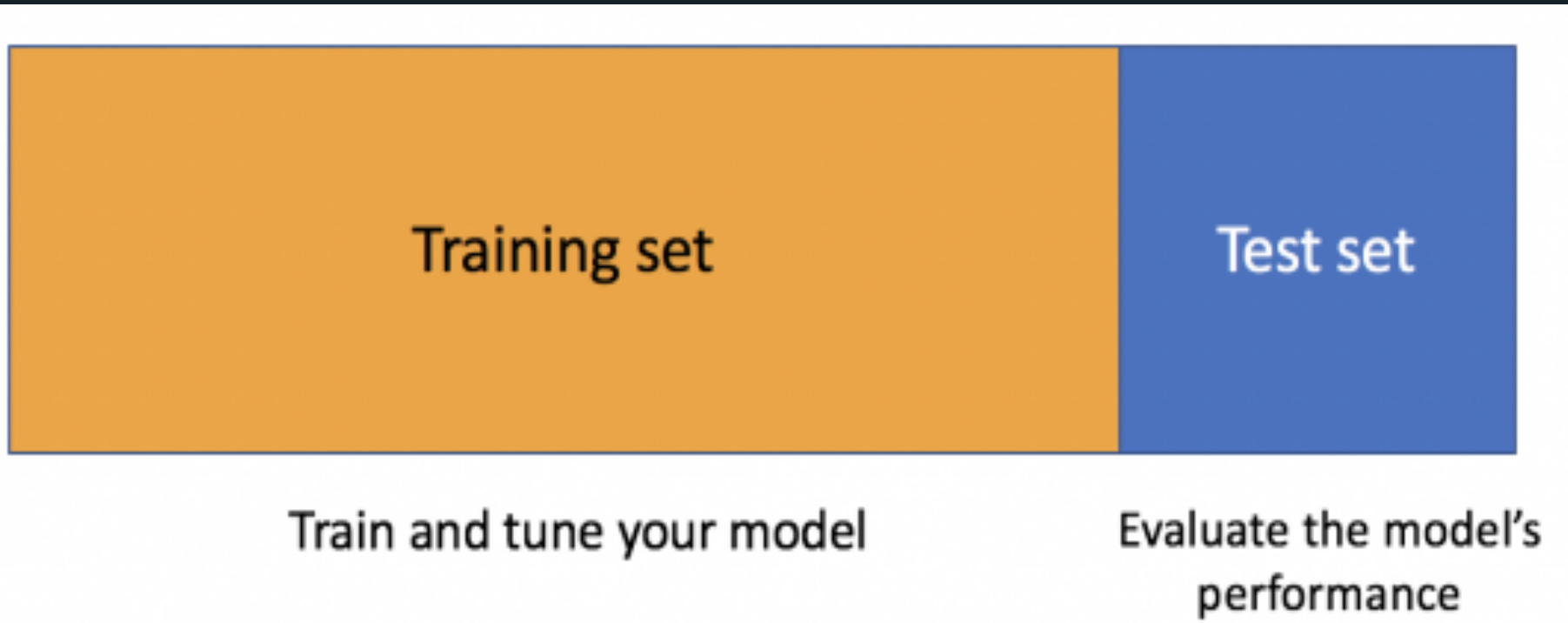
Estimate the test error rate by

holding out a subset of the training observations from the fitting process,

+ then **applying** the statistical learning method to those held out observations

Validation set approach

- Labeled data **randomly** into two parts: training and test (validation) sets.



Two concerns

- Arbitrariness of split
- Only use parts of the data for estimation

  $\rightarrow$ we tend to overestimate test MSE because our estimate of $f(x)$ is less precise

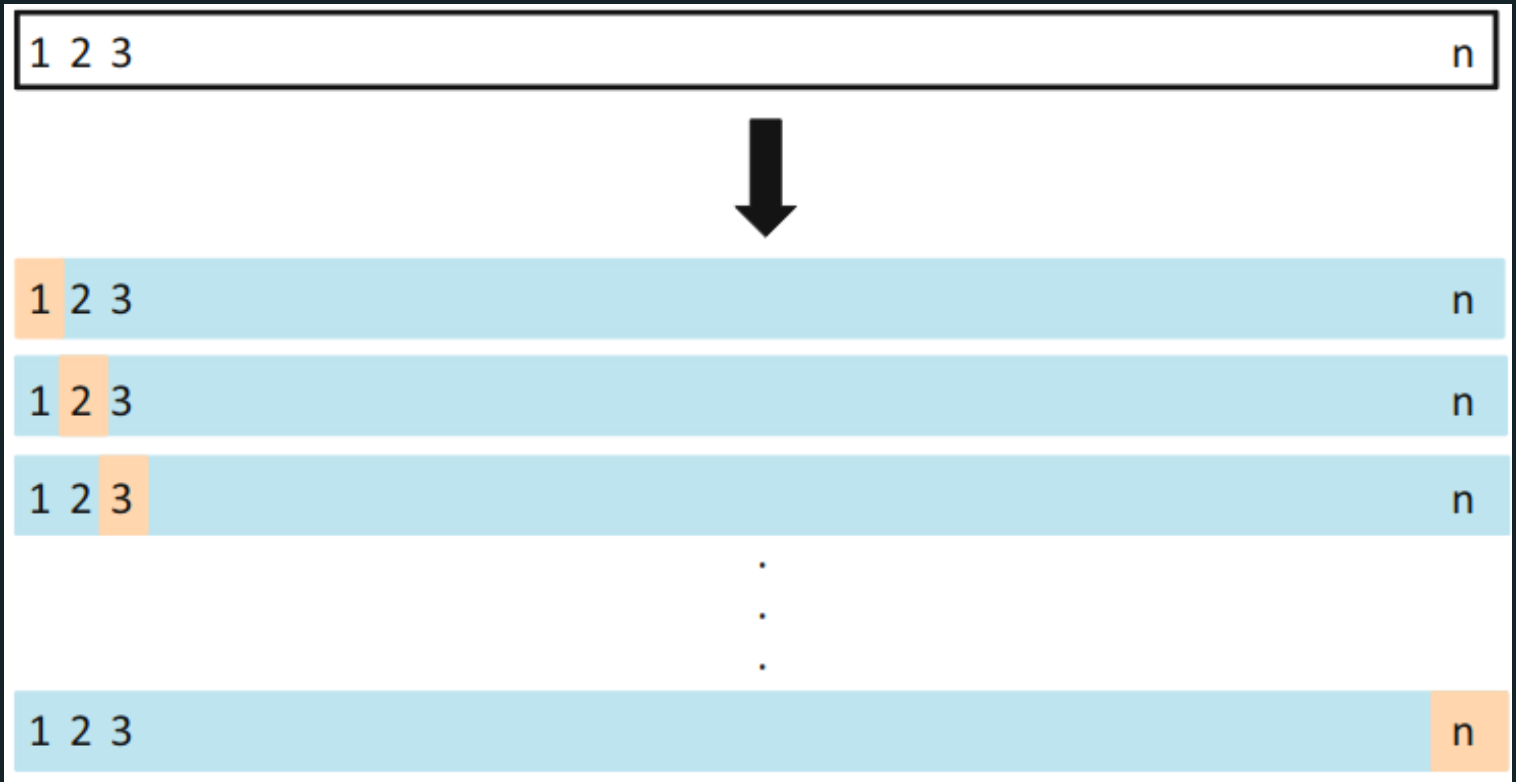
Leave-One-Out Cross-Validation (LOOC)

- Fit on $n - 1$ training observations, and a prediction the Last
- Iterate n times
- Assess the average model fit across each test set.

Estimate for the test MSE:

$$CV_n = \sum_{i=1}^{n} MSE_i$$

Leave-One-Out Cross-Validation (LOOC)



- less bias than the validation set approach
- always yield the same results
- potentially too expensive to implement

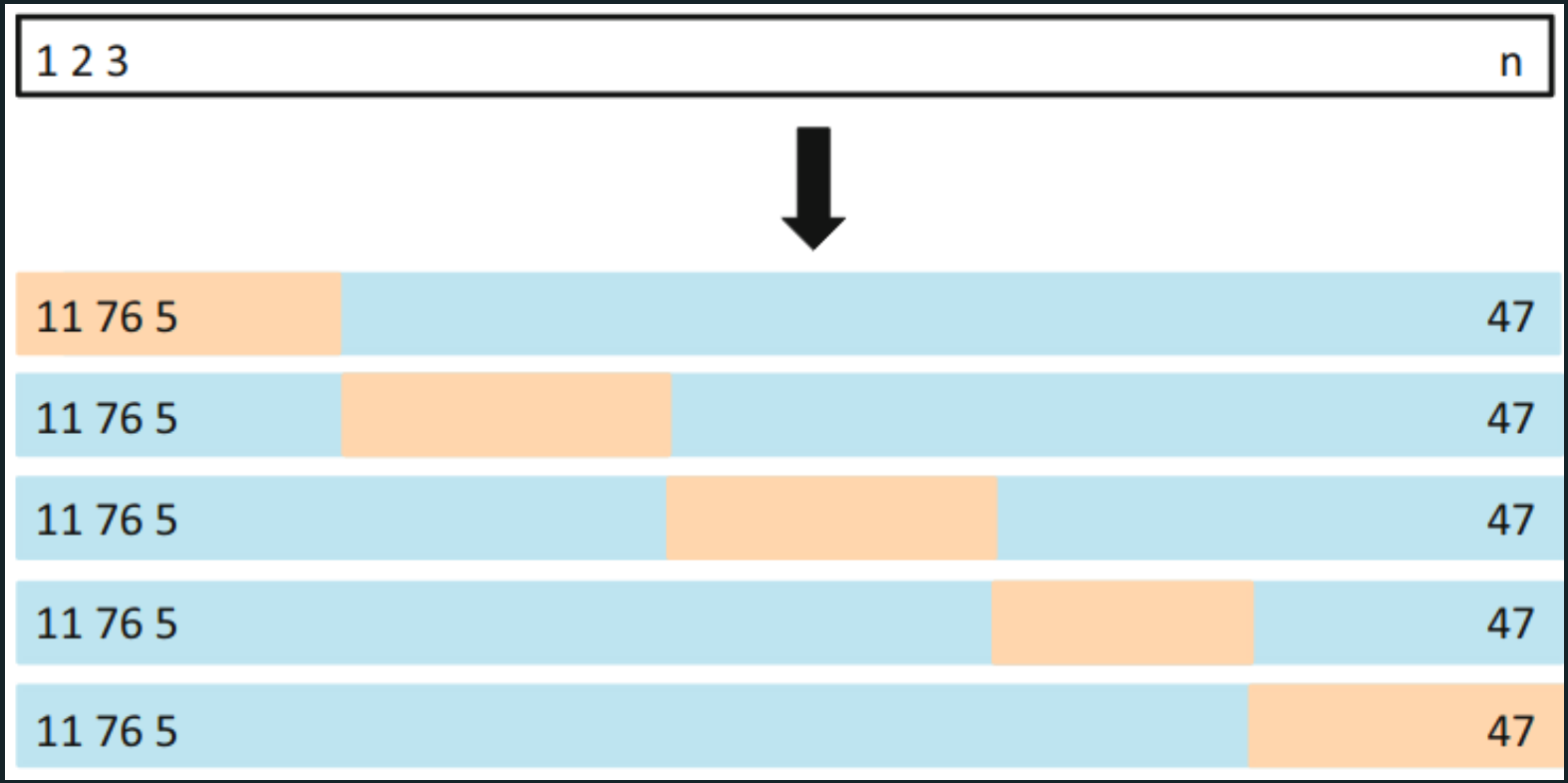
k-fold Cross-validation

- Leave-One-Out Cross-Validation with $k = 1$
- Randomly dividing the data into the set of observations into k groups
- 1st fold is treated as a validation set, and the method is fit on the remaining $k - 1$ folds
- Iterate k times

Estimate for the test MSE:

$$CV_k = \sum_{i=1}^{k} MSE_i$$

k-fold Cross-validation



⇒ Arguably the contribution to econom(etr)ics: Cross-validation (to estimate test MSE)!

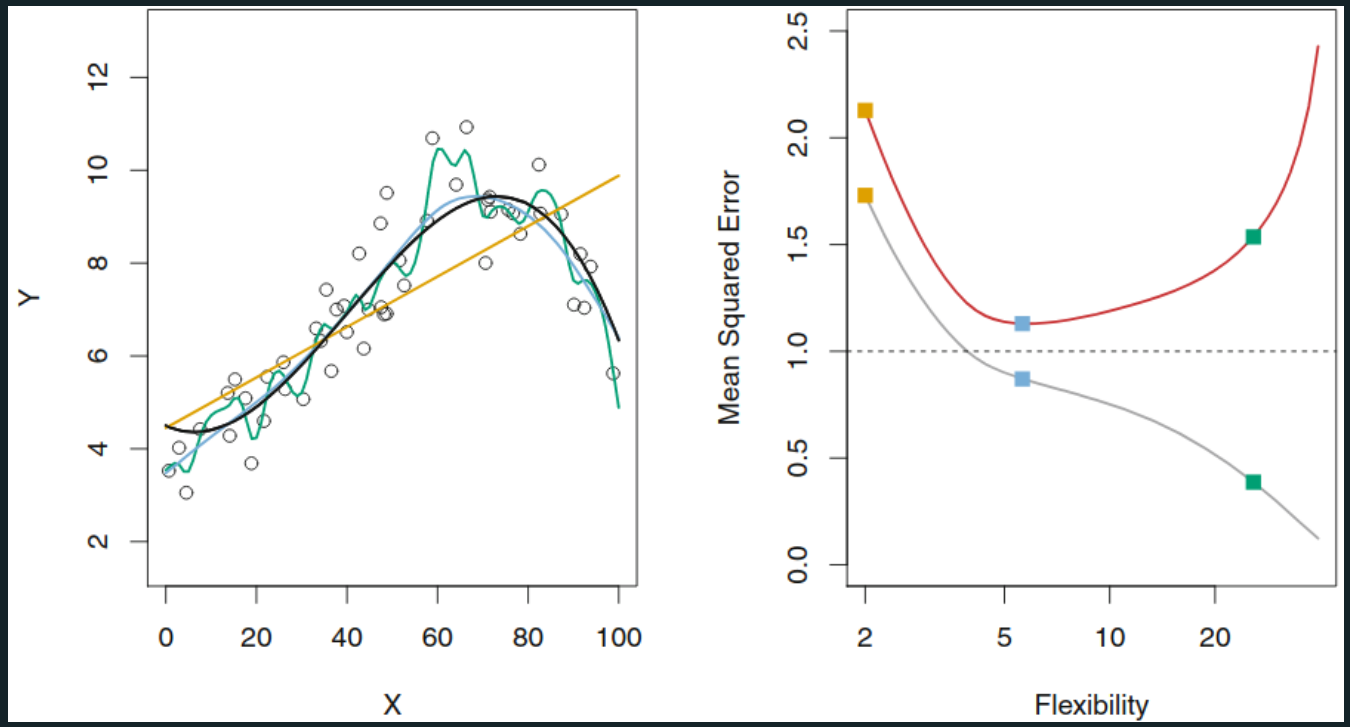
Measuring the fit

To estimate model fit we need to partition the data:

1. **Training set**: data used to **fit** the model
    - **Training MSE**: how well our model fits the training data.
1. **Test set**: data used to **test the fit**
    - **Test MSE**: how well our model fits new data

We are most concerned in **minimizing test MSE**

Training MSE, test MSE and model flexibility



Red (grey) curve is test (train) MSE

Increasing model flexibility tends to **decrease** training MSE but will eventually **increase** test MSE

Overfitting

- As model flexibility increases, training MSE will decrease, but the test MSE may not.
- When a given method yields a small training MSE but a large test MSE, we are said to be **overfitting** the data.
- (We almost always expect the training MSE to be smaller than the test MSE)
- Estimating test MSE is important, but requires training data...

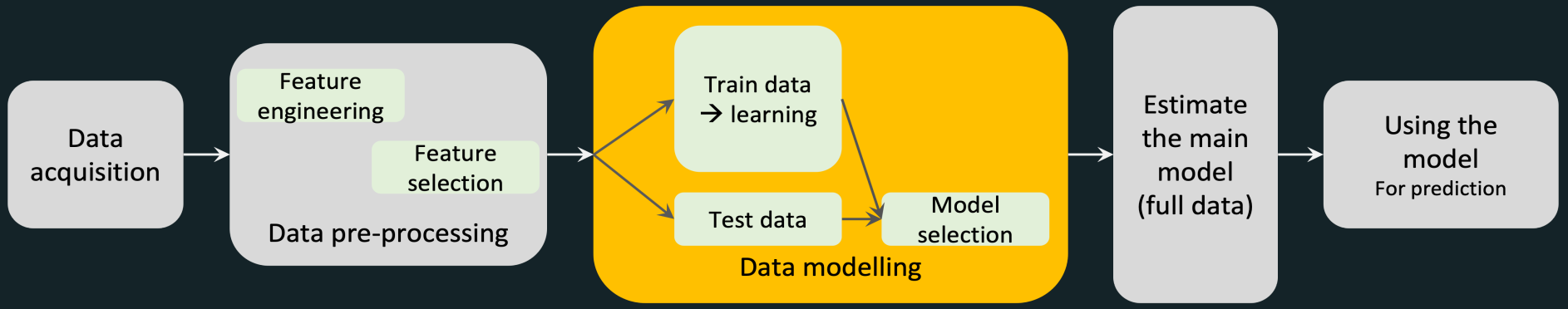
Conclusiong on model selection

= finding the best fit while avoiding overfitting and aiming for high external validity

1. Train the model on the training set
2. Evaluate predictive performances on using th test (validation) set
    - Evaluation criterion (loss function) = RMSE (root mean squared error)

Supervised ML: Regressions

What are the main categories of models?



Model building

- Deciding about the predictors to include in the model and their functional form.
- We have strong computers, cloud, etc - why could not we try out all possible models and pick the best one?
  - Too many models possible!
- 2 methods to build models:
  1. By hand -- mix domain knowledge and statistics
  2. By Smart algorithms = machine learning

1. Linear Regression as a predictive model

Linear Regression (Theory)

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$$

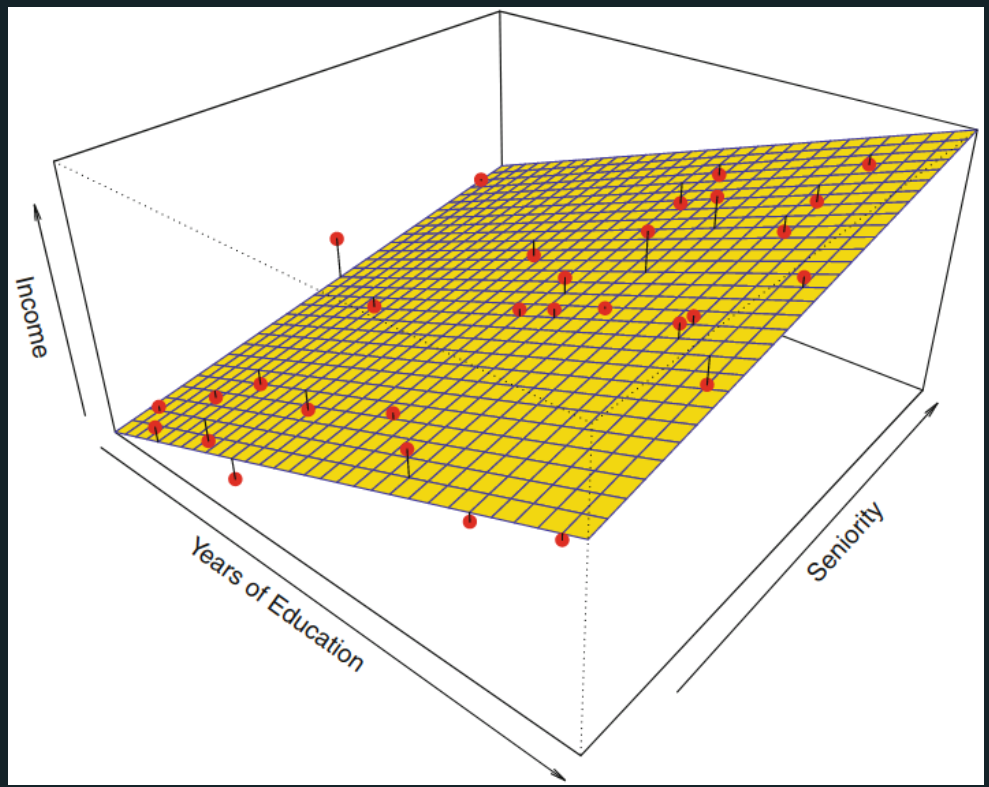
= one of the simplest algorithms for doing supervised learning

A good starting point before studying more complex learning methods

Notes:

Interpretation of β_j = the average effect on Y of a unit increase in X_j holding all other predictors fixed

Linear estimate



Implementation in python

Using statsmodel:

```
import statsmodels.formula.api as smf

reg1 = smf.ols(formula="lnw~female", data=df).fit(cov_type="HC1")
reg1.summary()
```

Extensions of the Linear Model

Going further model's assumptions:

- the **additive**: the effect of changes in a predictor x_j on the response Y is independent of the values of the other predictors
- **linearity**: the change in the response y due to a one-unit change in x_j is constant

Interactions

- Adding interacted variable can help
- Should respect the **hierarchy principle**:
  - if an interaction is included, the model should always include the main effects as well

Non Linearity

- Include transformed versions of the predictors in the model

$\Rightarrow$ Including polynomials in X may provide a better fit

Linear Models: pros and cons

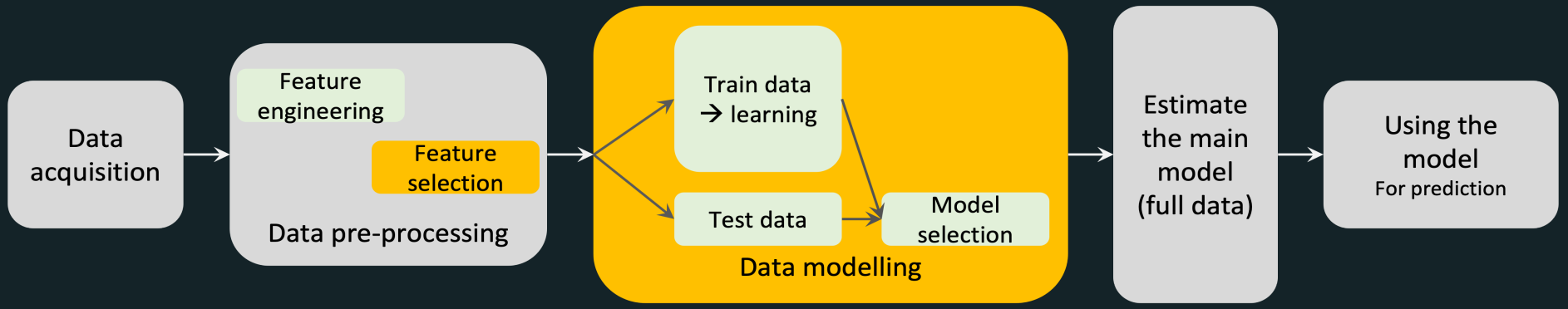
- Pros:
  - Interpretability
  - Good predictive performance
  - Accuracy measures for
    - coefficient estimates (standard errors and confidence intervals)
    - the model
- Cons:
  - When $p > n$
  - Tend to over-fit training data.
  - Cannot handle multicollinearity.

Generalization of the Linear Models

- **Regularized fitting**: ridge regression and lasso
- **Non-linearity**: nearest neighbor methods
- **Interactions**: Tree-based methods, random forests and boosting

2. Regularized linear regressions

Feature selection with regularization



Why Regularization?

- Key question: which features to enter into model, how to select?
  - There is room for an automatic selection process.
- Solution against **overfitting**
- Allow High-Dimensional Predictors
  - $p >> n$: OLS no longer has a unique solution
  - x_i "high-dimensional" i.e. very many regressors
    - pixels on a picture

LASSO

Least Absolute Shrinkage and Selection Operator

LASSO proposes a method to build a model which just **includes the most important predictors**.

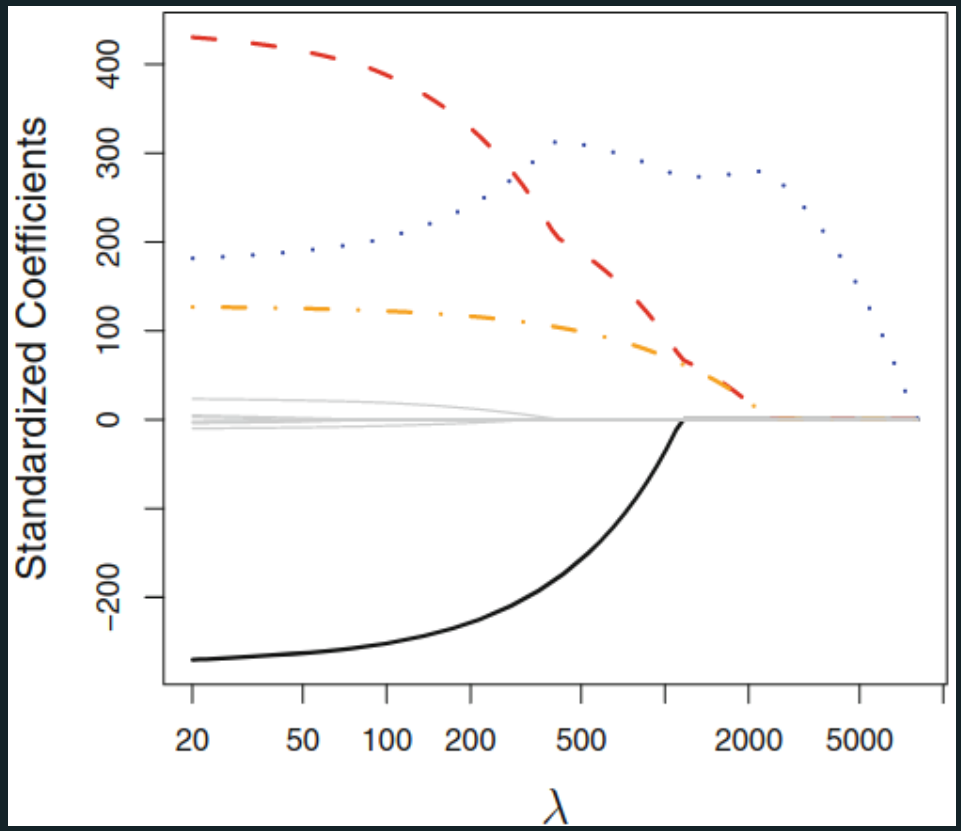
Modifies the way regression coefficients are estimated by adding a penalty term for too many coefficients

Lasso

$$\hat{\beta} = argmin_{\beta} \frac{1}{n} \sum_{i=1}^{n} \left(y_i - (\beta_0 + \sum_{j=1}^{k} \beta_j x_{ij}) \right)^2 + \lambda \sum_{j=1}^{k} |\beta_j|$$

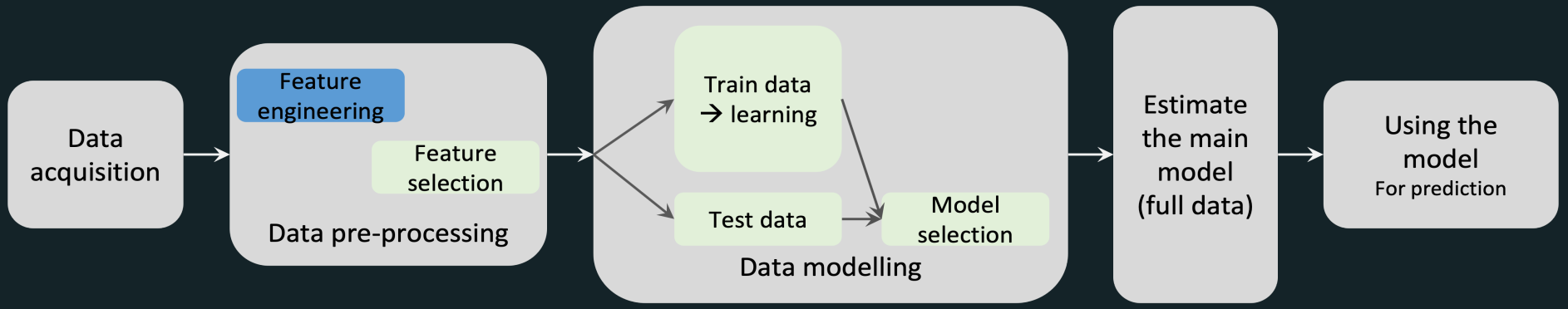
- $R(\beta) \sum_{j=1}^{k} |\beta_j|$ = **regularization function** with a L1 norm penalty function
- λ is a **hyperparameter** (or tunning) where higher values increase regularization

Lasso Coefficients



Working on the features: pre-processing

Pre-processing



Feature engineering: what x variables to have and in what functional form

Feature engineering typically includes

- deciding on what predictor variables to include
- exploring and addressing potential missing values and extreme values
- deciding about function forms of the predictor variables
- designing possible interactions of the predictor variables

What to do with different type of variables

- If binary (e.g, yes/no; male/female; 1/2)
  - create a 0/1 binary variable
- If string / factor
  - check values, and create a set of binaries.
- Continuous
  - Make sure it is stored as number
  - standardization/ normalization
- Text – Natural Language Processing.
  - Mining the text to get useful info.
  - May simply go and find some words, and create binaries

What's next?

- Classification
  - Logistic regression
  - Different measure of the performance
- Unsupervised learning
  - Dimensionality reduction
  - Clustering methods

Data Management

Supervised Machine Learning: Classifications

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Reference:

- JWHT: chap 2.2.3, 4

Classification Framework

- Response/target variable y is **qualitative** (or **categorical**):
  - 2 categories $\rightarrow$ binary classification
  - More than 2 categories $\rightarrow$ multi-class classification
- Features X:
  - can be high-dimensional
- We want to assign a class to a **quantitative response**

  $\rightarrow$ probability to belong to the class
- **Classifier**: An algorithm that maps the input data to a specific category.
- Performance measures specific to classification

Application examples

- In business:
  - Loan default prediction
  - Type of costumer
- In public economics:
  - Tax evasion prediction
- In political sciences:
  - political affiliation of author of texts
- In medical sciences:
  - Diagnostic diseases, drug choice
- Other:
  - email filtering, speech recognition...

Why not fitting a linear regression?

- **Technically possible** to fit a linear model using a categorical response variable but it implies
  - an **ordering** on the outcome
  - a **scale** in the class difference

→ If the response variable was coded differently, the results could be completely different

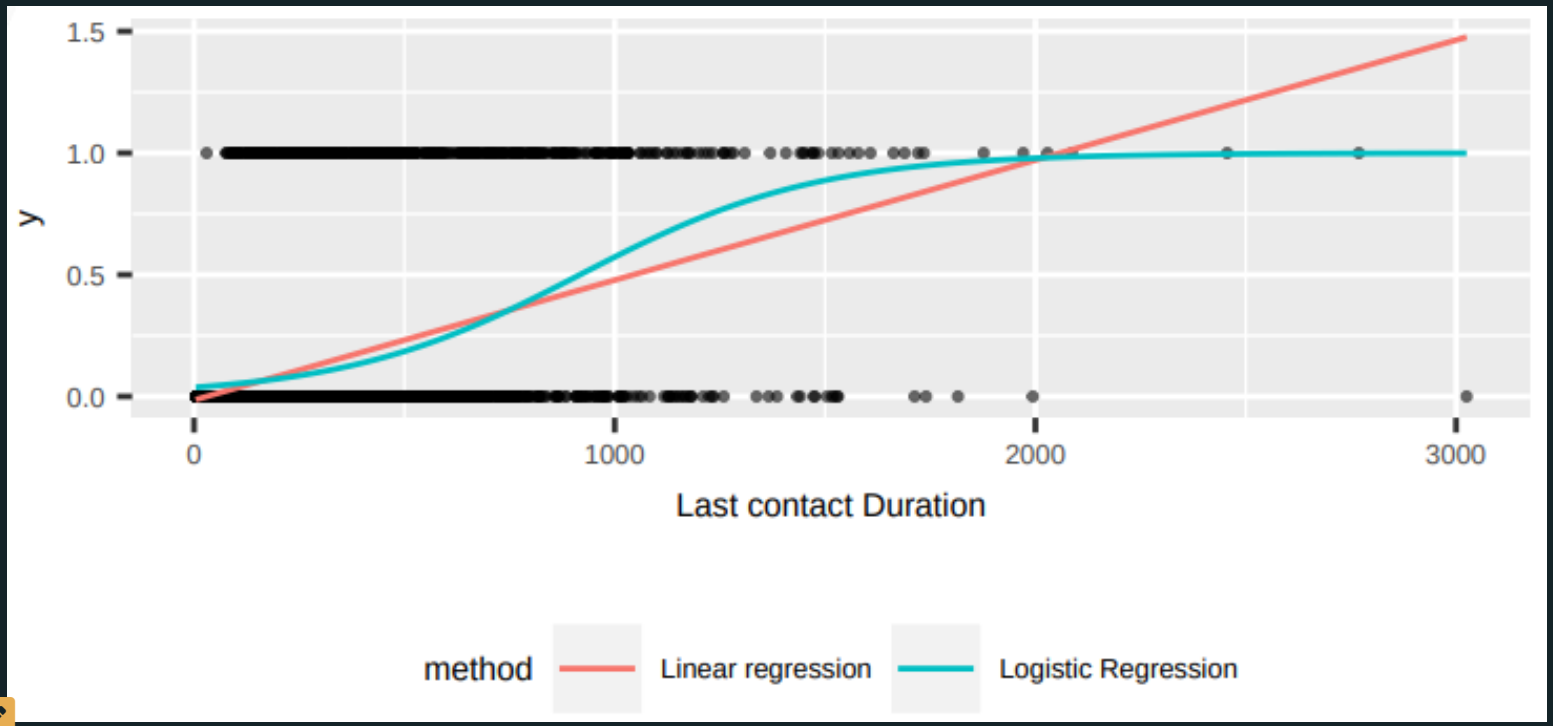
- Less problematic if the response variable is **binary**
  - The result of the model would be stable
  - But prediction may lie outside of $[0, 1]$: hard to interpret them in terms of probabilities

Example

- We predict y, the **occupation of individuals**:

$$y = \begin{cases} 0 & \text{if blue-collar} \\ 1 & \text{if white-collar} \end{cases}$$

- based on their characteristics X (gender, wage, contract duration, experience, age...)



Classification process

1. Model probability
   - Probability of belonging to a category

$$P(y = 1 \mid X)$$

2. Predict probability
   - rely on this probability to assign a class to the observation.
     - For example, we can assign the class 1 for all observations where $P(y = 1|x) > 0.5$
     - But we can also select a different **threshold**.
3. We can make errors
   - False negative
   - False positive

Binary Classifier

- Logistic Regressions
- Support Vector Machine

Logistic Regression

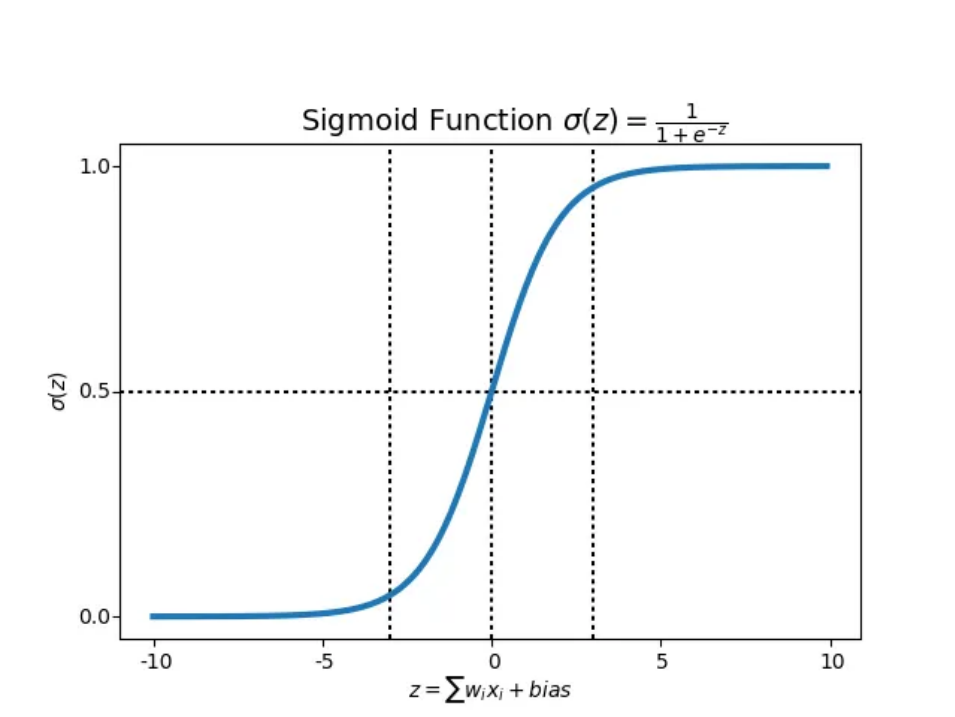
- Like OLS, logistic "regression" computes a weighted sum of the input features to predict the output.
  - But it transforms the sum using the **logistic function**.

$$\hat{p} = \Pr(Y_i = 1) = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_p x_p)$$

where $\sigma(\cdot)$ is the sigmoid function

$$\sigma(a) = \frac{1}{1 + \exp(-a)}$$

Logistic Regression



- Prediction:

$$\hat{y} = \begin{cases} 0 & \text{if } \hat{p} < .5 \\ 1 & \text{if } \hat{p} \geq .5 \end{cases}$$

Logistic Regression Cost Function

- The cost function to minimize is

$$J(\theta) = \underbrace{-\frac{1}{m}}_{\text{negative}} \sum_{i=1}^{m} [\underbrace{y_i}_{y_i=1} \underbrace{\log(\hat{p}_i)}_{\log \text{ proby}_i=1} + \underbrace{(1-y_i)}_{y_i=0} \underbrace{\log(1-\hat{p}_i)}_{\log \text{ proby}_i=0}]$$

  - this does not have a closed form solution
  - but it is convex, so gradient descent will find the global minimum.
- Just like linear models, logistic can be regulared with L1 or L2 penalties, e.g.:

$$J_2(\theta) = J(\theta) + \alpha_2 \frac{1}{2} \sum_{i=1}^{n} \theta_i^2$$

Support Vector Machine

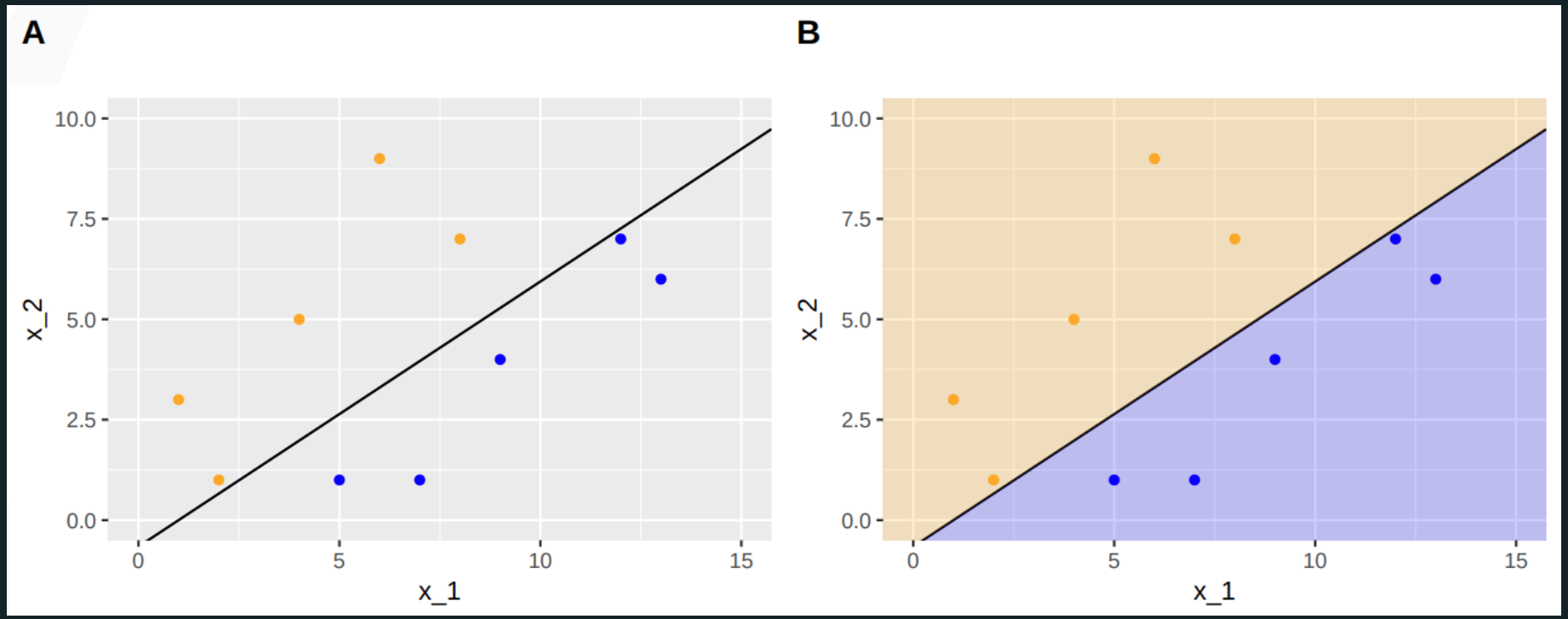
- Context: developed in the mid-1990s
- A generalization of the early logistic regression (1930s)
- One of the best “out of the box” classifiers
- Core idea: hyperplane that separates the data as well as possible, while allowing some violations to this separation

Support Vector Machine: context and concepts

- Pieces of the puzzle:
  1. A **maximal margin classifier**: requires that classes be separable by a linear boundary.
  2. A **support vector classifier**: extension of the maximal margin classifier.
  3. **Support vector machine**: further extension to accommodate non-linear class boundaries.
- For binary classification, can be extended to multiple classes

Classification and Hyperplane

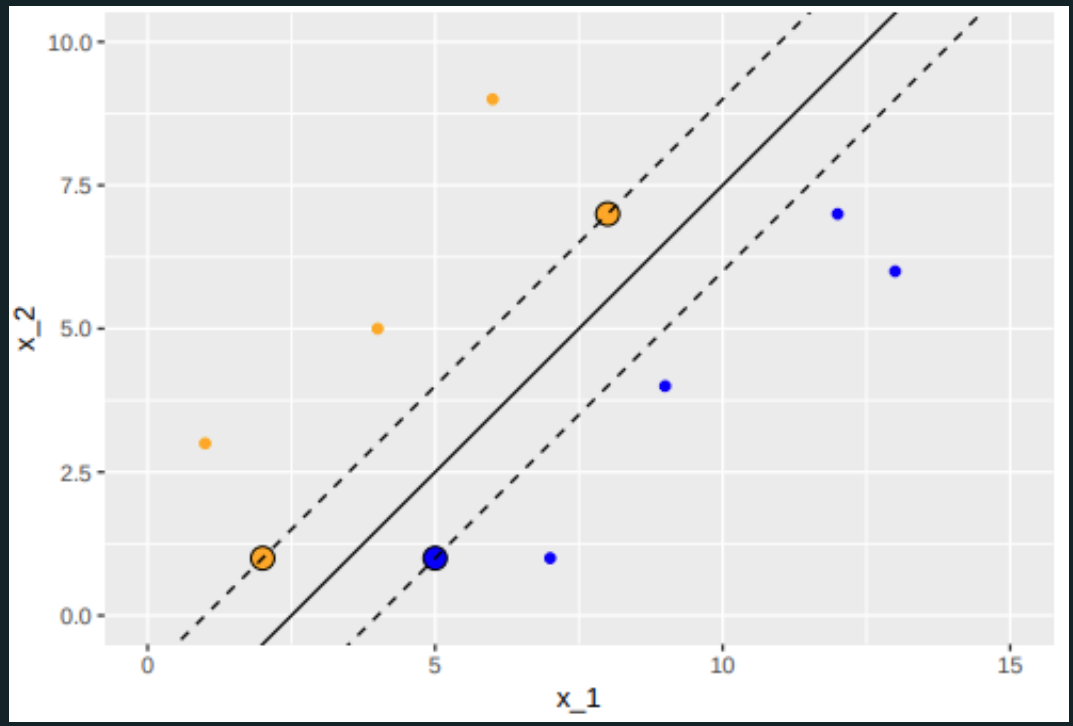
A perfectly separating linear hyperplan for a binary outcome



There are an infinity of such separating hyperplan
→ we need to choose one

Maximum Margin

Maximum margin classifier for a perfectly separable binary outcome variable



Criterium for optimal choice: the separating hyperplane for which the margin is the farthest from the observations
i.e., to select the **maximal margin hyperplane**

Support Vector

Support vector = the 3 observations from the training set that are equidistant from the maximal margin hyperplane

→ they “support” the maximal margin hyperplane (if they move, the the maximal margin hyperplane also moves)

Overcoming the perfectly separable hyperplan assumption

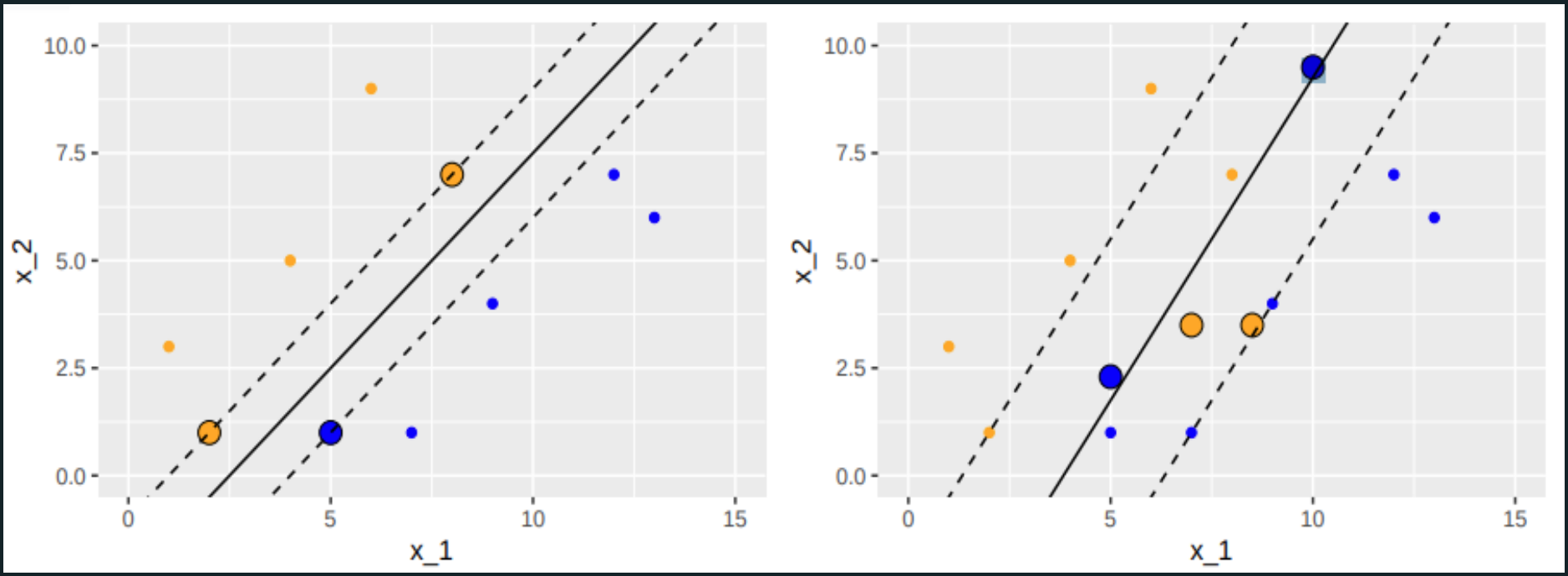
We allow some number of observations to violate the rules so that they can lie on the wrong side of the margin boundaries.

→ find a hyperplane that almost separates the classes

The **support vector classifier** generalizes the maximum margin classifier to the non-separable case.

Support Vector Classifiers

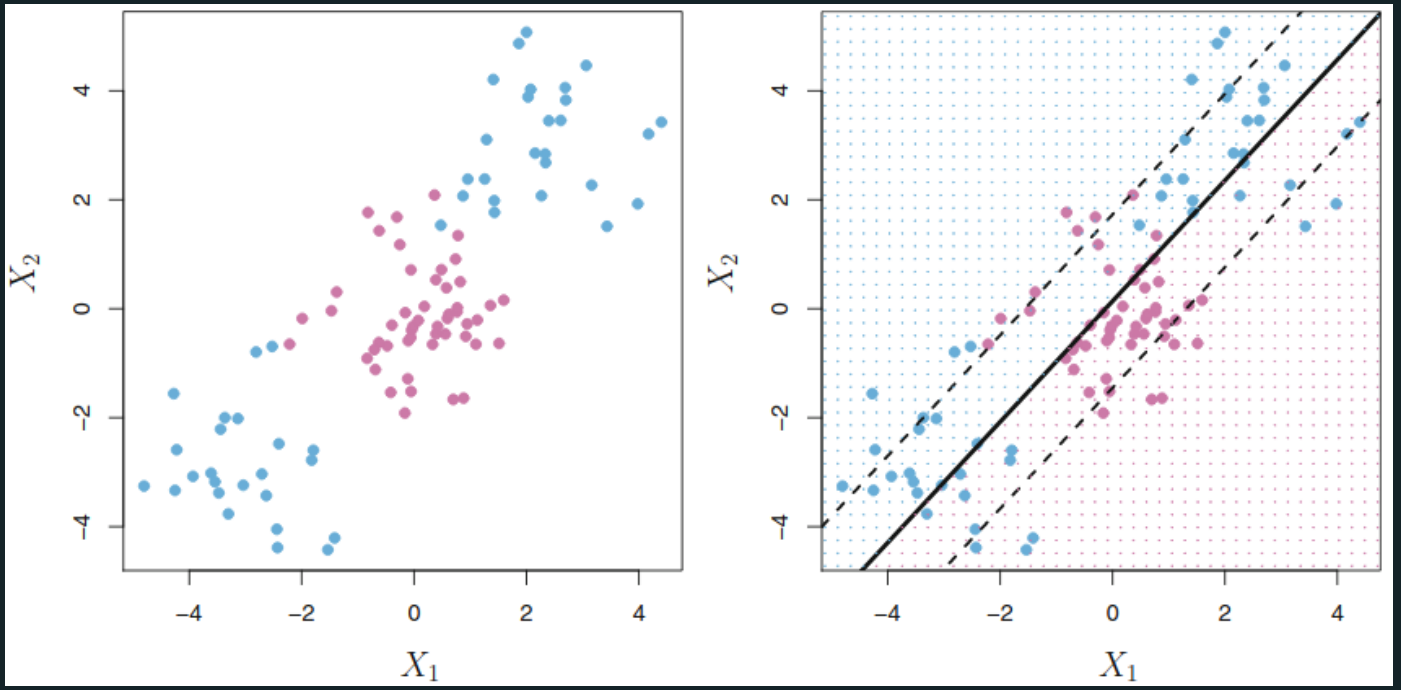
Maximal margin classifier (left) and support vector classifier (right)



Support Vector Classifiers: Details

- A **tuning parameter** C determines the severity of the violation ot the margin that the model tolerates
  - chosen by cross Validation
  - controls the bias-variance trade-off
- C small $\rightarrow$ narrow margins, rarely violated
- C large $\rightarrow$ wide margins, allow more violation
  - More bias classifier, but lower variance

Shortcomings of the linearity assumption:

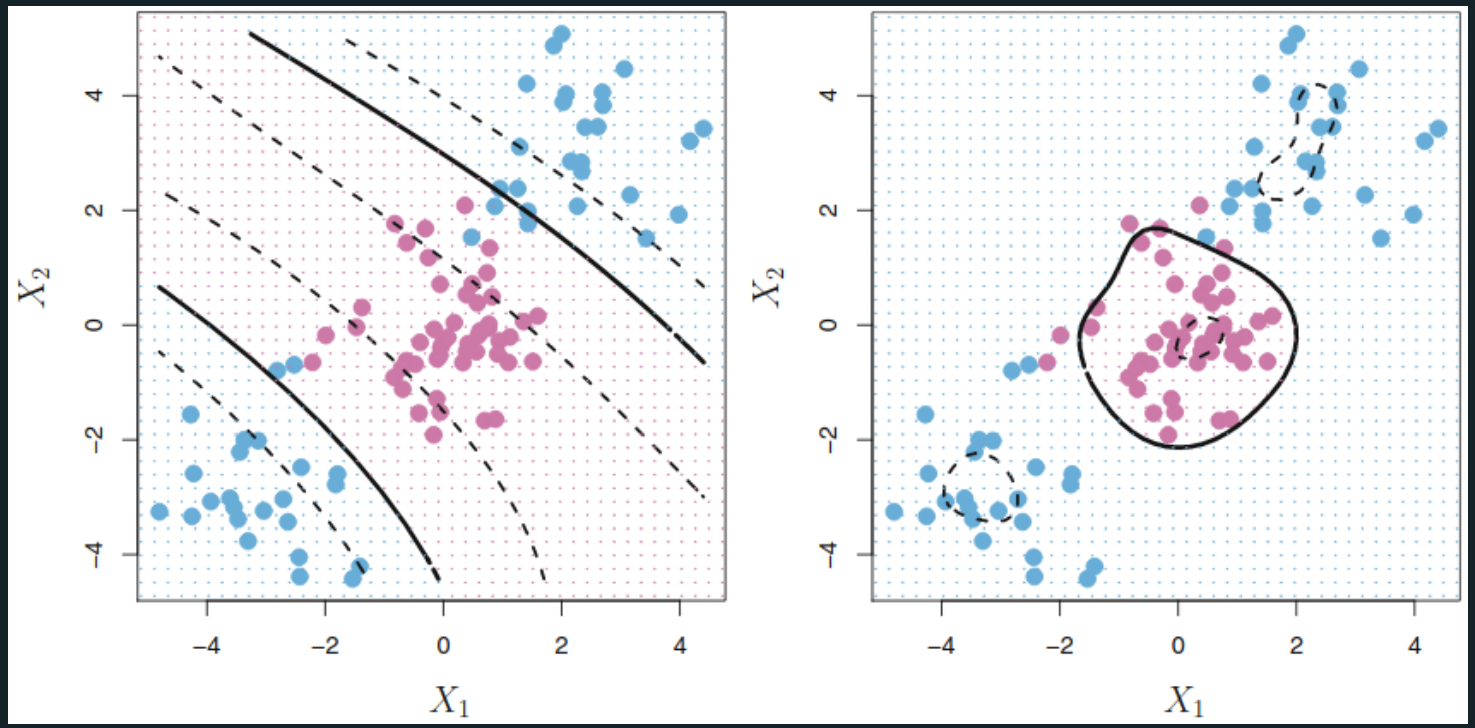


Overcoming the linearity assumption:

Support vector machines

- *Idea 1*: (polynomial) transformation of the features + StandardScaler + LinearSVC.
- *Idea 2*: convert a linear classifier into a classifier that produced **non-linear decision boundaries**. → using a **Kernel** such as:
  - Gaussian RBF kernel
  - Polynomial kernel
- **We do not open the kernel box.**
  - Just think as them as a way to construct non-linear hyperplans
  - Try out different kernel and distance specification

Support vector machines



- *Left*: polynomial kernel of degree 3;
- *Right*: radial kernel

Performance measures

Confusion Matrix

- For comparing the predictions of the fitted model to the actual classes.
- After applying a classifier to a data set with known labels *Yes* and *No*:

		Predicted class	
		no	yes
True class	no	TN	FP
	yes	FN	TP

Precision and Recall

- **Accuracy** : Proportion of rightly guessed observations
  - $\frac{\text{True Positives}+\text{True Negative}}{N}$
- **Precision**: accuracy of positive predictions
  - $\frac{\text{True Positives}}{\text{True Positives}+\text{False Positives}}$
  - decreases with false positives.
- **Recall**: proportion of true positives among all actual positives
  - $\frac{\text{True Positives}}{\text{True Positives}+\text{False Negatives}}$
  - decreases with false negatives.

F1 Score

- The F_1 score provides a single combined metric it is the **harmonic mean** of precision and recall

$$F_1 = \frac{2}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

$$= \frac{\text{Total Positives}}{\text{Total Positives} + \frac{1}{2}(\text{False Negatives} + \text{False Positives})}$$

- The harmonic mean gives **more weight to low values**.
- The F1 score values precision and recall **symmetrically**.

The Precision/Recall Trade-off

- F_1 favors classifiers with similar precision and recall,
- but sometimes you want **asymmetry**:

1. **low recall + high precision is better**
    - e.g. **deciding "guilty" in court**, you might prefer a model that
    - lets many actual-guilty go free (high false negatives ↔ low recall)...
    - ... but has very few actual-innocent put in jail (low false positives ↔ high precision
2. **high recall + low precision is better**

The Precision/Recall Trade-off

- F_1 favors classifiers with similar precision and recall,
- but sometimes you want **asymmetry**:

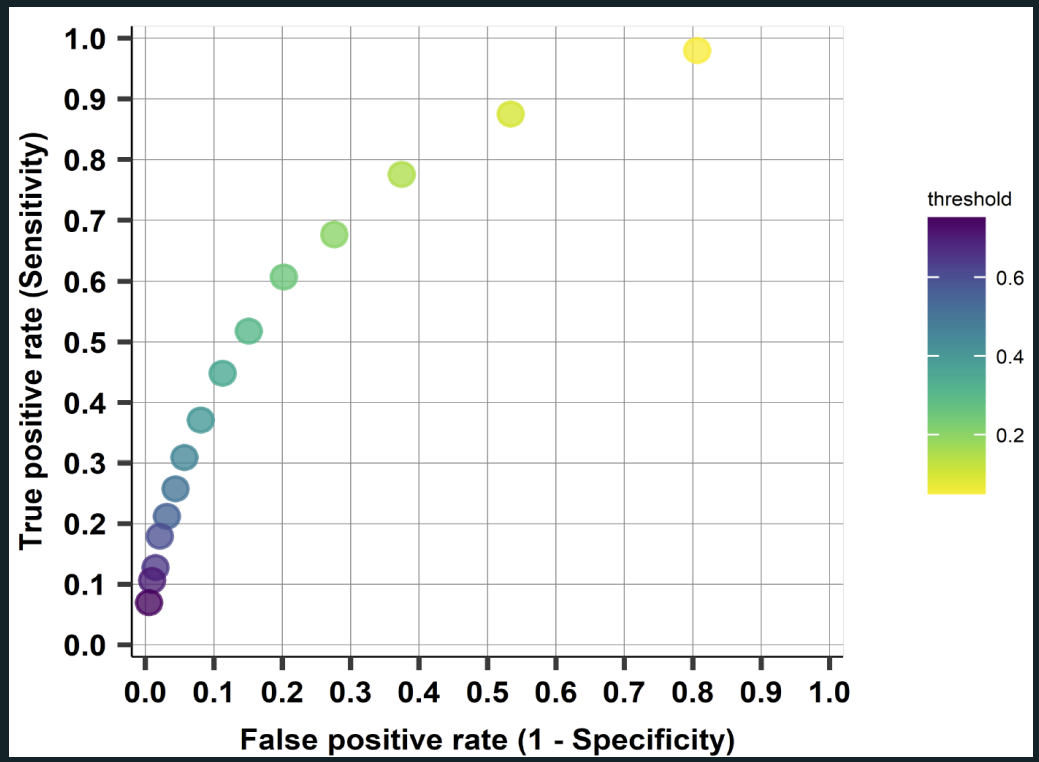
1. **low recall + high precision is better**
2. **high recall + low precision is better**
    - e.g classifier to **detect bombs during flight screening**, you might prefer a model that:
    - has many false alarms (low precision)...
    - ... to minimize the number of misses (high recall).

ROC Curve

- The ROC curve is a popular graphic for simultaneously displaying the types of errors for classification problems at various threshold settings
- For each threshold, we can compute confusion table –> calculate sensitivity and specificity
- The ROC curve of
  - a completely random probability prediction is the 45 degree line
  - a perfect probability prediction would jump from zero to one and stay at one

ROC Curve

- Plots *true positive rate* (recall) against the *false positive rate* ($\frac{FP}{FP+TN}$):



ROC Curve and AUC

- The Area Under (the ROC) Curve (AUC) is a popular metric ranging between:
  - 0.5
    - **random classification**
    - ROC curve = first diagonal
  - and 1
    - **perfect classification**
    - = area of the square
  - better classifier → ROC curve toward the top-left corner
- Good measure for model comparison